

Concept 1 | Arithmetic Sequences

English Student Edition • Focused formulas and guided practice

Formula opener

Term

$$a_n = a + (n - 1)d$$

Sum

$$S_n = \frac{n}{2}(2a + (n - 1)d)$$

Test

$$a_{n+1} - a_n = d$$

Classified Practice

Arithmetic Sequences

Q001 **Written**

When the general term of the sequence $\{a_n\}$ is $a_n = 2n - 3$, find the first term through the 3rd term.

Q002 Written

When the general term of the sequence $\{a_n\}$ is $a_n = n^2 - 7n$, find the first term through the 3rd term.

Classified Practice

Arithmetic Sequences

Q003 **Written**

When the general term of the sequence $\{a_n\}$ is $a_n = 5n - 8$, find the first term through the 3rd term.

Q004 Written

When the general term of the sequence $\{a_n\}$ is $a_n = -3n + 1$, find the first term through the 3rd term.

Classified Practice

Arithmetic Sequences

Q005 **Written**

When the general term of the sequence $\{a_n\}$ is $a_n = -2n - 4$, find the first term through the 3rd term.

Q006 Written

When the general term of the sequence $\{a_n\}$ is $a_n = n^2 - 1$, find the first term through the 3rd term.

Classified Practice

Arithmetic Sequences

Q007 **Written**

When the general term of the sequence $\{a_n\}$ is $a_n = n^2 + n + 1$, find the first term through the 3rd term.

Q008 Written

When the general term of the sequence $\{a_n\}$ is $a_n = n^2 + 2n + 4$, find the first term through the 3rd term.

Classified Practice

Arithmetic Sequences

Q009 MCQ

For $a_n = 3n - 5$, the first three terms (a_1, a_2, a_3) are:

Choose the correct option.

A $(-2, 1, 5)$

B $(-2, 1, 4)$

C $(-2, 0, 4)$

D $(2, -1, -4)$

Q010 MCQ

For $a_n = 4n + 7$, the first three terms (a_1, a_2, a_3) are:

Choose the correct option.

A (11, 15, 20)

B (11, 14, 19)

C (11, 15, 19)

D (-11, -15, -19)

Classified Practice

Arithmetic Sequences

Q011 MCQ

For $a_n = n^2 - 4n + 6$, the first three terms (a_1, a_2, a_3) are:

Choose the correct option.

A (3, 2, 4)

B (3, 1, 3)

C (-3, -2, -3)

D (3, 2, 3)

Q012 MCQ

For $a_n = 2n^2 + 3$, the first three terms (a_1, a_2, a_3) are:

Choose the correct option.

A (5, 11, 21)

B (5, 11, 22)

C (5, 10, 21)

D (−5, −11, −21)

Classified Practice

Arithmetic Sequences

Q013 MCQ

For $a_n = n^2 + 5n - 2$, the first three terms (a_1, a_2, a_3) are:

Choose the correct option.

A (4, 10, 17)

B (4, 10, 16)

C (4, 9, 16)

D (-4, -10, -16)

Q014 MCQ

For $a_n = 5 - 2n$, the first three terms (a_1, a_2, a_3) are:

Choose the correct option.

A (3, 1, 0)

B (3, 0, -1)

C (3, 1, -1)

D (-3, -1, 1)

Classified Practice

Arithmetic Sequences

Q015 MCQ

For $a_n = 2n^2 - 7n + 9$, the first three terms (a_1, a_2, a_3) are:

Choose the correct option.

A (4, 3, 7)

B (4, 2, 6)

C (-4, -3, -6)

D (4, 3, 6)

Q016 MCQ

For $a_n = n^2 - 3n - 1$, the first three terms (a_1, a_2, a_3) are:

Choose the correct option.

A $(-3, -3, -1)$

B $(-3, -3, 0)$

C $(-3, -4, -1)$

D $(3, 3, 1)$

Classified Practice

Arithmetic Sequences

Q017 MCQ

For $a_n = 3n^2 - 2$, the first three terms (a_1, a_2, a_3) are:

Choose the correct option.

A (1, 10, 26)

B (1, 10, 25)

C (1, 9, 25)

D (-1, -10, -25)

Q018 MCQ**For $a_n = n^2 + 2$, the first three terms (a_1, a_2, a_3) are:****Choose the correct option.****A** (3, 6, 12)**B** (3, 5, 11)**C** (3, 6, 11)**D** (−3, −6, −11)

Classified Practice

Arithmetic Sequences

Q019 MCQ

For $a_n = 7n - 10$, the first three terms (a_1, a_2, a_3) are:

Choose the correct option.

A $(-3, 4, 12)$

B $(-3, 3, 11)$

C $(3, -4, -11)$

D $(-3, 4, 11)$

Q020 MCQ

For $a_n = 2n^2 + n - 1$, the first three terms (a_1, a_2, a_3) are:

Choose the correct option.

A (2, 9, 20)

B (2, 9, 21)

C (2, 8, 20)

D (-2, -9, -20)

Classified Practice

Arithmetic Sequences

Q021 Written

When the general term of the sequence $\{a_n\}$ is $a_n = 2 \times 3^n$, find the first term through the 3rd term.

Q022 Written

When the general term of the sequence $\{a_n\}$ is $a_n = 3^n - 1$, find the first term through the 3rd term.

Classified Practice

Arithmetic Sequences

Q023 Written

When the general term of the sequence $\{a_n\}$ is $a_n = (-3)^n$, find the first term through the 3rd term.

Q024 Written

When the general term of the sequence $\{a_n\}$ is $a_n = 3 \times (-2)^n$, find the first term through the 3rd term.

Classified Practice

Arithmetic Sequences

Q025 MCQ

For $a_n = 2^n$, the first three terms (a_1, a_2, a_3) are:

Choose the correct option.

A (2, 4, 8)

B (2, 4, 9)

C (2, 3, 8)

D (-2, -4, -8)

Q026 MCQ**For $a_n = (-2)^n$, the first three terms (a_1, a_2, a_3) are:****Choose the correct option.****A** $(-2, 4, -7)$ **B** $(-2, 4, -8)$ **C** $(-2, 3, -8)$ **D** $(2, -4, 8)$

Classified Practice

Arithmetic Sequences

Q027 MCQ

For $a_n = 2 \times 3^n$, the first three terms (a_1, a_2, a_3) are:

Choose the correct option.

A (6, 18, 55)

B (6, 17, 54)

C (6, 18, 54)

D (-6, -18, -54)

Q028 MCQ**For $a_n = 5 \times (-3)^n$, the first three terms (a_1, a_2, a_3) are:****Choose the correct option.****A** $(-15, 45, -134)$ **B** $(-15, 44, -135)$ **C** $(15, -45, 135)$ **D** $(-15, 45, -135)$

Classified Practice

Arithmetic Sequences

Q029 MCQ

For $a_n = 4^n - 1$, the first three terms (a_1, a_2, a_3) are:

Choose the correct option.

A (3, 15, 63)

B (3, 15, 64)

C (3, 14, 63)

D (-3, -15, -63)

Q030 MCQ

For $a_n = 2 \times 3^{n-1}$, the first three terms (a_1, a_2, a_3) are:

Choose the correct option.

A (2, 6, 19)

B (2, 6, 18)

C (2, 5, 18)

D (-2, -6, -18)

Classified Practice

Arithmetic Sequences

Q031 MCQ

For $a_n = (-1)^{n-1}5^n$, the first three terms (a_1, a_2, a_3) are:

Choose the correct option.

A $(5, -25, 126)$

B $(5, -26, 125)$

C $(5, -25, 125)$

D $(-5, 25, -125)$

Q032 MCQ

For $a_n = (-3)^n + 2$, the first three terms (a_1, a_2, a_3) are:

Choose the correct option.

A $(-1, 11, -24)$

B $(-1, 10, -25)$

C $(1, -11, 25)$

D $(-1, 11, -25)$

Classified Practice

Arithmetic Sequences

Q033 Written

When the general term of the sequence $\{a_n\}$ is $a_n = \frac{n}{n+1}$, find the first term through the 3rd term.

Q034 Written

When the general term of the sequence $\{a_n\}$ is $a_n = \frac{n}{2n+1}$, find the first term through the 3rd term.

Classified Practice

Arithmetic Sequences

Q035 Written

When the general term of the sequence $\{a_n\}$ is $a_n = \frac{1}{3n-2}$, find the first term through the 3rd term.

Q036 Written

When the general term of the sequence $\{a_n\}$ is $a_n = \frac{2n}{n+3}$, find the first term through the 3rd term.

Classified Practice

Arithmetic Sequences

Q037 Written

When the general term of the sequence $\{a_n\}$ is $a_n = \frac{n+1}{2n-1}$, find the first term through the 3rd term.

Q038 MCQ

For $a_n = \frac{n+1}{n+2}$, the first three terms (a_1, a_2, a_3) are:

Choose the correct option.

A $(\frac{2}{3}, \frac{3}{4}, \frac{4}{6})$

B $(\frac{2}{3}, \frac{3}{3}, \frac{4}{5})$

C $(\frac{2}{3}, \frac{3}{4}, \frac{4}{5})$

D $(-\frac{2}{3}, -\frac{3}{4}, -\frac{4}{5})$

Classified Practice

Arithmetic Sequences

Q039 MCQ

For $a_n = \frac{2n-1}{3n+1}$, the first three terms (a_1, a_2, a_3) are:

Choose the correct option.

A $(\frac{1}{4}, \frac{3}{7}, \frac{1}{3})$

B $(\frac{1}{4}, \frac{3}{6}, \frac{1}{2})$

C $(-\frac{1}{4}, -\frac{3}{7}, -\frac{1}{2})$

D $(\frac{1}{4}, \frac{3}{7}, \frac{1}{2})$

Q040 MCQ

For $a_n = \frac{n^2+1}{n+1}$, the first three terms (a_1, a_2, a_3) are:

Choose the correct option.

A $(1, \frac{5}{3}, \frac{5}{2})$

B $(1, \frac{5}{3}, \frac{5}{3})$

C $(1, \frac{5}{2}, \frac{5}{2})$

D $(-1, -\frac{5}{3}, -\frac{5}{2})$

Classified Practice

Arithmetic Sequences

Q041 MCQ

For $a_n = \frac{3n}{2n-1}$, the first three terms (a_1, a_2, a_3) are:

Choose the correct option.

A $(3, 2, \frac{9}{6})$

B $(3, 2, \frac{9}{5})$

C $(3, 1, \frac{9}{5})$

D $(-3, -2, -\frac{9}{5})$

Q042 MCQ

For $a_n = \frac{2n+3}{n+4}$, the first three terms (a_1, a_2, a_3) are:

Choose the correct option.

A $(1, \frac{7}{6}, \frac{9}{8})$

B $(1, \frac{7}{5}, \frac{9}{7})$

C $(1, \frac{7}{6}, \frac{9}{7})$

D $(-1, -\frac{7}{6}, -\frac{9}{7})$

Classified Practice

Arithmetic Sequences

Q043 MCQ

For $a_n = \frac{n}{3n-2}$, the first three terms (a_1, a_2, a_3) are:

Choose the correct option.

A $(1, \frac{1}{4}, \frac{3}{8})$

B $(1, \frac{1}{3}, \frac{3}{7})$

C $(-1, -\frac{1}{4}, -\frac{3}{7})$

D $(1, \frac{1}{4}, \frac{3}{7})$

Q044 Written

For the sequence $\{a_n\}$: 0, 1, 4, 9, 16, ..., deduce the general term.

Classified Practice

Arithmetic Sequences

Q045 **Written**

For the sequence $\{a_n\} : 0, 1, 8, 27, 64, \dots$, deduce the general term.

Q046 Written

For the sequence $\{a_n\}$: 1, 4, 9, 16, 25, ..., deduce the general term.

Classified Practice

Arithmetic Sequences

Q047 **Written**

For the sequence $\{a_n\} : 7^2, 8^2, 9^2, 10^2, 11^2, \dots$, deduce the general term.

Q048 MCQ

Which formula gives the general term of $\{a_n\} : 1, 8, 27, 64, \dots$?:

Choose the correct option.

A n^2

B n^3

C n^5

D n^6

Classified Practice

Arithmetic Sequences

Q049 MCQ

Which formula gives the general term of $\{a_n\}$: 9, 16, 25, 36, ...?:

Choose the correct option.

A $(n + 2)^3$

B $(n + 2)^4$

C $(n + 2)^2$

D $(n + 2)^5$

Q050 MCQ

Which formula gives the general term of $\{a_n\} : -1, 0, 1, 8, \dots$?:

Choose the correct option.

A $(n - 2)^2$

B $(n - 2)^5$

C $(n - 2)^6$

D $(n - 2)^3$

Classified Practice

Arithmetic Sequences

Q051 MCQ

Which formula gives the general term of $\{a_n\}$: 36, 49, 64, 81, ...?:

Choose the correct option.

A $(n + 5)^2$

B $(n + 5)^3$

C $(n + 5)^4$

D $(n + 5)^5$

Q052 MCQ

Which formula gives the general term of $\{a_n\} : 1, 9, 25, 49, \dots$?:

Choose the correct option.

A $(2n + 1)^2$

B $(2n - 1)^2$

C $(2n - 1)^4$

D $(2n - 1)^5$

Classified Practice

Arithmetic Sequences

Q053 MCQ

Which formula gives the general term of $\{a_n\} : 0, 1, 16, 81, \dots$?:

Choose the correct option.

A $(n + 1)^4$

B $(n - 1)^6$

C $(n - 1)^4$

D $(n - 1)^7$

Q054 Written

For the sequence $\{a_n\}$: 2, 5, 8, 11, 14, ..., deduce the general term.

Classified Practice

Arithmetic Sequences

Q055 **Written**

For the sequence $\{a_n\} : 0, 2, 4, 6, 8, \dots$, deduce the general term.

Q056 Written

For the sequence $\{a_n\} : 0, 3, 6, 9, 12, \dots$, deduce the general term.

Classified Practice

Arithmetic Sequences

Q057 Written

For the sequence $\{a_n\} : 4, 9, 14, 19, 24, \dots$, deduce the general term.

Q058 Written

For the sequence $\{a_n\}$: 6, 13, 20, 27, 34, ..., deduce the general term.

Classified Practice

Arithmetic Sequences

Q059 MCQ

Which formula gives the general term of $\{a_n\}$: 7, 12, 17, 22, ...?:

Choose the correct option.

A $5n + 2$

B $(5n + 2) + 1$

C $(5n + 2) - 1$

D $-(5n + 2)$

Q060 MCQ**Which formula gives the general term of $\{a_n\} : -4, -1, 2, 5, \dots$?:****Choose the correct option.**

A $(3n - 7) + 1$

B $3n - 7$

C $(3n - 7) - 1$

D $-(3n - 7)$

Classified Practice

Arithmetic Sequences

Q061 MCQ

Which formula gives the general term of $\{a_n\}$: 10, 6, 2, -2 , ...?:

Choose the correct option.

A $(14 - 4n) + 1$

B $(14 - 4n) - 1$

C $14 - 4n$

D $-(14 - 4n)$

Q062 MCQ

Which formula gives the general term of $\{a_n\} : \frac{1}{2}, 1, \frac{3}{2}, 2, \dots$?:

Choose the correct option.

A $\left(\frac{n}{2}\right) + 1$

B $\left(\frac{n}{2}\right) - 1$

C $-\left(\frac{n}{2}\right)$

D $\frac{n}{2}$

Classified Practice

Arithmetic Sequences

Q063 Written

For the sequence $\{a_n\} : 2, -4, 6, -8, 10, \dots$, deduce the general term.

Q064 Written

For the sequence $\{a_n\}$: $-3, 6, -9, 12, -15, \dots$, deduce the general term.

Classified Practice

Arithmetic Sequences

Q065 **Written**

For the sequence $\{a_n\}$: $-5, 10, -15, 20, -25, \dots$, deduce the general term.

Q066 Written

For the sequence $\{a_n\}$: $-1, 4, -7, 10, -13, \dots$, deduce the general term.

Classified Practice

Arithmetic Sequences

Q067 Written

For the sequence $\{a_n\} : 1, \frac{2}{3}, \frac{3}{5}, \frac{4}{7}, \frac{5}{9}, \dots$, deduce the general term.

Q068 Written

For the sequence $\{a_n\} : 1, \frac{3}{4}, \frac{5}{9}, \frac{7}{16}, \frac{9}{25}, \dots$, deduce the general term.

Classified Practice

Arithmetic Sequences

Q069 Written

For the sequence $\{a_n\} : 1, -\frac{1}{3}, \frac{1}{5}, -\frac{1}{7}, \frac{1}{9}, \dots$, deduce the general term.

Q070 Written

For the sequence $\{a_n\} : \frac{1}{2}, \frac{2}{3}, \frac{3}{4}, \frac{4}{5}, \frac{5}{6}, \dots$, deduce the general term.

Classified Practice

Arithmetic Sequences

Q071 **Written**

For the sequence $\{a_n\}$: 1×3 , 2×9 , 3×27 , 4×81 , 5×243 , ..., deduce the general term.

Q072 Written

For the sequence $\{a_n\} : 1 \times \frac{1}{2}, 2 \times \frac{1}{3}, 3 \times \frac{1}{4}, 4 \times \frac{1}{5}, 5 \times \frac{1}{6}, \dots$, deduce the general term then find the 20th term.

Classified Practice

Arithmetic Sequences

Q073 **Written**

Find the first term and the common difference of the arithmetic sequences: (a) 2, 5, 8, 11, ...; (b) 3, -2, -7, -12, -17,

Q074 Written**Find the first term and common difference of: (a) 1, 5, 9, 13, 17, ...; (b) 23, 19, 15, 11, 7,**

Classified Practice

Arithmetic Sequences

Q075 MCQ

The listed terms $\{a_n\} = (4, 9, 14, 19, \dots)$ form an arithmetic sequence. Find the ordered pair (a, d) of first term and common difference select the correct ordered pair a, d :

Choose the correct option.

- A (4, 4)
- B (4, 5)
- C (5, 4)
- D (4, -5)

Q076 MCQ

The listed terms $\{a_n\} = (-7, -3, 1, 5, \dots)$ form an arithmetic sequence. Find the ordered pair (a, d) of first term and common difference select the correct ordered pair a, d :

Choose the correct option.

- A $(-7, 4)$
- B $(-7, -4)$
- C $(7, 4)$
- D $(-3, 4)$

Classified Practice

Arithmetic Sequences

Q077 MCQ

The listed terms $\{a_n\} = (18, 13, 8, 3, \dots)$ form an arithmetic sequence. Find the ordered pair (a, d) of first term and common difference select the correct ordered pair a, d :

Choose the correct option.

- A (18, 5)
 - B (13, -5)
 - C (18, -5)
 - D (-18, -5)
-
-
-

Q078 MCQ

The listed terms $\{a_n\} = (12, 8, 4, 0, \dots)$ form an arithmetic sequence. Find the ordered pair (a, d) of first term and common difference select the correct ordered pair a, d :

Choose the correct option.

A (12, 4)

B (8, -4)

C (-12, -4)

D (12, -4)

Classified Practice

Arithmetic Sequences

Q079 Written

Find the first term and common difference of: (a) 3, 5, 7, 9, 11, ...; (b) $-13, -9, -5, -1, 3, \dots$; (c) $-2, 1, 4, 7, 10, \dots$; (d) $10, 7, 4, 1, -2, \dots$; (e) $6, \frac{16}{3}, \frac{14}{3}, 4, \frac{10}{3}, \dots$; (f) $-4, -\frac{9}{2}, -5, -\frac{11}{2}, -6, \dots$

Classified Practice

Arithmetic Sequences

Q080 MCQ

The listed terms $\{a_n\} = (\frac{1}{2}, \frac{5}{6}, \frac{7}{6}, \frac{3}{2}, \dots)$ form an arithmetic sequence. Find the ordered pair (a, d) of first term and common difference select the correct ordered pair a, d :

Choose the correct option.

A $(\frac{1}{2}, \frac{2}{3})$

B $(\frac{1}{2}, \frac{1}{3})$

C $(\frac{1}{3}, \frac{1}{2})$

D $(-\frac{1}{2}, \frac{1}{3})$

Q081 MCQ

The listed terms $\{a_n\} = (-\frac{3}{4}, -\frac{1}{4}, \frac{1}{4}, \frac{3}{4}, \dots)$ form an arithmetic sequence. Find the ordered pair (a, d) of first term and common difference select the correct ordered pair a, d :

Choose the correct option.

A $(-\frac{3}{4}, \frac{1}{2})$

B $(-\frac{3}{4}, -\frac{1}{2})$

C $(\frac{3}{4}, \frac{1}{2})$

D $(-\frac{1}{4}, \frac{1}{2})$

Classified Practice

Arithmetic Sequences

Q082 MCQ

The listed terms $\{a_n\} = (\frac{5}{3}, 2, \frac{7}{3}, \frac{8}{3}, \dots)$ form an arithmetic sequence. Find the ordered pair (a, d) of first term and common difference select the correct ordered pair a, d :

Choose the correct option.

A $(\frac{5}{3}, \frac{2}{3})$

B $(2, \frac{1}{3})$

C $(\frac{5}{3}, \frac{1}{3})$

D $(-\frac{5}{3}, \frac{1}{3})$

Q083 MCQ

The listed terms $\{a_n\} = (-\frac{5}{2}, -1, \frac{1}{2}, 2, \dots)$ form an arithmetic sequence. Find the ordered pair (a, d) of first term and common difference select the correct ordered pair a, d :

Choose the correct option.

A $(-\frac{5}{2}, -\frac{3}{2})$

B $(-\frac{5}{2}, \frac{3}{2})$

C $(\frac{5}{2}, \frac{3}{2})$

D $(-1, \frac{3}{2})$

Classified Practice

Arithmetic Sequences

Q084 **Written**

The sequences are arithmetic. Find the common difference and the numbers that fit in (A) and (B): (a) $A, 2, -1, B, \dots$; (b) $13, A, B, -2, \dots$

Q085 Written

The sequences are arithmetic. Find the common difference and the missing numbers: (a) $2, 9, A, B, \dots$; (b) $A, 17, B, 5, \dots$

Classified Practice

Arithmetic Sequences

Q086 MCQ

The sequence $\{a_n\} = (6, A, 14, B, \dots)$ is arithmetic. Determine the common difference and every missing letter select the correct values of d and the missing terms:

Choose the correct option.

A $d = 4, A = 10, B = 18$

B $d = 8, A = 14, B = 22$

C $d = 4, A = 12, B = 18$

D $d = -4, A = 2, B = 10$

Q087 MCQ

The sequence $\{a_n\} = (A, 5, 9, B, \dots)$ is arithmetic. Determine the common difference and every missing letter select the correct values of d and the missing terms:

Choose the correct option.

A $d = 4, A = 9, B = 13$

B $d = -4, A = 9, B = 1$

C $d = 4, A = 1, B = 13$

D $d = 2, A = 3, B = 11$

Classified Practice

Arithmetic Sequences

Q088 MCQ

The sequence $\{a_n\} = (11, A, B, 23, \dots)$ is arithmetic. Determine the common difference and every missing letter select the correct values of d and the missing terms:

Choose the correct option.

A $d = 3, A = 14, B = 17$

B $d = 4, A = 15, B = 19$

C $d = 4, A = 14, B = 17$

D $d = -4, A = 7, B = 3$

Classified Practice

Arithmetic Sequences

Q089 Written

Find the common difference and missing terms: (a) $8, 15, A, B, \dots$; (b) $7, 13, A, B, \dots$; (c) $A, 9, 4, B, \dots$; (d) $A, 15, B, 9, \dots$; (e) $11, A, B, 20, \dots$; (f) $17, A, B, 5, C, \dots$

Classified Practice

Arithmetic Sequences

Q090 **Written**

For each arithmetic sequence, find the general term and the 7th term: (a) first term (4), common difference (3); (b) first term (2), common difference -3 .

Q091 Written

For each arithmetic sequence, find the general term and the 7th term: (a) $a = 5, d = 4$; (b) $a = -2, d = 3$; (c) $a = 10, d = -3$; (d) $a = -\frac{1}{2}, d = -\frac{1}{2}$.

Classified Practice

Arithmetic Sequences

Q092 **Written**

For the arithmetic sequence a_n with first term (8) and common difference -6 , find the general term and the 20th term.

Q093 Written

For each arithmetic sequence, find the general term and the 10th term: (a) $a = -7, d = 3$; (b) $a = -2, d = -3$; (c) $a = 7, d = \frac{1}{2}$; (d) $a = \frac{2}{3}, d = -\frac{1}{3}$.

Classified Practice

Arithmetic Sequences

Q094 Written

In an arithmetic sequence, $a_3 + a_7 = 38$ and $\frac{a_4}{a_{12}} = \frac{2}{5}$. Find the first term and common difference.

Q095 Written

Find the general term of the arithmetic sequence whose first term is -5 and whose 17th term is 75 .

Classified Practice

Arithmetic Sequences

Q096 Written

The first term is 6, and the 4th term is 18. Find a_n .

Q097 Written

The first term is 20, and the 8th term is -1 . Find a_n .

Classified Practice

Arithmetic Sequences

Q098 **Written**

The first term is -8 , and the 8th term is 55 . Find a_n .

Q099 Written

The first term is -18 , and the 10th term is -63 . Find a_n .

Classified Practice

Arithmetic Sequences

Q100 MCQ

An arithmetic sequence has $a = 4$ and $a_7 = 22$. Select the correct pair (a_n, d) :

Choose the correct option.

A $(3n - 1, 3)$

B $(4n, 4)$

C $(3n + 1, 4)$

D $(a_n, d) = (3n + 1, 3)$

Q101 MCQ

An arithmetic sequence has $a = -3$ and $a_9 = 29$. Select the correct pair (a_n, d) :

Choose the correct option.

A $(4n - 3, 4)$

B $(a_n, d) = (4n - 7, 4)$

C $(5n - 8, 5)$

D $(4n - 7, -4)$

Classified Practice

Arithmetic Sequences

Q102 MCQ

An arithmetic sequence has $a = 12$ and $a_5 = -4$. Select the correct pair (a_n, d) :

Choose the correct option.

A $(a_n, d) = (16 - 4n, -4)$

B $(12 - 4n, -4)$

C $(16 + 4n, 4)$

D $(16 - 3n, -3)$

Q103 MCQ

An arithmetic sequence has $a = 5$ and $a_{11} = 45$. Select the correct pair (a_n, d) :

Choose the correct option.

A $(5n, 5)$

B $(a_n, d) = (4n + 1, 4)$

C $(4n - 1, 4)$

D $(4n + 1, -4)$

Classified Practice

Arithmetic Sequences

Q104 **Written**

Find the general term of the arithmetic sequence whose common difference is 3 and whose 16th term is 81.

Q105 Written

The common difference is 2, and the 8th term is 4. Find a_n .

Classified Practice

Arithmetic Sequences

Q106 Written

The common difference is -4 , and the 5th term is -7 . Find a_n .

Q107 Written

The common difference is -7 , and the 6th term is -24 . Find a_n .

Classified Practice

Arithmetic Sequences

Q108 MCQ

An arithmetic sequence has common difference -2 and $a_4 = 1$. Select the correct pair (a_n, a) :

Choose the correct option.

A $(a_n, a) = (9 - 2n, 7)$

B $(7 - 2n, 9)$

C $(1 - 2n, 3)$

D $(9 + 2n, 7)$

Q109 MCQ

An arithmetic sequence has common difference 5 and $a_6 = 17$. Select the correct pair (a_n, a) :

Choose the correct option.

A $(5n - 3, -2)$

B $(17 + 5n, 2)$

C $(13 - 5n, 8)$

D $(a_n, a) = (5n - 13, -8)$

Classified Practice

Arithmetic Sequences

Q110 MCQ

An arithmetic sequence has common difference 4 and $a_{10} = 38$. Select the correct pair (a_n, a) :

Choose the correct option.

A $(4n + 2, -2)$

B $(2n + 4, 6)$

C $(a_n, a) = (4n - 2, 2)$

D $(4n - 2, 4)$

Q111 MCQ

An arithmetic sequence has common difference -3 and $a_8 = -18$. Select the correct pair (a_n, a) :

Choose the correct option.

A $(3 - 3n, 6)$

B $(6 + 3n, 3)$

C $(6 - 2n, 2)$

D $(a_n, a) = (6 - 3n, 3)$

Classified Practice

Arithmetic Sequences

Q112 **Written**

Find the general term of the arithmetic sequence $\{a_n\}$ whose 3rd term is 10 and the 8th term is 50.

Q113 Written

Find the general term of the arithmetic sequence whose 6th term is 2 and whose 15th term is -25 .

Classified Practice

Arithmetic Sequences

Q114 **Written****The 4th term is 14, and the 10th term is 62. Find a_n .**

Q115 Written

The 3rd term is 30, and the 10th term is 2. Find a_n .

Classified Practice

Arithmetic Sequences

Q116 **Written**

The 10th term is 49, and the 23rd term is 10. Find a_n .

Q117 Written

The 3rd term is 6, and the 10th term is 62. Find a_n .

Classified Practice

Arithmetic Sequences

Q118 MCQ

An arithmetic sequence has $a_3 = 5$ and $a_9 = 17$. Select the correct pair (a_n, a) :

Choose the correct option.

A $(2n + 1, -1)$

B $(n + 3, 2)$

C $(a_n, a) = (2n - 1, 1)$

D $(2n - 1, 5)$

Q119 MCQ

An arithmetic sequence has $a_4 = 7$ and $a_{10} = -5$. Select the correct pair (a_n, a) :

Choose the correct option.

A $(13 - 2n, 15)$

B $(15 + 2n, 5)$

C $(a_n, a) = (15 - 2n, 13)$

D $(2n - 15, -13)$

Classified Practice

Arithmetic Sequences

Q120 MCQ

An arithmetic sequence has $a_7 = 31$ and $a_{12} = 56$. Select the correct pair (a_n, a) :

Choose the correct option.

A $(5n + 4, -1)$

B $(a_n, a) = (5n - 4, 1)$

C $(4n - 5, 2)$

D $(5n - 4, 31)$

Q121 MCQ

An arithmetic sequence has $a_2 = -4$ and $a_8 = 14$. Select the correct pair (a_n, a) :

Choose the correct option.

A $(a_n, a) = (3n - 10, -7)$

B $(3n - 4, -1)$

C $(2n - 7, -5)$

D $(10 - 3n, 7)$

Classified Practice

Arithmetic Sequences

Q122 MCQ

An arithmetic sequence has $a_5 = 3$ and $a_{11} = 9$. Select the correct pair (a_n, a) :

Choose the correct option.

- A $\left(a_n = \frac{n-2}{2}, -\frac{1}{2}\right)$
B $\left(a_n = \frac{n+2}{2}, -\frac{5}{2}\right)$
C $\left(a_n = \frac{n-2}{2}, \frac{1}{2}\right)$
D $\left(a_n = \frac{2n-1}{3}, -\frac{1}{3}\right)$

Q123 MCQ

An arithmetic sequence has $a_2 = 6$ and $a_9 = 20$. Select the correct pair (a_n, a) :

Choose the correct option.

A $(a_n = 2n + 4, 2)$

B $(a_n = 2n + 2, 4)$

C $(a_n = n + 4, 5)$

D $(a_n = 2n - 2, 8)$

Classified Practice

Arithmetic Sequences

Q124 **Written**

The common difference is $\frac{2}{5}$, and the 30th term is $\frac{23}{5}$. Find a_n .

Q125 MCQ

An arithmetic sequence has $d = -\frac{3}{2}$ and $a_5 = -9$. Select the correct pair (a_n, a) :

Choose the correct option.

A $\left(a_n = -\frac{3(n+1)}{2}, -3\right)$

B $\left(a_n = -\frac{3(n-1)}{2}, -6\right)$

C $\left(a_n = \frac{3(n+1)}{2}, 3\right)$

D $\left(a_n = -\frac{n+1}{2}, -3\right)$

Classified Practice

Arithmetic Sequences

Q126 MCQ

An arithmetic sequence has $d = \frac{2}{3}$ and $a_{12} = 10$. Select the correct pair (a_n, a) :

Choose the correct option.

A $\left(a_n = \frac{2n+4}{3}, 2\right)$

B $\left(a_n = \frac{2n-6}{3}, -\frac{8}{3}\right)$

C $\left(a_n = \frac{n+6}{3}, \frac{8}{3}\right)$

D $\left(a_n = \frac{2n+6}{3}, \frac{8}{3}\right)$

Q127 Written

For the arithmetic sequence $\{a_n\}$ with common difference 4 and $a_8 = 33$, which term is 113?

Classified Practice

Arithmetic Sequences

Q128 **Written**

For the arithmetic sequence with first term 1 and common difference 3, determine which term is 40.

Q129 Written

For the arithmetic sequence with common difference -5 and 13th term -7 , determine which term is -52 .

Classified Practice

Arithmetic Sequences

Q130 **Written**

For the arithmetic sequence whose 10th term is -20 and 20th term is -50 , determine which term is -110 .

Q131 MCQ**For $a = 3, d = 4$, which term of the arithmetic sequence is 59?:****Choose the correct option.****A $n = 14$** **B $n = 16$** **C $n = 15$** **D $n = 13$**

Classified Practice

Arithmetic Sequences

Q132 MCQ

For $a = -5$, $d = 6$, which term of the arithmetic sequence is 55?:

Choose the correct option.

A $n = 10$

B $n = 12$

C $n = 9$

D $n = 11$

Q133 MCQ

For $a = 20$, $d = -3$, which term of the arithmetic sequence is -16 ?:

Choose the correct option.

A $n = 13$

B $n = 12$

C $n = 14$

D $n = 11$

Classified Practice

Arithmetic Sequences

Q134 MCQ

For $a = 1, d = 5$, which term of the arithmetic sequence is 86?:

Choose the correct option.

A $n = 17$

B $n = 18$

C $n = 19$

D $n = 16$

Q135 Written

For the arithmetic sequence $\{a_n\}$ with common difference 4 and $a_8 = 33$, find the first term greater than 500.

Classified Practice

Arithmetic Sequences

Q136 **Written**

For the arithmetic sequence with first term 1 and common difference 3, find the first term greater than 100.

Q137 MCQ**For $a = 2, d = 7$, find the first term greater than 100:****Choose the correct option.****A $n = 16$** **B $n = 15$** **C $n = 17$** **D $n = 14$**

Classified Practice

Arithmetic Sequences

Q138 MCQ

For $a = -9$, $d = 4$, find the first term greater than 50:**Choose the correct option.**A $n = 15$ B $n = 17$ C $n = 14$ D $n = 16$

Q139 MCQ**For $a = 6$, $d = 3$, find the first term greater than 200:****Choose the correct option.**

A $n = 65$

B $n = 67$

C $n = 64$

D $n = 66$

Classified Practice

Arithmetic Sequences

Q140 MCQ

For $a = -1$, $d = 9$, find the first term greater than 300:

Choose the correct option.

A $n = 34$

B $n = 35$

C $n = 36$

D $n = 33$

Q141 Written

For the arithmetic sequence with common difference -5 and 13th term -7 , find the first term smaller than -300 .

Classified Practice

Arithmetic Sequences

Q142 **Written**

For the arithmetic sequence whose 10th term is -20 and 20th term is -50 , find the first term smaller than -500 .

Q143 MCQ**For $a = 10$, $d = -4$, find the first term smaller than -100 :****Choose the correct option.**

A $n = 28$

B $n = 29$

C $n = 30$

D $n = 27$

Classified Practice

Arithmetic Sequences

Q144 MCQ

For $a = 25$, $d = -6$, find the first term smaller than -250 :

Choose the correct option.

A $n = 46$

B $n = 47$

C $n = 48$

D $n = 45$

Q145 MCQ**For $a = -3, d = -5$, find the first term smaller than -200 :****Choose the correct option.**

A $n = 40$

B $n = 42$

C $n = 41$

D $n = 39$

Classified Practice

Arithmetic Sequences

Q146 MCQ

For $a = 4$, $d = 6$, find the first term that is at least 100:

Choose the correct option.

A $n = 17$

B $n = 16$

C $n = 18$

D $n = 15$

Q147 MCQ

For the expression For $a = 5, d = 4$, does the sequence contain the term 42?, the required value is:

Choose the correct option.

- A $n = 9$
- B No term
- C $n = 10$
- D $n = 11$

Classified Practice

Arithmetic Sequences

Q148 **Written**

When $a, 6, 2a$ form an arithmetic sequence in this order, find a .

Q149 Written

When the three numbers $x, 6, x^2$ form an arithmetic sequence in this order, find x .

Classified Practice

Arithmetic Sequences

Q150 **Written**

When x^2 , x , -8 form an arithmetic sequence in this order, find x .

Q151 Written

When $-3, x, x^2$ form an arithmetic sequence in this order, find x .

Classified Practice

Arithmetic Sequences

Q152 Challenge

When $2, x, y, z$ form an arithmetic sequence in this order, and $2x, y + 1, z - y$ also form an arithmetic sequence in this order, find x, y, z .

Classified Practice

Arithmetic Sequences

Q153 Written

Find the sum (S_7) of a sequence with first term (8), last term (22), and (7) terms.

Q154 Written

Find (S_{10}) for a sequence with first term (7), last term (13), and (10) terms.

Classified Practice

Arithmetic Sequences

Q155 Written

Find (S_{10}) for a sequence with first term (3), last term (21), and (10) terms.

Q156 Written

Find (S_8) for a sequence with first term (5), last term (33), and (8) terms.

Classified Practice

Arithmetic Sequences

Q157 **Written****Find (S_{10}) for a sequence with first term (2), last term (29), and (10) terms.**

Q158 MCQ**A sequence has first term 6, last term 30, and 7 terms. Find its sum:****Choose the correct option.**

A 105

B 126

C 132

D 168

Classified Practice

Arithmetic Sequences

Q159 MCQ

A sequence has first term -4 , last term 20 , and 9 terms. Find its sum:

Choose the correct option.

A 64 B 72 C 81 D 144

Q160 MCQ**Find the sum of 10 terms whose first term is 15 and last term is 3:****Choose the correct option.**

A 45

B 90

C 150

D 180

Classified Practice

Arithmetic Sequences

Q161 MCQ

The first term is 7, the last term is 42, and there are 8 terms. Find S_8 :

Choose the correct option.

A 168

B 196

C 224

D 392

Q162 MCQ**Find the sum of 6 terms with first term -12 and last term 8 :****Choose the correct option.**A -120 B -12 C 12 D -2

Classified Practice

Arithmetic Sequences

Q163 MCQ

A sequence has $S_9 = 72$ and first term -4 . Find its last term:

Choose the correct option.

A 12

B 16

C 20

D 24

Q164 Written

Find (S_{13}) when the first term is (20), the common difference is (-5) , and the number of terms is (13).

Classified Practice

Arithmetic Sequences

Q165 **Written**

Find (S_{12}) when the first term is (-6) , the common difference is (3) , and there are (12) terms.

Q166 Written

Find (S_{15}) when the first term is (10), the common difference is (-3) , and there are (15) terms.

Classified Practice

Arithmetic Sequences

Q167 **Written**

Find (S_{11}) when the first term is (-4) , the common difference is (4) , and there are (11) terms.

Q168 Written

Find (S_{18}) when the first term is (3), the common difference is (2), and there are (18) terms.

Classified Practice

Arithmetic Sequences

Q169 MCQ

For $a = 4, d = 3, n = 12$, find S_{12} :**Choose the correct option.**

A 198

B 228

C 246

D 264

Q170 MCQ**For $a = 9, d = -2, n = 10$, find S_{10} :****Choose the correct option.****A** -90 **B** 0 **C** 9 **D** 90

Classified Practice

Arithmetic Sequences

Q171 MCQ

For $a = -5, d = 4, n = 9$, find S_9 :**Choose the correct option.**A -88 B 44 C 88 D 99

Q172 MCQ**For $a = 2, d = -3, n = 15$, find S_{15} :****Choose the correct option.****A** -315 **B** -285 **C** -270 **D** 285

Classified Practice

Arithmetic Sequences

Q173 MCQ

For $a = 11, d = 0, n = 8$, find S_8 :**Choose the correct option.**

A 11

B 44

C 88

D 96

Q174 MCQ**For $a = -3, d = -2, n = 7$, find S_7 :****Choose the correct option.****A** -42 **B** -54 **C** 54 **D** -63

Classified Practice

Arithmetic Sequences

Q175 Written

Find the sum (S) of a sequence with first term (2), common difference (3), and last term (47).

Q176 Written

Find (S) for a sequence with first term (5), common difference (3), and last term (44).

Classified Practice

Arithmetic Sequences

Q177 Written

Find (S) for a sequence with first term (3), common difference (4), and last term (75).

Q178 Written

Find (S) for a sequence with first term (2), common difference (5), and last term (57).

Classified Practice

Arithmetic Sequences

Q179 Written

Find (S) for a sequence with first term (4), common difference (3), and last term (100).

Q180 MCQ**A sequence has $a = 5, d = 4, l = 41$. Find its sum:****Choose the correct option.**

A 205

B 225

C 230

D 246

Classified Practice

Arithmetic Sequences

Q181 MCQ

A sequence has $a = -7$, $d = 5$, $l = 48$. Find its sum:

Choose the correct option.

A 228

B 240

C 246

D 264

Q182 MCQ**A sequence has $a = 12, d = 2, l = 50$. Find its sum:****Choose the correct option.****A** 580**B** 600**C** 620**D** 640

Classified Practice

Arithmetic Sequences

Q183 MCQ

A sequence has $a = -10$, $d = 6$, $l = 50$. Find its sum:

Choose the correct option.

A 198

B 210

C 220

D 240

Q184 MCQ**A sequence has $a = 18, d = -3, l = -9$. Find its sum:****Choose the correct option.****A** -45 **B** 36 **C** 45 **D** 90

Classified Practice

Arithmetic Sequences

Q185 MCQ

A sequence has $a = 40, d = -4, l = -12$. Find its sum:**Choose the correct option.**

A 168

B 196

C 224

D 280

Q186 Written

For the arithmetic sequence with first term (48) and common difference (-2) , find the sum from the 20th term to the 30th term.

Classified Practice

Arithmetic Sequences

Q187 **Written**

For the arithmetic sequence with first term (-53) and common difference (4) , find the sum from the 10th term to the 25th term.

Q188 MCQ

For $a = 3, d = 4$, find the sum from a_5 to a_{12} :

Choose the correct option.

A 252

B 264

C 276

D 288

Classified Practice

Arithmetic Sequences

Q189 MCQ

For $a = 20$, $d = -3$, find the sum from a_4 to a_{10} :

Choose the correct option.

A -14

B 7

C 14

D 28

Q190 MCQ**For $a = 60$, $d = -3$, find the sum from a_8 to a_{16} :****Choose the correct option.****A 216****B 234****C 243****D 270**

Classified Practice

Arithmetic Sequences

Q191 MCQ

For $a = 1, d = 7$, find the sum from a_3 to a_9 :**Choose the correct option.**

A 224

B 245

C 252

D 266

Q192 MCQ**For $a = 60, d = -3$, find the sum from a_9 to a_{17} :****Choose the correct option.****A 198****B 216****C 225****D 234**

Classified Practice

Arithmetic Sequences

Q193 MCQ

For $a = 12$, $d = 5$, find the sum from a_2 to a_8 :

Choose the correct option.

A 196

B 210

C 224

D 238

Q194 Written

For the arithmetic sequence $(90, 84, 78, \dots)$, find the sum from the 10th term to the 20th term.

Classified Practice

Arithmetic Sequences

Q195 **Written**

For the arithmetic sequence $(-9, -3, 3, \dots)$, find the sum from the 11th term to the 20th term.

Q196 Written

For the arithmetic sequence $(74, 69, 64, \dots)$, find the sum from the 8th term to the 18th term.

Classified Practice

Arithmetic Sequences

Q197 MCQ

For $a = 4$, $d = 5$, find the sum from a_6 to a_{13} :

Choose the correct option.

A 248

B 336

C 372

D 384

Q198 MCQ**For $a = -25$, $d = 6$, find the sum from a_5 to a_{11} :****Choose the correct option.****A** 105**B** 112**C** 119**D** 126

Classified Practice

Arithmetic Sequences

Q199 **Written****Find (S_n) for a sequence with first term (1) and common difference (4).**

Q200 Written**Find (S_n) for a sequence with first term (2) and common difference (-3) .**

Classified Practice

Arithmetic Sequences

Q201 **Written****Find (S_n) for a sequence with first term (0) and common difference (5).**

Q202 Written**Find (S_n) for a sequence with first term (3) and common difference (-4) .**

Classified Practice

Arithmetic Sequences

Q203 **Written**

Find (S_n) for a sequence with first term (-4) and common difference (-2) .

Classified Practice

Arithmetic Sequences

Q204 Challenge

For the arithmetic sequence $(94, 89, 84, \dots)$, find the sum of the odd-numbered terms from the first term to the 11th term.

Classified Practice

Arithmetic Sequences

Q205 **Written****Find the sum of the natural numbers from 1 to 100.**

Q206 Written**Find the sum of the natural numbers from 1 to 200.**

Classified Practice

Arithmetic Sequences

Q207 Written

Find the sum of the natural numbers from 1 to 150.

Q208 Written**Find the sum of the natural numbers from 1 to 70.**

Classified Practice

Arithmetic Sequences

Q209 Written

Find the sum of the natural numbers from 1 to 100.

Q210 Written

Find the sum of the natural numbers from 1 to 400.

Classified Practice

Arithmetic Sequences

Q211 **Written****Find the sum of the natural numbers from 1 to 250.**

Q212 Written**Find the sum of the natural numbers from 1 to 81.**

Classified Practice

Arithmetic Sequences

Q213 **Written****Find the sum of the natural numbers from 1 to 301.**

Q214 Written**Find the sum of the natural numbers from 1 to 99.**

Classified Practice

Arithmetic Sequences

Q215 MCQ

Find the sum of the natural numbers from 1 to 30:**Choose the correct option.**

A 435

B 465

C 495

D 525

Q216 MCQ**Find the sum of the natural numbers from 1 to 45:****Choose the correct option.**

- A 990
 - B 1035
 - C 1080
 - D 1125
-
-
-

Classified Practice

Arithmetic Sequences

Q217 MCQ

Find the sum of the natural numbers from 1 to 60:**Choose the correct option.**

A 1770

B 1830

C 1890

D 1950

Q218 MCQ**Find the sum of the natural numbers from 1 to 75:****Choose the correct option.****A** 2775**B** 2850**C** 2925**D** 3000

Classified Practice

Arithmetic Sequences

Q219 MCQ

Find the sum of the natural numbers from 1 to 90:**Choose the correct option.**

A 4005

B 4095

C 4185

D 4275

Q220 MCQ**Find the sum of the natural numbers from 1 to 125:****Choose the correct option.****A** 7750**B** 7875**C** 8000**D** 8125

Classified Practice

Arithmetic Sequences

Q221 MCQ

Find the sum of the natural numbers from 1 to 180:**Choose the correct option.**

A 16110

B 16290

C 16470

D 16650

Q222 MCQ**Find the sum of the natural numbers from 1 to 220:****Choose the correct option.****A** 24090**B** 24310**C** 24530**D** 24750

Classified Practice

Arithmetic Sequences

Q223 MCQ

Find the sum of the natural numbers from 1 to 275:**Choose the correct option.**

A 37675

B 37950

C 38225

D 38500

Q224 MCQ**Find the sum of the natural numbers from 1 to 350:****Choose the correct option.****A** 61075**B** 61425**C** 61775**D** 62125

Classified Practice

Arithmetic Sequences

Q225 **Written****Find the sum of the natural numbers from 101 to 200.**

Q226 Written**Find the sum of the natural numbers from 201 to 300.**

Classified Practice

Arithmetic Sequences

Q227 **Written****Find the sum of all two-digit natural numbers.**

Q228 MCQ**A sequence has first term 5, last term 35, and 7 terms. Select its sum:****Choose the correct option.****A** 133**B** 140**C** 147**D** 154

Classified Practice

Arithmetic Sequences

Q229 MCQ

A sequence has first term 12, last term 68, and 9 terms. Select its sum:

Choose the correct option.

A 351

B 360

C 369

D 378

Q230 MCQ**A sequence has first term -4, last term 20, and 9 terms. Select its sum:****Choose the correct option.****A** 63**B** 72**C** 81**D** 90

Classified Practice

Arithmetic Sequences

Q231 MCQ

A sequence has first term 7, last term 42, and 8 terms. Select its sum:

Choose the correct option.

A 188

B 196

C 204

D 212

Q232 MCQ**A sequence has first term 15, last term 3, and 10 terms. Select its sum:****Choose the correct option.****A** 80**B** 90**C** 100**D** 110

Classified Practice

Arithmetic Sequences

Q233 MCQ

A sequence has first term 10, last term 70, and 13 terms. Select its sum:

Choose the correct option.

A 507

B 520

C 533

D 546

Q234 MCQ**A sequence has first term 2, last term 57, and 12 terms. Select its sum:****Choose the correct option.****A** 342**B** 354**C** 366**D** 378

Classified Practice

Arithmetic Sequences

Q235 MCQ

A sequence has first term 25, last term 100, and 16 terms. Select its sum:

Choose the correct option.

- A 984
- B 1000
- C 1016
- D 1032

Q236 MCQ**A sequence has first term 101, last term 200, and 100 terms. Select its sum:****Choose the correct option.****A** 14950**B** 15050**C** 15150**D** 15250

Classified Practice

Arithmetic Sequences

Q237 MCQ

A sequence has first term 31, last term 91, and 16 terms. Select its sum:

Choose the correct option.

A 960

B 976

C 992

D 1008

Q238 Written**Find the sum of the multiples of 3 among the two-digit natural numbers.**

Classified Practice

Arithmetic Sequences

Q239 **Written****Find the sum of all two-digit natural numbers that are multiples of 5.**

Q240 Written**Find the sum of the multiples of 4 from 1 to 100.**

Classified Practice

Arithmetic Sequences

Q241 **Written****Find the sum of all multiples of 6 from 10 to 100.**

Q242 MCQ**Select the sum of the multiples of 3 from 3 to 99:****Choose the correct option.****A** 1680**B** 1683**C** 1686**D** 1689

Classified Practice

Arithmetic Sequences

Q243 MCQ

Select the sum of the multiples of 4 from 4 to 100:**Choose the correct option.**

A 1296

B 1300

C 1304

D 1308

Q244 Written

Find the sum of the natural numbers from 1 to 100 that leave a remainder of 1 when divided by 4.

Classified Practice

Arithmetic Sequences

Q245 **Written**

Find the sum of all two-digit natural numbers that leave a remainder of 3 when divided by 5.

Q246 Written**Find the sum of all two-digit odd natural numbers.**

Classified Practice

Arithmetic Sequences

Q247 **Written**

Find the sum of all natural numbers from 10 to 100 that are not divisible by 6.

Classified Practice

Arithmetic Sequences

Q248 **Challenge****Find the sum of the numbers from 1 to 100 which are not divisible by 4.**

Classified Practice

Arithmetic Sequences

Q249 **Written**

In an arithmetic sequence with first term 4 and common difference 3, find how many terms from the first term have sum 50.

Q250 Written

Find n when $a = 3$, $d = 2$ and $S_n = 63$.

Classified Practice

Arithmetic Sequences

Q251 MCQ

Select the correct option: Find n if $S_n = 48$ for $a = 3, d = 2$:**Choose the correct option.**

A 5

B 6

C 7

D 8

Q252 MCQ

Select the correct option: Find n if $S_n = 80$ for $a = 3, d = 2$:

Choose the correct option.

A 7

B 8

C 9

D 10

Classified Practice

Arithmetic Sequences

Q253 MCQ

Select the correct option: Find n if $S_n = 120$ for $a = 3, d = 2$:**Choose the correct option.**

A 9

B 10

C 11

D 12

Q254 MCQ

Select the correct option: Find n if $S_n = 168$ for $a = 3, d = 2$:**Choose the correct option.**

A 11

B 12

C 13

D 14

Classified Practice

Arithmetic Sequences

Q255 MCQ

Select the correct option: Find n if $S_n = 255$ for $a = 3, d = 2$:**Choose the correct option.**

A 14

B 15

C 16

D 17

Q256 MCQ**Select the correct option: Find n if $S_n = 360$ for $a = 3, d = 2$:****Choose the correct option.****A** 17**B** 18**C** 19**D** 20

Classified Practice

Arithmetic Sequences

Q257 MCQ

Select the correct option: Find n if $S_n = 440$ for $a = 3, d = 2$:**Choose the correct option.**

A 19

B 20

C 21

D 22

Q258 MCQ

Select the correct option: Find n if $S_n = 624$ for $a = 3, d = 2$:**Choose the correct option.**

A 23

B 24

C 25

D 26

Classified Practice

Arithmetic Sequences

Q259 MCQ

Select the correct option: Find n if $S_n = 840$ for $a = 3, d = 2$:**Choose the correct option.**

A 27

B 28

C 29

D 30

Q260 MCQ**Select the correct option: Find n if $S_n = 1088$ for $a = 3, d = 2$:****Choose the correct option.****A** 31**B** 32**C** 33**D** 34

Classified Practice

Arithmetic Sequences

Q261 MCQ

Select the correct option: Find n if $S_n = 1368$ for $a = 3, d = 2$:**Choose the correct option.**

A 35

B 36

C 37

D 38

Q262 MCQ**Select the correct option: Find n if $S_n = 1680$ for $a = 3, d = 2$:****Choose the correct option.****A** 39**B** 40**C** 41**D** 42

Classified Practice

Arithmetic Sequences

Q263 **Written**

For first term 17 and common difference -2 , find how many terms have sum -63 .

Q264 Written

Find n when $a = 24$, $d = -4$ and $S_n = -28$.

Classified Practice

Arithmetic Sequences

Q265 MCQ

Select the correct option: For $a = 10$, $d = 3$, find n when $S_n = 80$:**Choose the correct option.**

A 3

B 4

C 5

D 6

Q266 MCQ**Select the correct option: For $a = 10, d = 3$, find n when $S_n = 133$:****Choose the correct option.****A** 5**B** 6**C** 7**D** 8

Classified Practice

Arithmetic Sequences

Q267 MCQ

Select the correct option: For $a = 10$, $d = 3$, find n when $S_n = 198$:**Choose the correct option.**

A 7

B 8

C 9

D 10

Q268 MCQ**Select the correct option: For $a = 10, d = 3$, find n when $S_n = 275$:****Choose the correct option.****A** 9**B** 10**C** 11**D** 12

Classified Practice

Arithmetic Sequences

Q269 Written

Find the general term of $\{a_n\}$ with first term -60 and sum of the first 15 terms -60 .

Q270 Written

Find the general term with first term 8 and sum of the first 13 terms -130 .

Classified Practice

Arithmetic Sequences

Q271 **Written****Find the general term with first term 4 and $S_6 = 114$.**

Q272 Written**Find the general term when $a_3 = 10$ and $S_{10} = 175$.**

Classified Practice

Arithmetic Sequences

Q273 **Written**

For first term 65 and common difference -4 , find the number of terms giving the maximum sum and the sum.

Q274 Written

For first term 70 and common difference -4 , find the number of terms giving the maximum sum and the sum.

Classified Practice

Arithmetic Sequences

Q275 Written

For $a = 20$, $d = -6$, find n giving the maximum sum and the sum.

Q276 Written

For $a = 126, d = -13$, find n giving the maximum sum and the sum.

Classified Practice

Arithmetic Sequences

Q277 Written

Find the general term where $S_{10} = 10$ and $S_{20} = 40$.

Q278 Written**Find the general term where $S_6 = 3$ and $S_{10} = 65$.**

Classified Practice

Arithmetic Sequences

Q279 **Written****Find the general term where $S_{20} = 400$ and $S_{30} = 900$.**

Classified Practice

Arithmetic Sequences

Q280 Challenge

The 15th term is 33 and the 50th term is 103. Find the sum of the first 79 terms.

Classified Practice

Arithmetic Sequences

Quiz MCQ 1 **Concept Quiz · MCQ**

For $a_n = \frac{2n+1}{n+2}$, what is a_5 ?:

Choose the correct option.

- A $\frac{11}{7}$
B $\frac{10}{7}$
C $\frac{11}{9}$
D $\frac{9}{7}$

Quiz MCQ 2 Concept Quiz · MCQ

An arithmetic sequence has $a_3 = 10$ and $a_8 = 30$. Select its general term:

Choose the correct option.

A $a_n = 4n + 2$

B $a_n = 4n - 2$

C $a_n = 5n - 5$

D $a_n = 2n + 4$

Classified Practice

Arithmetic Sequences

Quiz MCQ 3 [Concept Quiz](#) · MCQ

For $a = 12$, $d = -2$, select the sum of odd-numbered terms through a_9 :

Choose the correct option.

A 16

B 18

C 20

D 24

Quiz MCQ 4 Concept Quiz · MCQ

Select the correct option: If $S_{10} = 10$ and $S_{20} = 40$, find d :

Choose the correct option.

- A $\frac{1}{10}$
- B $\frac{1}{5}$
- C $\frac{5}{2}$
- D $\frac{1}{2}$

Classified Practice

Arithmetic Sequences

Quiz WRITTEN 5 **Concept Quiz · Written**

If $a_n = 3n^2 - 2n + 1$, find a_1, a_2, a_3 .

Quiz WRITTEN 6 Concept Quiz · Written

Find a_n for an arithmetic sequence with $a_4 = 14$ and $a_{10} = 62$.

Classified Practice

Arithmetic Sequences

Quiz WRITTEN 7 Concept Quiz · Written

For $a = 9, d = 4$, find the sum of odd-numbered terms from a_1 to a_{15} .

Quiz WRITTEN 8 Concept Quiz · Written

Find the general term when $S_{20} = 400$ and $S_{30} = 900$.

Answer key

Use the question number shown on each page.

Q001 $a_1 = -1, a_2 = 1, a_3 = 3$

Q002 $a_1 = -6, a_2 = -10, a_3 = -12$

Q003 $a_1 = -3, a_2 = 2, a_3 = 7$

Q004 $a_1 = -2, a_2 = -5, a_3 = -8$

Q005 $a_1 = -6, a_2 = -8, a_3 = -10$

Q006 $a_1 = 0, a_2 = 3, a_3 = 8$

Q007 $a_1 = 3, a_2 = 7, a_3 = 13$

Q008 $a_1 = 7, a_2 = 12, a_3 = 19$

Q009 B

Q010 C

Q011 D

Q012 A

Q013 B

Q014 C

Q015 D

Q016 A

Q017 B

Q018 C

Q019 D

Q020 A

Q021 $a_1 = 6, a_2 = 18, a_3 = 54$

Q022 $a_1 = 2, a_2 = 8, a_3 = 26$

Q023 $a_1 = -3, a_2 = 9, a_3 = -27$

Q024 $a_1 = -6, a_2 = 12, a_3 = -24$

Q025 A

Q026 B

Q027 C

Q028 D

Q029 A

Q030 B

Q031 C

Q032 D

Q033 $a_1 = \frac{1}{2}, a_2 = \frac{2}{3}, a_3 = \frac{3}{4}$

Q034 $a_1 = \frac{1}{3}, a_2 = \frac{2}{5}, a_3 = \frac{3}{7}$

Q035 $a_1 = 1, a_2 = \frac{1}{4}, a_3 = \frac{1}{7}$

Q036 $a_1 = \frac{1}{2}, a_2 = \frac{4}{5}, a_3 = 1$

Q037 $a_1 = 2, a_2 = 1, a_3 = \frac{4}{5}$

Q038 C

Q039 D

Q040 A

Q041 B

Q042 C

Q043 D

Q044 $a_n = (n-1)^2$

Q045 $a_n = (n-1)^3$

Q046 $a_n = n^2$

Q047 $a_n = (n+6)^2$

Q048 B

Q049 C

Q050 D

Q051 A

Q052 B

Q053 C

Q054 $a_n = 3n - 1$

Q055 $a_n = 2n - 2$

Q056 $a_n = 3n - 3$

Q057 $a_n = 5n - 1$

Q058 $a_n = 7n - 1$

Q059 A

Q060 B

Q061 C

Q062 D

Q063 $a_n = (-1)^{n-1} 2n$

Q064 $a_n = (-1)^n 3n$

Q065 $a_n = (-1)^n 5n$

Q066 $a_n = (-1)^n (3n - 2)$

Q067 $a_n = \frac{n}{2n-1}$

Q068 $a_n = \frac{2n-1}{n^2}$

Q069 $a_n = \frac{(-1)^{n-1}}{2n-1}$

Q070 $a_n = \frac{n}{n+1}$

Q071 $a_n = n 3^n$

Q072 $a_n = \frac{n}{n+1}; a_{20} = \frac{20}{21}$

Q073 (a) $a = 2, d = 3(b) a = 3, d = -5$

Q074 (a) $a = 1, d = 4(b) a = 23, d = -4$

Q075 B

Q076 A

Q077 C

Q078 D

Q079 (a)(3, 2); (b)(-13, 4); (c)(-2, 3); (d)(10, -3); (e)(6, $-\frac{2}{3}$); (f)(-4, $-\frac{1}{2}$).

Q080 B

Q081 A

Q082 C

Q083 B

Q084 (a) $d = -3, A = 5, B = -4(b) d = -5, A = 8, B = 3$

Q085 (a) $d = 7, A = 16, B = 23(b) d = -6, A = 23, B = 11$

Q086 A

Q087 C

Q088 B

Q089 (a) $d = 7, A = 22, B = 28; (b) d = 6, A = 19, B = 25; (c) d = -5, A = 14, B = -1; (d) d = -2, A = 11, B = 13; (e) d = 5, A = 14, B = 17; (f) d = -4, A = 11, B = 9, C = 1.$

Q090 (a) $a_n = 3n + 1, a_7 = 22(b) a_n = 5 - 3n, a_7 = -16$

Q091 (a) $a_n = 4n + 1, a_7 = 29; (b) a_n = 3n - 5, a_7 = 16; (c) a_n = 13 - 3n, a_7 = -8; (d) a_n = -\frac{n}{2}, a_7 = -\frac{7}{2}.$

Q092 $a_n = 8 + (n-1)(-6) = 14 - 6n; a_{20} = -106$

Q093 (a) $a_n = 3n - 10, a_{10} = 20; (b) a_n = 1 - 3n, a_{10} = -29; (c) a_n = \frac{n+13}{2}, a_{10} = \frac{23}{2}; (d) a_n = \frac{3-n}{3}, a_{10} = -\frac{7}{3}.$

Q094 $a = 7, d = 3.$

Q095 $a_n = 5n - 10$

Q096 $a_n = 4n + 2$

Q097 $a_n = 23 - 3n$

Q098 $a_n = 9n - 17$

Q099 $a_n = -5n - 13$

Q100 D

Q101 B	Q133 A	Q165 144	Q197 C
Q102 A	Q134 B	Q166 -165	Q198 C
Q103 B	Q135 $n = 125$, the 125th term	Q167 176	Q199 $2n^2 - n$
Q104 $a_n = 3n + 33$	Q136 $n = 35$, the 35th term	Q168 360	Q200 $\frac{n(7-3n)}{2}$
Q105 $a_n = 2n - 12$	Q137 A	Q169 C	Q201 $\frac{5n(n-1)}{2}$
Q106 $a_n = 13 - 4n$	Q138 D	Q170 B	Q202 $5n - 2n^2$
Q107 $a_n = 18 - 7n$	Q139 D	Q171 C	Q203 $-n^2 - 3n$
Q108 A	Q140 B	Q172 B	Q204 414
Q109 D	Q141 $n = 72$, the 72nd term	Q173 C	Q205 5050
Q110 C	Q142 $n = 171$, the 171st term	Q174 B	Q206 20100
Q111 D	Q143 B	Q175 392	Q207 11325
Q112 $a_n = 8n - 14$	Q144 B	Q176 343	Q208 2485
Q113 $a_n = 20 - 3n$	Q145 C	Q177 741	Q209 5050
Q114 $a_n = 8n - 18$	Q146 A	Q178 354	Q210 80200
Q115 $a_n = 42 - 4n$	Q147 B	Q179 1716	Q211 31375
Q116 $a_n = 79 - 3n$	Q148 $a = 4$	Q180 C	Q212 8085
Q117 $a_n = 8n - 18$	Q149 $x = -4$ or $x = 3$	Q181 C	Q213 24535
Q118 C	Q150 $x = -2$ or $x = 4$	Q182 C	Q214 4950
Q119 C	Q151 $x = -1$ or $x = 3$	Q183 C	Q215 B
Q120 B	Q152 $x = 0, y = -2, z = -4$	Q184 C	Q216 B
Q121 A	Q153 105	Q185 B	Q217 B
Q122 A	Q154 100	Q186 0	Q218 B
Q123 B	Q155 120	Q187 208	Q219 B
Q124 $a_n = \frac{2n-37}{5}$	Q156 152	Q188 B	Q220 B
Q125 A	Q157 155	Q189 C	Q221 B
Q126 D	Q158 B	Q190 C	Q222 B
Q127 $n = 28$, the 28th term	Q159 B	Q191 C	Q223 B
Q128 $n = 14$, the 14th term	Q160 B	Q192 B	Q224 B
Q129 $n = 22$, the 22nd term	Q161 B	Q193 C	Q225 15050
Q130 $n = 40$, the 40th term	Q162 B	Q194 66	Q226 25050
Q131 C	Q163 C	Q195 750	Q227 4905
Q132 D	Q164 -130	Q196 154	

Q228 B	Q244 1225	Q260 B	Q276 $n = 10, S = 675$
Q229 B	Q245 999	Q261 B	Q277 $a_n = \frac{2n-1}{10}$
Q230 B	Q246 2475	Q262 B	Q278 $a_n = 3n - 10$
Q231 B	Q247 4195	Q263 21	Q279 $a_n = 2n - 1$
Q232 B	Q248 3750	Q264 14	Q280 6557
Q233 B	Q249 5	Q265 C	Quiz MCQ 1 A
Q234 B	Q250 7	Q266 C	Quiz MCQ 2 B
Q235 B	Q251 B	Q267 C	Quiz MCQ 3 C
Q236 B	Q252 B	Q268 C	Quiz MCQ 4 B
Q237 B	Q253 B	Q269 $a_n = 8n - 68$	Quiz WRITTEN 5 $a_1 = 2, a_2 = 9, a_3 = 22$
Q238 1665	Q254 B	Q270 $a_n = 11 - 3n$	Quiz WRITTEN 6 $a_n = 8n - 18$
Q239 945	Q255 B	Q271 $a_n = 6n - 2$	Quiz WRITTEN 7 296
Q240 1300	Q256 B	Q272 $a_n = 3n + 1$	Quiz WRITTEN 8 $a_n = 2n - 1$
Q241 810	Q257 B	Q273 $n = 17, S = 561$	
Q242 B	Q258 B	Q274 $n = 18, S = 648$	
Q243 B	Q259 B	Q275 $n = 4, S = 44$	

Concept 2 | Geometric Sequences

English Student Edition • Focused formulas and guided practice

Formula opener

Term

$$a_n = ar^{n-1}$$

Finite sum

$$S_n = a \frac{r^n - 1}{r - 1}$$

Infinite sum

$$S_{\infty} = \frac{a}{1-r} \quad (|r| < 1)$$

Classified Practice

Geometric Sequences

Q001 **Written**

Find the general term of a geometric sequence whose first term is 2 and common ratio is 3.

Q002 Written

Find the general term when the first term is 3 and the common ratio is 4.

Classified Practice

Geometric Sequences

Q003 **Written**

Find the general term of the geometric sequence in part a. The first term is 5, common ratio 2.

Q004 Written

Find the general term of the geometric sequence in part b. The first term is 2, common ratio 5.

Classified Practice

Geometric Sequences

Q005 Written

Find the general term of the geometric sequence in part c. The first term is 7, common ratio 4.

Q006 MCQ

Select the correct option: For first term $a = 2$, common ratio $r = 3$, and $n = 5$, which expression gives a_n ?:

Choose the correct option.

A $a_n = 2(3)^4$

B $a_n = 2(3)^5$

C $a_n = 2(3)^3$

D $a_n = 2 + (3)^4$

Classified Practice

Geometric Sequences

Q007 MCQ

Select the correct option: For first term $a = 3$, common ratio $r = 2$, and $n = 6$, which expression gives a_n ?:

Choose the correct option.

A $a_n = 3(2)^5$

B $a_n = 3(2)^6$

C $a_n = 3(2)^4$

D $a_n = 3 + (2)^5$

Q008 MCQ

Select the correct option: For first term $a = 5$, common ratio $r = 4$, and $n = 4$, which expression gives a_n ?:

Choose the correct option.

A $a_n = 5(4)^3$

B $a_n = 5(4)^4$

C $a_n = 5(4)^2$

D $a_n = 5 + (4)^3$

Classified Practice

Geometric Sequences

Q009 MCQ

Select the correct option: For first term $a = 4$, common ratio $r = 5$, and $n = 3$, which expression gives a_n ?:

Choose the correct option.

A $a_n = 4(5)^2$

B $a_n = 4(5)^3$

C $a_n = 4(5)^1$

D $a_n = 4 + (5)^2$

Q010 MCQ

Select the correct option: For first term $a = 5$, common ratio $r = 2$, and $n = 7$, which expression gives a_n ?:

Choose the correct option.

A $a_n = 5(2)^6$

B $a_n = 5(2)^7$

C $a_n = 5(2)^5$

D $a_n = 5 + (2)^6$

Classified Practice

Geometric Sequences

Q011 MCQ

Select the correct option: For first term $a = 7$, common ratio $r = 4$, and $n = 3$, which expression gives a_n ?:

Choose the correct option.

A $a_n = 7(4)^2$

B $a_n = 7(4)^3$

C $a_n = 7(4)^1$

D $a_n = 7 + (4)^2$

Q012 Written

Find the general term when the first term is 5 and the common ratio is -2 .

Classified Practice

Geometric Sequences

Q013 Written

Find the general term of the geometric sequence in part d. The first term is 2, common ratio -3 .

Q014 Written

Find the general term of the geometric sequence in part e. The first term is 3, common ratio -5 .

Classified Practice

Geometric Sequences

Q015 Written

Find the general term of the geometric sequence in part f. The first term is 4, common ratio -2 .

Q016 MCQ

Select the correct option: For first term $a = 2$, common ratio $r = -3$, and $n = 5$, which expression gives a_n ?:

Choose the correct option.

A $a_n = 2(-3)^4$

B $a_n = 2(-3)^5$

C $a_n = 2(-3)^3$

D $a_n = 2 + (-3)^4$

Classified Practice

Geometric Sequences

Q017 MCQ

Select the correct option: For first term $a = 3$, common ratio $r = -2$, and $n = 6$, which expression gives a_n ?

Choose the correct option.

A $a_n = 3(-2)^5$

B $a_n = 3(-2)^6$

C $a_n = 3(-2)^4$

D $a_n = 3 + (-2)^5$

Q018 MCQ

Select the correct option: For first term $a = 6$, common ratio $r = -1$, and $n = 4$, which expression gives a_n ?:

Choose the correct option.

A $a_n = 6(-1)^3$

B $a_n = 6(-1)^4$

C $a_n = 6(-1)^2$

D $a_n = 6 + (-1)^3$

Classified Practice

Geometric Sequences

Q019 MCQ

Select the correct option: For first term $a = 8$, common ratio $r = -2$, and $n = 3$, which expression gives a_n ?

Choose the correct option.

A $a_n = 8(-2)^2$

B $a_n = 8(-2)^3$

C $a_n = 8(-2)^1$

D $a_n = 8 + (-2)^2$

Q020 MCQ

Select the correct option: For first term $a = 9$, common ratio $r = -2$, and $n = 5$, which expression gives a_n ?:

Choose the correct option.

A $a_n = 9(-2)^4$

B $a_n = 9(-2)^5$

C $a_n = 9(-2)^3$

D $a_n = 9 + (-2)^4$

Classified Practice

Geometric Sequences

Q021 Written

Find the general term when the first term is 8 and the common ratio is $-\frac{1}{3}$.

Q022 Written

Find the general term of the geometric sequence in part g. The first term is 4, common ratio $\frac{1}{3}$.

Classified Practice

Geometric Sequences

Q023 Written

Find the general term of the geometric sequence in part h. The first term is 3, common ratio $-\frac{1}{2}$.

Q024 Written

Find the general term of the geometric sequence in part i. The first term is 6, common ratio $\frac{1}{2}$.

Classified Practice

Geometric Sequences

Q025 MCQ

Select the correct option: For first term $a = 4$, common ratio $r = \frac{1}{3}$, and $n = 5$, which expression gives a_n ?:

Choose the correct option.

A $a_n = 4\left(\frac{1}{3}\right)^4$

B $a_n = 4\left(\frac{1}{3}\right)^5$

C $a_n = 4\left(\frac{1}{3}\right)^3$

D $a_n = 4 + \left(\frac{1}{3}\right)^4$

Q026 MCQ

Select the correct option: For first term $a = 3$, common ratio $r = -\frac{1}{2}$, and $n = 6$, which expression gives a_n ?:

Choose the correct option.

A $a_n = 3\left(-\frac{1}{2}\right)^5$

B $a_n = 3\left(-\frac{1}{2}\right)^6$

C $a_n = 3\left(-\frac{1}{2}\right)^4$

D $a_n = 3 + \left(-\frac{1}{2}\right)^5$

Classified Practice

Geometric Sequences

Q027 MCQ

Select the correct option: For first term $a = 6$, common ratio $r = \frac{1}{2}$, and $n = 4$, which expression gives a_n ?:

Choose the correct option.

A $a_n = 6 \left(\frac{1}{2}\right)^3$

B $a_n = 6 \left(\frac{1}{2}\right)^4$

C $a_n = 6\left(\frac{1}{2}\right)^2$

D $a_n = 6 + \left(\frac{1}{2}\right)^3$

Q028 MCQ

Select the correct option: For first term $a = 2$, common ratio $r = \frac{3}{2}$, and $n = 5$, which expression gives a_n ?

Choose the correct option.

A $a_n = 2 \left(\frac{3}{2}\right)^4$

B $a_n = 2 \left(\frac{3}{2}\right)^5$

C $a_n = 2 \left(\frac{3}{2}\right)^3$

D $a_n = 2 + \left(\frac{3}{2}\right)^4$

Classified Practice

Geometric Sequences

Q029 MCQ

Select the correct option: For first term $a = 8$, common ratio $r = -\frac{3}{2}$, and $n = 4$, which expression gives a_n ?:

Choose the correct option.

A $a_n = 8 \left(-\frac{3}{2}\right)^3$

B $a_n = 8 \left(-\frac{3}{2}\right)^4$

C $a_n = 8 \left(-\frac{3}{2}\right)^2$

D $a_n = 8 + \left(-\frac{3}{2}\right)^3$

Q030 Written**Find the general term of 5, 10, 20, 40,**

Classified Practice

Geometric Sequences

Q031 **Written****Find the general term of 2, 6, 18, 54,**

Q032 Written

Find the general term of the geometric sequence in part j. The sequence is 3, 6, 12, 24,

Classified Practice

Geometric Sequences

Q033 MCQ

Select the correct option: Find the general term of the geometric sequence 3, 6, 12, 24, ...:**Choose the correct option.**

A $3 \times 2^{n-1}$

B 3×2^n

C $3 \times 2^{n-2}$

D 0

Q034 MCQ

Select the correct option: Find the general term of the geometric sequence 2, 10, 50, 250, ...:**Choose the correct option.**

A $2 \times 5^{n-1}$

B 2×5^n

C $2 \times 5^{n-2}$

D 0

Classified Practice

Geometric Sequences

Q035 MCQ

Select the correct option: Find the general term of the geometric sequence 4, 8, 16, 32, ...:**Choose the correct option.**

A $4 \times 2^{n-1}$

B 4×2^n

C $4 \times 2^{n-2}$

D 0

Q036 MCQ

Select the correct option: Find the general term of the geometric sequence 7, 21, 63, 189, ...:**Choose the correct option.**

A $7 \times 3^{n-1}$

B 7×3^n

C $7 \times 3^{n-2}$

D 0

Classified Practice

Geometric Sequences

Q037 MCQ

Select the correct option: Find the general term of the geometric sequence 1, 4, 16, 64, ...:**Choose the correct option.**

A 4^{n-1}

B 4^n

C 4^{n-2}

D 0

Q038 Written

Find the general term of the geometric sequence in part k. The sequence is $2, -6, 18, -54, \dots$

Classified Practice

Geometric Sequences

Q039 Written

Find the general term of the geometric sequence in part I. The sequence is $5, -10, 20, -40, \dots$

Q040 MCQ**Select the correct option: Find the general term of the geometric sequence $5, -10, 20, -40, \dots$:****Choose the correct option.**

A $5(-2)^{n-1}$

B $5(-2)^n$

C $5(-2)^{n-2}$

D 0

Classified Practice

Geometric Sequences

Q041 MCQ

Select the correct option: Find the general term of the geometric sequence $2, -6, 18, -54, \dots$:

Choose the correct option.

A $2(-3)^{n-1}$

B $2(-3)^n$

C $2(-3)^{n-2}$

D 0

Q042 MCQ

Select the correct option: Find the general term of the geometric sequence $-4, 8, -16, 32, \dots$:**Choose the correct option.**

A $-4(-2)^{n-1}$

B $-4(-2)^n$

C $-4(-2)^{n-2}$

D 0

Classified Practice

Geometric Sequences

Q043 MCQ

Select the correct option: Find the general term of the geometric sequence $6, -18, 54, -162, \dots$:**Choose the correct option.**

A $6(-3)^{n-1}$

B $6(-3)^n$

C $6(-3)^{n-2}$

D 0

Q044 Written**Find the general term of $-4, 6, -9, \frac{27}{2}, \dots$**

Classified Practice

Geometric Sequences

Q045 **Written****Find the general term of $-8, 12, -18, 27, \dots$**

Q046 Written**Find the general term of $2, \frac{2}{3}, \frac{2}{9}, \frac{2}{27}, \dots$**

Classified Practice

Geometric Sequences

Q047 **Written**

Find the general term of the geometric sequence in part m. The sequence is 48, 24, 12, 6,

Q048 Written

Find the general term of the geometric sequence in part n. The sequence is $\frac{1}{6}, \frac{1}{4}, \frac{3}{8}, \frac{9}{16}, \dots$

Classified Practice

Geometric Sequences

Q049 Written

Find the general term of the geometric sequence in part o. The sequence is $8, 16\sqrt{2}, 64, 128\sqrt{2}, \dots$

Q050 Written

Find the general term of the geometric sequence in part p. The sequence is $3, -\frac{15}{2}, \frac{75}{4}, -\frac{375}{8}, \dots$

Classified Practice

Geometric Sequences

Q051 Written

Find the general term of the geometric sequence in part q. The sequence is $9, -3\sqrt{3}, 3, -\sqrt{3}, \dots$

Q052 Written

Find the general term of the geometric sequence in part r. The sequence is $5, -\frac{10}{3}, \frac{20}{9}, -\frac{40}{27}, \dots$

Classified Practice

Geometric Sequences

Q053 MCQ

Select the correct option: Find the general term of the geometric sequence $-8, 12, -18, 27, \dots$:**Choose the correct option.**

A $-8 \left(-\frac{3}{2}\right)^{n-1}$

B $-8 \left(-\frac{3}{2}\right)^n$

C $-8 \left(-\frac{3}{2}\right)^{n-2}$

D 0

Q054 MCQ

Select the correct option: Find the general term of the geometric sequence 48, 24, 12, 6, ...:**Choose the correct option.**

A $48 \left(\frac{1}{2}\right)^{n-1}$

B $48 \left(\frac{1}{2}\right)^n$

C $48 \left(\frac{1}{2}\right)^{n-2}$

D 0

Classified Practice

Geometric Sequences

Q055 MCQ

Select the correct option: Find the general term of the geometric sequence $\frac{1}{6}, \frac{1}{4}, \frac{3}{8}, \dots$:

Choose the correct option.

- A $\frac{1}{6} \left(\frac{3}{2}\right)^{n-1}$
B $\frac{1}{6} \left(\frac{3}{2}\right)^n$
C $\frac{1}{6} \left(\frac{3}{2}\right)^{n-2}$
D 0

Q056 MCQ

Select the correct option: Find the general term of the geometric sequence $8, 16\sqrt{2}, 64, \dots$:

Choose the correct option.

A $8(2\sqrt{2})^{n-1}$

B $8(2\sqrt{2})^n$

C $8(2\sqrt{2})^{n-2}$

D 0

Classified Practice

Geometric Sequences

Q057 MCQ

Select the correct option: Find the general term of the geometric sequence $9, -3\sqrt{3}, 3, \dots$:

Choose the correct option.

A $9\left(-\frac{\sqrt{3}}{3}\right)^{n-1}$

B $9\left(-\frac{\sqrt{3}}{3}\right)^n$

C $9\left(-\frac{\sqrt{3}}{3}\right)^{n-2}$

D 0

Q058 MCQ

Select the correct option: Find the general term of the geometric sequence $3, -\frac{3}{2}, \frac{3}{4}, \dots$:

Choose the correct option.

A $3\left(-\frac{1}{2}\right)^{n-1}$

B $3\left(-\frac{1}{2}\right)^n$

C $3\left(-\frac{1}{2}\right)^{n-2}$

D 0

Classified Practice

Geometric Sequences

Q059 MCQ

Select the correct option: Find the general term of the geometric sequence $10, 5, \frac{5}{2}, \dots$:

Choose the correct option.

- A $10\left(\frac{1}{2}\right)^{n-1}$
- B $10\left(\frac{1}{2}\right)^n$
- C $10\left(\frac{1}{2}\right)^{n-2}$
- D 0

Q060 Written**Find the common ratio and missing values in $A, -18, 6, B, \dots$**

Classified Practice

Geometric Sequences

Q061 MCQ

Select the correct option: The sequence $A, 12, 6, B, \dots$ is geometric. Find r, A, B :

Choose the correct option.

A $r = \frac{1}{2}, A = 24, B = 3$

B $r = \frac{2}{3}, A = 1, B = 2$

C $r = -1, A = 0, B = 0$

D $r = 3, A = 3, B = 9$

Q062 Written

Find the common ratio and the missing values in $A, 24, 12, B, \dots$

Classified Practice

Geometric Sequences

Q063 Written

Find the common ratio and missing values in $A, -3, 9, B$.

Q064 Written

Find the common ratio and missing values in $A, \frac{1}{3}, \frac{1}{9}, B$.

Classified Practice

Geometric Sequences

Q065 Written

Find the common ratio and missing values in $A, -\frac{1}{4}, \frac{1}{8}, B$.

Classified Practice

Geometric Sequences

Q066 Challenge

In an increasing geometric sequence, if $a_3 + a_5 = 80$ and $a_4 = 24$, find a_7 .

Classified Practice

Geometric Sequences

Q067 Written

Find the general term of a geometric sequence whose 4th term is 24 and 6th term is 96.

Q068 Written**Find the general term when the 3rd term is 18 and the 5th term is 162.**

Classified Practice

Geometric Sequences

Q069 Written

Find the requested values. The 2nd term is 6 and the 4th term is 54.

Q070 Written

Find the requested values. The 3rd term is 27 and the 5th term is 243.

Classified Practice

Geometric Sequences

Q071 Written

Find the requested values. The 2nd term is 14 and the 5th term is 112.

Q072 Written**Find the general term when the 3rd term is 12 and the 6th term is -96 .**

Classified Practice

Geometric Sequences

Q073 Written

Find the general term when the 2nd term is -6 and the 5th term is 48 .

Q074 Written**Find the general term when the 3rd term is 12 and the 7th term is 192.**

Classified Practice

Geometric Sequences

Q075 **Written**

Find the requested values. The 3rd term is -28 and the 5th term is -112 .

Q076 Written

Find the requested values. The 4th term is 16 and the 7th term is -128 .

Classified Practice

Geometric Sequences

Q077 Written

Find the requested values. The 3rd term is -20 and the 6th term is 160 .

Q078 Written

Find the requested values. The 2nd term is 3 and the 6th term is 768.

Classified Practice

Geometric Sequences

Q079 **Written**

Find the requested values. The 3rd term is 9 and the 7th term is $729/16$.

Q080 Written

Find the requested values. The 4th term is $\frac{2}{9}$ and the 8th term is 18.

Classified Practice

Geometric Sequences

Q081 MCQ

Select the correct option: For $a = 2$, $r = 3$, which expression gives the general term a_n :

Choose the correct option.

A $a_n = 2(3)^{n-1}$

B $a_n = 2(3)^n$

C $a_n = 2(3)^{n+1}$

D $a_n = 2 + 3^{n-1}$

Q082 MCQ

Select the correct option: For $a = 3$, $r = 2$, which expression gives the general term a_n :

Choose the correct option.

A $a_n = 3(2)^{n-1}$

B $a_n = 3(2)^n$

C $a_n = 3(2)^{n+1}$

D $a_n = 3 + 2^{n-1}$

Classified Practice

Geometric Sequences

Q083 MCQ

Select the correct option: For $a = 5$, $r = 4$, which expression gives the general term a_n :

Choose the correct option.

A $a_n = 5(4)^{n-1}$

B $a_n = 5(4)^n$

C $a_n = 5(4)^{n+1}$

D $a_n = 5 + 4^{n-1}$

Q084 MCQ

Select the correct option: For $a = 4$, $r = 5$, which expression gives the general term a_n :

Choose the correct option.

A $a_n = 4(5)^{n-1}$

B $a_n = 4(5)^n$

C $a_n = 4(5)^{n+1}$

D $a_n = 4 + 5^{n-1}$

Classified Practice

Geometric Sequences

Q085 MCQ

Select the correct option: For $a = 2$, $r = -3$, which expression gives the general term a_n :

Choose the correct option.

A $a_n = 2(-3)^{n-1}$

B $a_n = 2(-3)^n$

C $a_n = 2(-3)^{n+1}$

D $a_n = 2 - 3^{n-1}$

Q086 MCQ

Select the correct option: For $a = 3$, $r = -2$, which expression gives the general term a_n :

Choose the correct option.

A $a_n = 3(-2)^{n-1}$

B $a_n = 3(-2)^n$

C $a_n = 3(-2)^{n+1}$

D $a_n = 3 - 2^{n-1}$

Classified Practice

Geometric Sequences

Q087 MCQ

Select the correct option: For $a = 6$, $r = -1$, which expression gives the general term a_n :

Choose the correct option.

A $a_n = 6(-1)^{n-1}$

B $a_n = 6(-1)^n$

C $a_n = 6(-1)^{n+1}$

D $a_n = 6 - 1^{n-1}$

Q088 MCQ

Select the correct option: For $a = 8$, $r = -2$, which expression gives the general term a_n :

Choose the correct option.

A $a_n = 8(-2)^{n-1}$

B $a_n = 8(-2)^n$

C $a_n = 8(-2)^{n+1}$

D $a_n = 8 - 2^{n-1}$

Classified Practice

Geometric Sequences

Q089 MCQ

Select the correct option: For $a = 4$, $r = \frac{1}{3}$, which expression gives the general term a_n :

Choose the correct option.

- A $a_n = 4\left(\frac{1}{3}\right)^{n-1}$
- B $a_n = 4\left(\frac{1}{3}\right)^n$
- C $a_n = 4\left(\frac{1}{3}\right)^{n+1}$
- D $a_n = 4 + \frac{1}{3}^{n-1}$

Q090 MCQ

Select the correct option: For $a = 3$, $r = -\frac{1}{2}$, which expression gives the general term a_n :

Choose the correct option.

A $a_n = 3 \left(-\frac{1}{2}\right)^{n-1}$

B $a_n = 3 \left(-\frac{1}{2}\right)^n$

C $a_n = 3 \left(-\frac{1}{2}\right)^{n+1}$

D $a_n = 3 - \frac{1}{2}^{n-1}$

Classified Practice

Geometric Sequences

Q091 MCQ

Select the correct option: For $a = 6$, $r = \frac{1}{2}$, which expression gives the general term a_n :

Choose the correct option.

- A $a_n = 6 \left(\frac{1}{2}\right)^{n-1}$
- B $a_n = 6 \left(\frac{1}{2}\right)^n$
- C $a_n = 6 \left(\frac{1}{2}\right)^{n+1}$
- D $a_n = 6 + \frac{1}{2}^{n-1}$

Q092 MCQ

Select the correct option: For $a = 2$, $r = \frac{3}{2}$, which expression gives the general term a_n :

Choose the correct option.

A $a_n = 2 \left(\frac{3}{2}\right)^{n-1}$

B $a_n = 2 \left(\frac{3}{2}\right)^n$

C $a_n = 2 \left(\frac{3}{2}\right)^{n+1}$

D $a_n = 2 + \frac{3^{n-1}}{2}$

Classified Practice

Geometric Sequences

Q093 MCQ

Select the correct option: For $a = 8$, $r = -\frac{3}{2}$, which expression gives the general term a_n :

Choose the correct option.

- A $a_n = 8 \left(-\frac{3}{2}\right)^{n-1}$
- B $a_n = 8 \left(-\frac{3}{2}\right)^n$
- C $a_n = 8 \left(-\frac{3}{2}\right)^{n+1}$
- D $a_n = 8 - \frac{3^{n-1}}{2}$

Q094 MCQ

Select the correct option: For $a = 5$, $r = 2$, which expression gives the general term a_n :

Choose the correct option.

A $a_n = 5(2)^{n-1}$

B $a_n = 5(2)^n$

C $a_n = 5(2)^{n+1}$

D $a_n = 5 + 2^{n-1}$

Classified Practice

Geometric Sequences

Q095 MCQ

Select the correct option: For $a = 7$, $r = 4$, which expression gives the general term a_n :

Choose the correct option.

A $a_n = 7(4)^{n-1}$

B $a_n = 7(4)^n$

C $a_n = 7(4)^{n+1}$

D $a_n = 7 + 4^{n-1}$

Q096 MCQ

Select the correct option: For $a = 9$, $r = -2$, which expression gives the general term a_n :

Choose the correct option.

A $a_n = 9(-2)^{n-1}$

B $a_n = 9(-2)^n$

C $a_n = 9(-2)^{n+1}$

D $a_n = 9 - 2^{n-1}$

Classified Practice

Geometric Sequences

Q097 Written

When 2, x , 8 form a geometric sequence in this order, find x .

Q098 Written

Find the requested values. When 1, x , 25 form a geometric sequence, find x .

Classified Practice

Geometric Sequences

Q099 Written

Find the requested values. When $1/2, x, 8$ form a geometric sequence, find x .

Q100 Written

When 1, x , 12 form a geometric sequence in this order, find x .

Classified Practice

Geometric Sequences

Q101 MCQ

Select the correct option: If x is the middle term of the geometric triple $2, x, 8$, find x :

Choose the correct option.

A ± 4

B $+4$

C -4

D 0

Q102 MCQ

Select the correct option: If x is the middle term of the geometric triple $1, x, 25$, find x :

Choose the correct option.

A ± 5

B $+5$

C -5

D 0

Classified Practice

Geometric Sequences

Q103 MCQ

Select the correct option: If x is the middle term of the geometric triple $\frac{1}{2}, x, 8$, find x :

Choose the correct option.

- A ± 2
- B $+2$
- C -2
- D 0

Q104 MCQ

Select the correct option: If x is the middle term of the geometric triple 3, x , 12, find x :

Choose the correct option.

A ± 6

B $+6$

C -6

D 0

Classified Practice

Geometric Sequences

Q105 MCQ

Select the correct option: If x is the middle term of the geometric triple 4, x , 9, find x :

Choose the correct option.

A ± 6 B $+6$ C -6

D 0

Q106 MCQ**Select the correct option: If x is the middle term of the geometric triple $2, x, 18$, find x :****Choose the correct option.**A ± 6 B $+6$ C -6 D 0

Classified Practice

Geometric Sequences

Q107 Written

The numbers $6, a, b$ form an arithmetic sequence and $a, b, 16$ form a geometric sequence. Find a, b .

Q108 Written

Find the requested values. The numbers $a, b, 2$ form an arithmetic sequence and $a, 2, b$ form a geometric sequence. Find a, b .

Classified Practice

Geometric Sequences

Q109 Written

The numbers $-2, a, b$ form an arithmetic sequence and $a, b, 18$ form a geometric sequence. Find a, b .

Classified Practice

Geometric Sequences

Q110 Challenge

A planning model gives $-5, a, b$ as an arithmetic sequence, while $a, b, 45$ form a geometric sequence. Find a, b .

Classified Practice

Geometric Sequences

Q111 Written

Find the sum S of a geometric sequence with first term 4, common ratio 3, and 5 terms.

Q112 Written

Find the sum S of a geometric sequence with first term 3, common ratio 2, and 5 terms.

Classified Practice

Geometric Sequences

Q113 Written

Find the sum S of a geometric sequence with first term 5, common ratio 3, and 4 terms.

Q114 Written

Find the sum S of a geometric sequence with first term 3, common ratio 2, and 10 terms.

Classified Practice

Geometric Sequences

Q115 MCQ

For a geometric sequence with $a = 4$, $r = 3$, and $n = 4$, calculate S_n :

Choose the correct option.

A 161

B 159

C 162

D 160

Q116 MCQ

For a geometric sequence with $a = 5$, $r = 2$, and $n = 6$, calculate S_n :

Choose the correct option.

A 316

B 314

C 315

D 317

Classified Practice

Geometric Sequences

Q117 MCQ

For a geometric sequence with $a = 2$, $r = 4$, and $n = 5$, calculate S_n :

Choose the correct option.

A 683

B 682

C 681

D 684

Q118 MCQ**For a geometric sequence with $a = 3$, $r = 5$, and $n = 4$, calculate S_n :****Choose the correct option.****A** 468**B** 469**C** 467**D** 470

Classified Practice

Geometric Sequences

Q119 MCQ

For a geometric sequence with $a = 7$, $r = 2$, and $n = 5$, calculate S_n :

Choose the correct option.

A 218

B 216

C 219

D 217

Q120 MCQ**For a geometric sequence with $a = 6$, $r = 3$, and $n = 3$, calculate S_n :****Choose the correct option.****A** 79**B** 77**C** 78**D** 80

Classified Practice

Geometric Sequences

Q121 Written

Find the sum S of a geometric sequence with first term 8, common ratio $\frac{1}{2}$, and 6 terms.

Q122 Written

Find the sum S of a geometric sequence with first term 6, common ratio $\frac{1}{2}$, and 6 terms.

Classified Practice

Geometric Sequences

Q123 Written

Find the sum S of a geometric sequence with first term 1, common ratio $\frac{1}{3}$, and 3 terms.

Q124 Written

Find the sum S of a geometric sequence with first term 1, common ratio $\frac{1}{2}$, and 8 terms.

Classified Practice

Geometric Sequences

Q125 MCQ

For a geometric sequence with $a = 8$, $r = \frac{1}{2}$, and $n = 4$, calculate S_n :

Choose the correct option.

A 16

B 15

C 14

D 17

Q126 MCQ

For a geometric sequence with $a = 9$, $r = \frac{1}{3}$, and $n = 3$, calculate S_n :

Choose the correct option.

- A 13
- B 14
- C 12
- D 15

Classified Practice

Geometric Sequences

Q127 MCQ

For a geometric sequence with $a = 12$, $r = \frac{1}{2}$, and $n = 5$, calculate S_n :

Choose the correct option.

A $\frac{47}{2}$

B $\frac{23}{95}$

C $\frac{4}{93}$

D $\frac{93}{4}$

Q128 MCQ

For a geometric sequence with $a = 5$, $r = \frac{1}{4}$, and $n = 3$, calculate S_n :

Choose the correct option.

A $\frac{53}{8}$

B $\frac{13}{2}$

C $\frac{105}{16}$

D $\frac{107}{16}$

Classified Practice

Geometric Sequences

Q129 MCQ

For a geometric sequence with $a = 10$, $r = \frac{1}{2}$, and $n = 6$, calculate S_n :

Choose the correct option.

A $\frac{79}{4}$

B $\frac{315}{16}$

C $\frac{157}{8}$

D $\frac{317}{16}$

Q130 MCQ

For a geometric sequence with $a = 4$, $r = \frac{1}{3}$, and $n = 4$, calculate S_n :

Choose the correct option.

- A $\frac{160}{27}$
- B $\frac{161}{27}$
- C $\frac{53}{9}$
- D 6

Classified Practice

Geometric Sequences

Q131 Written

Find the sum S of a geometric sequence with first term 6, common ratio -2 , and 4 terms.

Q132 Written

Find the sum S of a geometric sequence with first term 1, common ratio -3 , and 4 terms.

Classified Practice

Geometric Sequences

Q133 Written

Find the sum S of a geometric sequence with first term 2, common ratio -3 , and 6 terms.

Q134 MCQ**For a geometric sequence with $a = 4$, $r = -2$, and $n = 5$, calculate S_n :****Choose the correct option.****A** 45**B** 43**C** 46**D** 44

Classified Practice

Geometric Sequences

Q135 MCQ

For a geometric sequence with $a = 3$, $r = -3$, and $n = 4$, calculate S_n :

Choose the correct option.

A -59

B -61

C -60

D -58

Q136 MCQ

For a geometric sequence with $a = 2$, $r = -1$, and $n = 6$, calculate S_n :

Choose the correct option.

A 1

B 0

C -1

D 2

Classified Practice

Geometric Sequences

Q137 MCQ

For a geometric sequence with $a = 5$, $r = -2$, and $n = 6$, calculate S_n :

Choose the correct option.

A -105 B -104 C -106 D -103

Q138 MCQ

For a geometric sequence with $a = 7$, $r = -1$, and $n = 5$, calculate S_n :

Choose the correct option.

A 8

B 6

C 9

D 7

Classified Practice

Geometric Sequences

Q139 MCQ

For a geometric sequence with $a = 6$, $r = -3$, and $n = 3$, calculate S_n :

Choose the correct option.

A 43

B 41

C 42

D 44

Q140 Written

Find S_n from the first term to the n -th term when the first term is 3 and the common ratio is -2 .

Classified Practice

Geometric Sequences

Q141 Written

Find S_n from the first term to the n -th term when the first term is 7 and the common ratio is 2.

Q142 Written

Find S_n from the first term to the n -th term when the first term is 2 and the common ratio is $\frac{1}{3}$.

Classified Practice

Geometric Sequences

Q143 **Written**

Find S_n from the first term to the n -th term when the first term is 4 and the common ratio is 2.

Q144 MCQ

For the geometric sequence with $a = 3, r = 2$, find the sum S_n from the first term to the n -th term:

Choose the correct option.

A $3(2^{n-1} - 1)$

B $6(2^n - 1)$

C $3(2^n - 1)$

D $3(1 - 2^n)$

Classified Practice

Geometric Sequences

Q145 MCQ

For the geometric sequence with $a = 5, r = \frac{1}{2}$, find the sum S_n from the first term to the n -th term:

Choose the correct option.

A $5\{1 - (\frac{1}{2})^n\}$

B $10\{1 - (\frac{1}{2})^{n-1}\}$

C $10\{1 - (\frac{1}{2})^n\}$

D $10\{1 - (\frac{1}{2})^n\}$

Q146 MCQ

For the geometric sequence with $a = 4, r = -2$, find the sum S_n from the first term to the n -th term:

Choose the correct option.

A $\frac{4\{1 - (-2)^n\}}{3}$

B $\frac{4\{1 - (-2)^{n-1}\}}{3}$

C $\frac{4\{1 + (-2)^n\}}{3}$

D $\frac{4\{1 - 2^n\}}{3}$

Classified Practice

Geometric Sequences

Q147 Written

Find S_n from the first term to the n -th term of 2, 6, 18, 54,

Q148 Written

Find S_n from the first term to the n -th term of 2, -6 , 18, -54 ,

Classified Practice

Geometric Sequences

Q149 **Written****Find S_n from the first term to the n -th term of $8, -4, 2, -1, \dots$**

Q150 Written

Find S_n from the first term to the n -th term of $1, -3, 9, -27, \dots$

Classified Practice

Geometric Sequences

Q151 Written

Find S_n from the first term to the n -th term when the first term is 16 and the common ratio is $-\frac{1}{2}$.

Q152 Written

Find S_n from the first term to the n -th term of $1, -2, 4, -8, \dots$

Classified Practice

Geometric Sequences

Q153 MCQ

A culture grows by 3, 9, 27, 81 units. Find the four-stage total:

Choose the correct option.

A 120

B 121

C 119

D 122

Classified Practice

Geometric Sequences

Q154 Challenge

A filtration scale is geometric: $3, 1, \frac{1}{3}, \frac{1}{9}, \dots$ mg/L removed per stage. Calculate the total removed in 4 stages.

Classified Practice

Geometric Sequences

Q155 MCQ

Given $S_3 = 7$ and the sum from the 3rd to the 5th terms is 28, find all real pairs (a, r) :

Choose the correct option.

A $(a, r) = (2, 2)$

B $(a, r) = (1, 3)$

C $(a, r) = \left(\frac{10}{3}, -2\right)$

D $(a, r) = (1, 2)$ or $(a, r) = \left(\frac{7}{3}, -2\right)$

Classified Practice

Geometric Sequences

Q156 MCQ

Given $S_3 = 26$ and the sum from the 3rd to the 5th terms is 234, find all real pairs (a, r) :

Choose the correct option.

A $(a, r) = (3, 3)$

B $(a, r) = (2, 4)$

C $(a, r) = (2, 3)$ or $(a, r) = \left(\frac{26}{7}, -3\right)$

D $(a, r) = \left(\frac{33}{7}, -3\right)$

Classified Practice

Geometric Sequences

Q157 MCQ

Given $S_3 = 21$ and the sum from the 3rd to the 5th terms is 84, find all real pairs (a, r) :

Choose the correct option.

A $(a, r) = (4, 2)$

B $(a, r) = (3, 2)$ or $(a, r) = (7, -2)$

C $(a, r) = (3, 3)$

D $(a, r) = (8, -2)$

Classified Practice

Geometric Sequences

Q158 MCQ

Given $S_3 = 28$ and the sum from the 3rd to the 5th terms is 112, find all real pairs (a, r) :

Choose the correct option.

A $(a, r) = (4, 2)$ or $(a, r) = \left(\frac{28}{3}, -2\right)$

B $(a, r) = (5, 2)$

C $(a, r) = (4, 3)$

D $(a, r) = \left(\frac{31}{3}, -2\right)$

Classified Practice

Geometric Sequences

Q159 MCQ

Given $S_3 = 13$ and the sum from the 3rd to the 5th terms is 117, find all real pairs (a, r) :

Choose the correct option.

A $(a, r) = (2, 3)$

B $(a, r) = (1, 4)$

C $(a, r) = \left(\frac{20}{7}, -3\right)$

D $(a, r) = (1, 3)$ or $(a, r) = \left(\frac{13}{7}, -3\right)$

Classified Practice

Geometric Sequences

Q160 MCQ

Given $S_3 = 35$ and the sum from the 3rd to the 5th terms is 140, find all real pairs (a, r) :

Choose the correct option.

A $(a, r) = (6, 2)$

B $(a, r) = (5, 3)$

C $(a, r) = (5, 2)$ or $(a, r) = \left(\frac{35}{3}, -2\right)$

D $(a, r) = \left(\frac{38}{3}, -2\right)$

Classified Practice

Geometric Sequences

Q161 MCQ

Given $S_3 = \frac{19}{2}$ and the sum from the 3rd to the 5th terms is $\frac{171}{8}$, find all real pairs (a, r) :

Choose the correct option.

A $(a, r) = \left(3, \frac{3}{2}\right)$

B $(a, r) = \left(2, \frac{5}{2}\right)$

C $(a, r) = \left(\frac{45}{7}, -\frac{3}{2}\right)$

D $(a, r) = \left(2, \frac{3}{2}\right)$ or $(a, r) = \left(\frac{38}{7}, -\frac{3}{2}\right)$

Classified Practice

Geometric Sequences

Q162 **Written**

Find the first term and common ratio of a geometric sequence if $S_3 = 9$ and the sum from the 4th to the 6th term is -72 .

Classified Practice

Geometric Sequences

Q163 MCQ

Given $S_3 = 7$ and the sum from the 3rd to the 5th terms is $\frac{7}{4}$, find all real pairs (a, r) :

Choose the correct option.

A $(a, r) = \left(5, \frac{1}{2}\right)$

B $(a, r) = \left(4, \frac{3}{2}\right)$

C $(a, r) = \left(4, \frac{1}{2}\right)$ or $(a, r) = \left(\frac{28}{3}, -\frac{1}{2}\right)$

D $(a, r) = \left(\frac{31}{3}, -\frac{1}{2}\right)$

Classified Practice

Geometric Sequences

Q164 **Written**

Find the first term and common ratio of a geometric sequence if $S_3 = 35$ and the sum from the 3rd to the 5th term is 140.

Q165 Written

Find the first term and common ratio of a geometric sequence if $S_3 = 21$ and the sum from the 3rd to the 5th term is 84.

Classified Practice

Geometric Sequences

Q166 Written

Find the first term and common ratio of a geometric sequence if $S_3 = 9$ and the sum from the 3rd to the 5th term is 36.

Q167 Written

Find the first term and common ratio of a geometric sequence if $S_3 = 78$ and the sum from the 3rd to the 5th term is 702.

Classified Practice

Geometric Sequences

Q168 MCQ

Given $S_3 = 6$ and the sum from the 3rd to the 5th terms is 24, find all real pairs (a, r) :

Choose the correct option.

A $(a, r) = (3, -2)$

B $(a, r) = (2, -2)$ or $(a, r) = \left(\frac{6}{7}, 2\right)$

C $(a, r) = (2, -1)$

D $(a, r) = \left(\frac{13}{7}, 2\right)$

Classified Practice

Geometric Sequences

Q169 MCQ

Given $S_3 = 9$ and the sum from the 3rd to the 5th terms is 36, find all real pairs (a, r) :

Choose the correct option.

A $(a, r) = (3, -2)$ or $(a, r) = \left(\frac{9}{7}, 2\right)$

B $(a, r) = (4, -2)$

C $(a, r) = (3, -1)$

D $(a, r) = \left(\frac{16}{7}, 2\right)$

Classified Practice

Geometric Sequences

Q170 **Written**

Find the first term and common ratio of a geometric sequence if the 2nd term is 3 and $S_3 = 13$.

Q171 Written

Find the first term and common ratio of a geometric sequence if the 2nd term is 6, and $S_3 = 21$.

Classified Practice

Geometric Sequences

Q172 MCQ

The second term is 6 and $S_3 = 21$. Find all real pairs (a, r) :

Choose the correct option.

A $(a, r) = (4, 2)$

B $(a, r) = (3, 3)$

C $(a, r) = (3, 2)$ or $(a, r) = \left(12, \frac{1}{2}\right)$

D $(a, r) = \left(13, \frac{1}{2}\right)$

Classified Practice

Geometric Sequences

Q173 MCQ

The second term is 6 and $S_3 = 26$. Find all real pairs (a, r) :

Choose the correct option.

A $(a, r) = (3, 3)$

B $(a, r) = (2, 4)$

C $(a, r) = \left(19, \frac{1}{3}\right)$

D $(a, r) = (2, 3)$ or $(a, r) = \left(18, \frac{1}{3}\right)$

Classified Practice

Geometric Sequences

Q174 MCQ

The second term is 8 and $S_3 = 28$. Find all real pairs (a, r) :

Choose the correct option.

A $(a, r) = (5, 2)$

B $(a, r) = (4, 2)$ or $(a, r) = \left(16, \frac{1}{2}\right)$

C $(a, r) = (4, 3)$

D $(a, r) = \left(17, \frac{1}{2}\right)$

Classified Practice

Geometric Sequences

Q175 MCQ

The second term is 3 and $S_3 = 13$. Find all real pairs (a, r) :

Choose the correct option.

A $(a, r) = (2, 3)$

B $(a, r) = (1, 4)$

C $(a, r) = (1, 3)$ or $(a, r) = \left(9, \frac{1}{3}\right)$

D $(a, r) = \left(10, \frac{1}{3}\right)$

Classified Practice

Geometric Sequences

Q176 MCQ

The second term is $\frac{5}{2}$ and $S_3 = \frac{35}{4}$. Find all real pairs (a, r) :

Choose the correct option.

A $(a, r) = \left(6, \frac{1}{2}\right)$

B $(a, r) = \left(5, \frac{3}{2}\right)$

C $(a, r) = \left(\frac{9}{4}, 2\right)$

D $(a, r) = \left(5, \frac{1}{2}\right)$ or $(a, r) = \left(\frac{5}{4}, 2\right)$

Classified Practice

Geometric Sequences

Q177 Written

Find the first term and common ratio of a geometric sequence if the 3rd term is 4 and $S_3 = 7$.

Q178 Written

Find the first term and common ratio of a geometric sequence if the 3rd term is 12, and $S_3 = 9$.

Classified Practice

Geometric Sequences

Q179 **Written**

Find the first term and common ratio of a geometric sequence if the 3rd term is 20, and $S_3 = 35$.

Q180 Written

Find the first term and common ratio of a geometric sequence if $S_3 = 21$ and $S_6 = 189$.

Classified Practice

Geometric Sequences

Q181 **Written**

Find the first term and common ratio of a geometric sequence if $S_3 = 26$ and $S_6 = 728$.

Q182 Written

Find the first term and common ratio of a geometric sequence if $S_3 = 65$ and $S_6 = 1820$.

Classified Practice

Geometric Sequences

Q183 **Written**

Find the first term and common ratio of a geometric sequence if the sum of the first and second terms is 3, and $S_4 = 51$.

Q184 Written

Find the first term and common ratio of a geometric sequence if the sum of the first and second terms is -8 , and $S_4 = -80$.

Classified Practice

Geometric Sequences

Q185 MCQ

A geometric sequence has $S_6 = 63$, $S_7 = 127$, and $S_8 = 255$. Find a_{10} :

Choose the correct option.

A 537

B 487

C 562

D 512

Classified Practice

Geometric Sequences

Q186 Challenge

If S_n is the sum of the first n terms of a geometric sequence, $S_8 = 255$, $S_6 = 63$, and $a_8 - a_7 = 64$, find a_{10} .

Classified Practice

Geometric Sequences

Quiz MCQ 1 [Concept Quiz](#) · MCQ

Select the correct option: For $a = 3, r = 2$, find a_6 :

Choose the correct option.

A 48

B 96

C 192

D 64

Quiz MCQ 2 Concept Quiz · MCQ

Select the correct option: If 2, x , 8 are consecutive geometric terms, find x :

Choose the correct option.

- A 4
- B -4
- C ± 4
- D 16

Classified Practice

Geometric Sequences

Quiz MCQ 3 Concept Quiz · MCQ

For 1, 3, 9, ..., find S_n :

Choose the correct option.

A $\frac{3^{n-1} - 1}{2}$

B $\frac{3^n - 1}{2}$

C $\frac{3^n + 1}{2}$

D $\frac{3^n - 1}{3}$

Quiz MCQ 4 Concept Quiz · MCQ

A geometric sequence has $S_6 = 63$, $S_7 = 127$, and $S_8 = 255$. Find a_{10} :

Choose the correct option.

A 528

B 496

C 544

D 512

Classified Practice

Geometric Sequences

Quiz WRITTEN 5 Concept Quiz · Written**Find the general term of $-4, 8, -16, 32, \dots$**

Quiz WRITTEN 6 Concept Quiz · Written

If 1, x , 25 are consecutive geometric terms, find x .

Classified Practice

Geometric Sequences

Quiz WRITTEN 7 Concept Quiz · Written

A four-stage process has outputs 5, 15, 45, 135. Find the total output.

Quiz WRITTEN 8 Concept Quiz · Written

A geometric sequence has $S_6 = 63$, $S_7 = 127$, and $S_8 = 255$. Find a_{10} .

Answer key

Use the question number shown on each page.

Q001 $2 \times 3^{n-1}$

Q002 $3 \times 4^{n-1}$

Q003 $5 \times 2^{n-1}$

Q004 $2 \times 5^{n-1}$

Q005 $7 \times 4^{n-1}$

Q006 A

Q007 A

Q008 A

Q009 A

Q010 A

Q011 A

Q012 $5(-2)^{n-1}$

Q013 $2(-3)^{n-1}$

Q014 $3(-5)^{n-1}$

Q015 $4(-2)^{n-1}$

Q016 A

Q017 A

Q018 A

Q019 A

Q020 A

Q021 $8\left(-\frac{1}{3}\right)^{n-1}$

Q022 $4\left(\frac{1}{3}\right)^{n-1}$

Q023 $3\left(-\frac{1}{2}\right)^{n-1}$

Q024 $6\left(\frac{1}{2}\right)^{n-1}$

Q025 A

Q026 A

Q027 A

Q028 A

Q029 A

Q030 $5 \times 2^{n-1}$

Q031 $2 \times 3^{n-1}$

Q032 $3 \times 2^{n-1}$

Q033 A

Q034 A

Q035 A

Q036 A

Q037 A

Q038 $2(-3)^{n-1}$

Q039 $5(-2)^{n-1}$

Q040 A

Q041 A

Q042 A

Q043 A

Q044 $-4\left(-\frac{3}{2}\right)^{n-1}$

Q045 $-8\left(-\frac{3}{2}\right)^{n-1}$

Q046 $2\left(\frac{1}{3}\right)^{n-1}$

Q047 $48\left(\frac{1}{2}\right)^{n-1}$

Q048 $\frac{1}{6}\left(\frac{3}{2}\right)^{n-1}$

Q049 $8(2\sqrt{2})^{n-1}$

Q050 $3\left(-\frac{5}{2}\right)^{n-1}$

Q051 $9\left(-\frac{\sqrt{3}}{3}\right)^{n-1}$

Q052 $5\left(-\frac{2}{3}\right)^{n-1}$

Q053 A

Q054 A

Q055 A

Q056 A

Q057 A

Q058 A

Q059 A

Q060 $r = -\frac{1}{3}, A = 54, B = -2$

Q061 A

Q062 $r = \frac{1}{2}, A = 48, B = 6$

Q063 $r = -3, A = 1, B = -27$

Q064 $r = \frac{1}{3}, A = 1, B = \frac{1}{27}$

Q065 $r = -\frac{1}{2}, A = \frac{1}{2}, B = -\frac{1}{16}$

Q066 $a_7 = 648$

Q067 $a_n = 3(2)^{n-1}$ or $-3(-2)^{n-1}$

Q068 $a_n = 2(3)^{n-1}$ or $2(-3)^{n-1}$

Q069 $a_n = 2(3)^{n-1}$ or $-2(-3)^{n-1}$

Q070 $a_n = 3(3)^{n-1}$ or $3(-3)^{n-1}$

Q071 $a_n = 7(2)^{n-1}$

Q072 $a_n = 3(-2)^{n-1}$

Q073 $a_n = 3(-2)^{n-1}$

Q074 $a_n = 3(2)^{n-1}$ or $3(-2)^{n-1}$

Q075 $a_n = -7(2)^{n-1}$ or $-7(-2)^{n-1}$

Q076 $a_n = -2(-2)^{n-1}$

Q077 $a_n = -5(-2)^{n-1}$

Q078 $a_n = \frac{3}{4}(4)^{n-1}$ or $-\frac{3}{4}(-4)^{n-1}$

Q079 $a_n = 4\left(\frac{3}{2}\right)^{n-1}$ or $4\left(-\frac{3}{2}\right)^{n-1}$

Q080 $a_n = \frac{2}{243}(3)^{n-1}$ or $-\frac{2}{243}(-3)^{n-1}$

Q081 A

Q082 A

Q083 A

Q084 A

Q085 A

Q086 A

Q087 A

Q088 A

Q089 A

Q090 A

Q091 A

Q092 A

Q093 A

Q094 A

Q095 A	Q122 $\frac{189}{16}$	Q148 $\frac{1 - (-3)^n}{2}$	Q171 $(a, r) = (3, 2)$ or $(a, r) = \left(12, \frac{1}{2}\right)$
Q096 A	Q123 $\frac{13}{9}$	Q149 $\frac{16}{3}\{1 - (-\frac{1}{2})^n\}$	Q172 C
Q097 $x = \pm 4$	Q124 $\frac{255}{128}$	Q150 $\frac{1 - (-3)^n}{4}$	Q173 D
Q098 $x = \pm 5$	Q125 B	Q151 $\frac{32}{3}\{1 - (-\frac{1}{2})^n\}$	Q174 B
Q099 $x = \pm 2$	Q126 A	Q152 $\frac{1 - (-2)^n}{3}$	Q175 C
Q100 $x = \pm 2\sqrt{3}$	Q127 D	Q153 A	Q176 D
Q101 A	Q128 C	Q154 $\frac{40}{9}$	Q177 $(a, r) = (1, 2)$ or $(a, r) = \left(9, -\frac{2}{3}\right)$
Q102 A	Q129 B	Q155 D	Q178 $(a, r) = (3, -2)$
Q103 A	Q130 A	Q156 C	Q179 $(a, r) = (5, 2)$ or $(a, r) = \left(45, -\frac{2}{3}\right)$
Q104 A	Q131 -30	Q157 B	Q180 $(a, r) = (3, 2)$
Q105 A	Q132 -20	Q158 A	Q181 $(a, r) = (2, 3)$
Q106 A	Q133 -364	Q159 D	Q182 $(a, r) = (5, 3)$
Q107 $(a, b) = (9, 12)$ or $(1, -4)$	Q134 D	Q160 C	Q183 $(a, r) = \left(\frac{3}{5}, 4\right)$ or $(a, r) = (-1, -4)$
Q108 $(a, b) = (2, 2)$ or $(-4, -1)$	Q135 C	Q161 D	Q184 $(a, r) = (-2, 3)$ or $(a, r) = (4, -3)$
Q109 $(a, b) = (2, 6)$ or $(\frac{1}{2}, 3)$	Q136 B	Q162 $(a, r) = (3, -2)$	Q185 D
Q110 $(a, b) = (5, 15)$ or $(\frac{5}{4}, \frac{15}{2})$	Q137 A	Q163 C	Q186 $a_7 = 64, a_8 = 128, r = 2, a_{10} = 512$
Q111 484	Q138 D	Q164 $(a, r) = (5, 2)$ or $(a, r) = \left(\frac{35}{3}, -2\right)$	Quiz MCQ 1 B
Q112 93	Q139 C	Q165 $(a, r) = (3, 2)$ or $(a, r) = (7, -2)$	Quiz MCQ 2 C
Q113 200	Q140 $1 - (-2)^n$	Q166 $(a, r) = \left(\frac{9}{7}, 2\right)$ or $(a, r) = (3, -2)$	Quiz MCQ 3 B
Q114 3069	Q141 $7(2^n - 1)$	Q167 $(a, r) = (6, 3)$ or $(a, r) = \left(\frac{78}{7}, -3\right)$	Quiz MCQ 4 D
Q115 D	Q142 $3\{1 - (\frac{1}{3})^n\}$	Q168 B	Quiz WRITTEN 5 $-4(-2)^{n-1}$
Q116 C	Q143 $4(2^n - 1)$	Q169 A	Quiz WRITTEN 6 $x = \pm 5$
Q117 B	Q144 C	Q170 $(a, r) = (1, 3)$ or $(a, r) = \left(9, \frac{1}{3}\right)$	Quiz WRITTEN 7 200
Q118 A	Q145 D		Quiz WRITTEN 8 512
Q119 D	Q146 A		
Q120 C	Q147 $3^n - 1$		
Q121 $\frac{63}{4}$			

Concept 3 | Summation

English Student Edition • Focused formulas and guided practice

Formula opener

Linearity

$$\sum (au_k + bv_k) = a \sum u_k + b \sum v_k$$

Standard sums

$$\sum k = \frac{n(n+1)}{2}$$

Telescoping

$$\sum_{k=1}^n (u_k - u_{k+1}) = u_1 - u_{n+1}$$

Classified Practice

Summation

Q001 Written

Write $\sum_{k=1}^5 2k$ as a sum of terms.

Q002 Written

Write $\sum_{k=1}^5 (2k + 1)$ as a sum of terms.

Classified Practice

Summation

Q003 Written

Write $\sum_{k=1}^5 3k$ as a sum of terms.

Q004 Written

Write $\sum_{k=1}^4 (4k - 2)$ as a sum of terms.

Classified Practice

Summation

Q005 Written

Write $\sum_{k=1}^5 (2k - 3)$ as a sum of terms.

Q006 MCQ

Select the correct option: Expand $\sum_{k=1}^3 4k$ into a sum of terms:

Choose the correct option.

A $4 + 8 + 12$

B $4 + 8 + 16$

C $8 + 12 + 16$

D $4 + 12 + 20$

Classified Practice

Summation

Q007 MCQ

Select the correct option: Expand $\sum_{k=1}^4 2^k$ into a sum of terms:

Choose the correct option.

A $2 + 4 + 8 + 16$

B $1 + 2 + 4 + 8$

C $2 + 6 + 18 + 54$

D $4 + 8 + 16 + 32$

Q008 MCQ

Select the correct option: Expand $\sum_{k=1}^3 (5^k - 1)$ into a sum of terms:

Choose the correct option.

A $4 + 24 + 124$

B $5 + 25 + 125$

C $4 + 25 + 124$

D $1 + 4 + 24$

Classified Practice

Summation

Q009 MCQ

Select the correct option: Expand $\sum_{k=1}^5 (k + 4)$ into a sum of terms:

Choose the correct option.

A $5 + 6 + 7 + 8 + 9$

B $4 + 5 + 6 + 7 + 8$

C $1 + 2 + 3 + 4 + 5$

D $6 + 7 + 8 + 9 + 10$

Q010 MCQ

Select the correct option: Expand $\sum_{k=2}^5 (2k + 3)$ into a sum of terms:

Choose the correct option.

A $7 + 9 + 11 + 13$

B $5 + 7 + 9 + 11$

C $4 + 7 + 10 + 13$

D $8 + 10 + 12 + 14$

Classified Practice

Summation

Q011 MCQ

Select the correct option: Expand $\sum_{k=1}^3 (2^k + 1)$ into a sum of terms:

Choose the correct option.

- A $3 + 5 + 9$
- B $2 + 4 + 8$
- C $1 + 3 + 5$
- D $4 + 6 + 10$

Q012 MCQ

Select the correct option: Expand $\sum_{k=1}^3 (k^2 + 3k)$ into a sum of terms:

Choose the correct option.

A $4 + 10 + 18$

B $3 + 8 + 15$

C $1 + 4 + 9$

D $6 + 12 + 18$

Classified Practice

Summation

Q013 MCQ

Select the correct option: Expand $\sum_{k=2}^5 (k^2 + 1)$ into a sum of terms:

Choose the correct option.

A $5 + 10 + 17 + 26$

B $2 + 5 + 10 + 17$

C $4 + 9 + 16 + 25$

D $6 + 11 + 18 + 27$

Q014 Written

Write $\sum_{k=1}^3 (k^2 - 3)$ as a sum of terms.

Classified Practice

Summation

Q015 Written

Write $\sum_{k=1}^3 (3k^2 + 2)$ as a sum of terms.

Q016 Written

Write $\sum_{k=1}^4 k^2$ as a sum of terms.

Classified Practice

Summation

Q017 Written

Write $\sum_{k=1}^3 (2k^2 + 3)$ as a sum of terms.

Q018 Written

Write $\sum_{k=1}^4 (k^2 - 2k)$ as a sum of terms.

Classified Practice

Summation

Q019 MCQ

Select the correct option: Expand $\sum_{k=1}^4 (2k + 1)$ into a sum of terms:

Choose the correct option.

A $3 + 5 + 7 + 9$

B $2 + 4 + 6 + 8$

C $3 + 6 + 9 + 12$

D $1 + 3 + 5 + 7$

Q020 MCQ

Select the correct option: Expand $\sum_{k=2}^5 (3k - 1)$ into a sum of terms:

Choose the correct option.

A $5 + 8 + 11 + 14$

B $2 + 5 + 8 + 11$

C $5 + 9 + 13 + 17$

D $4 + 7 + 10 + 13$

Classified Practice

Summation

Q021 MCQ

Select the correct option: Expand $\sum_{k=1}^4 (2k^2 - 3)$ into a sum of terms:

Choose the correct option.

A $-1 + 5 + 15 + 29$

B $-1 + 1 + 5 + 13$

C $2 + 5 + 10 + 17$

D $-3 + 5 + 13 + 29$

Q022 MCQ

Select the correct option: Expand $\sum_{k=1}^4 (3k^2 + 1)$ into a sum of terms:

Choose the correct option.

A $4 + 13 + 28 + 49$

B $3 + 12 + 27 + 48$

C $4 + 10 + 19 + 28$

D $1 + 4 + 9 + 16$

Classified Practice

Summation

Q023 MCQ

Select the correct option: Expand $\sum_{k=1}^4 (k^2 - k)$ into a sum of terms:

Choose the correct option.

A $0 + 2 + 6 + 12$

B $1 + 4 + 9 + 16$

C $0 + 1 + 4 + 9$

D $2 + 6 + 12 + 20$

Q024 MCQ

Select the correct option: Expand $\sum_{k=2}^4 5k$ into a sum of terms:

Choose the correct option.

A $10 + 15 + 20$

B $5 + 10 + 15$

C $2 + 4 + 6$

D $15 + 20 + 25$

Classified Practice

Summation

Q025 MCQ

Select the correct option: Expand $\sum_{k=1}^4 3$ into a sum of terms:

Choose the correct option.

A $3 + 3 + 3 + 3$

B $3 + 6 + 9 + 12$

C $1 + 1 + 1 + 1$

D $4 + 4 + 4 + 4$

Q026 MCQ

Select the correct option: Expand $\sum_{k=1}^4 (5k + 1)$ into a sum of terms:

Choose the correct option.

A $6 + 11 + 16 + 21$

B $5 + 10 + 15 + 20$

C $1 + 6 + 11 + 16$

D $7 + 12 + 17 + 22$

Classified Practice

Summation

Q027 Written

Write $\sum_{k=1}^4 3^k$ as a sum of terms.

Q028 Written

Write $\sum_{k=1}^3 2$ as a sum of terms.

Classified Practice

Summation

Q029 Written

Write $\sum_{k=1}^4 2^k$ as a sum of terms.

Q030 Written

Write $\sum_{k=1}^6 3$ as a sum of terms.

Classified Practice

Summation

Q031 Written

Write $\sum_{k=1}^3 3^k$ as a sum of terms.

Q032 Written

Write $\sum_{k=1}^3 (5^k + 3)$ as a sum of terms.

Classified Practice

Summation

Q033 Written

Write $\sum_{k=1}^4 (2^k - 3)$ as a sum of terms.

Q034 Written

Write $\sum_{k=1}^4 5$ as a sum of terms.

Classified Practice

Summation

Q035 Written

Write $\sum_{k=1}^3 3$ as a sum of terms.

Q036 Written

Write $\sum_{k=1}^5 4$ as a sum of terms.

Classified Practice

Summation

Q037 MCQ

Select the correct option: Expand $\sum_{k=1}^3 (k^2 + 2)$ into a sum of terms:

Choose the correct option.

A $3 + 6 + 11$

B $2 + 5 + 10$

C $3 + 5 + 9$

D $1 + 4 + 9$

Q038 MCQ

Select the correct option: Expand $\sum_{k=1}^5 7$ into a sum of terms:

Choose the correct option.

A $7 + 7 + 7 + 7 + 7$

B $7 + 14 + 21 + 28 + 35$

C $7 + 7 + 7 + 7$

D $5 + 5 + 5 + 5 + 5$

Classified Practice

Summation

Q039 MCQ

Select the correct option: Expand $\sum_{k=3}^6 k^2$ into a sum of terms:

Choose the correct option.

A $9 + 16 + 25 + 36$

B $1 + 4 + 9 + 16$

C $9 + 25 + 49 + 81$

D $3 + 4 + 5 + 6$

Q040 MCQ

Select the correct option: Expand $\sum_{k=1}^3 6^k$ into a sum of terms:

Choose the correct option.

A $6 + 36 + 216$

B $1 + 6 + 36$

C $6 + 12 + 18$

D $36 + 72 + 108$

Classified Practice

Summation

Q041 MCQ

Select the correct option: Expand $\sum_{k=1}^5 (4k - 3)$ into a sum of terms:

Choose the correct option.

A $1 + 5 + 9 + 13 + 17$

B $4 + 8 + 12 + 16 + 20$

C $1 + 4 + 7 + 10 + 13$

D $-1 + 3 + 7 + 11 + 15$

Q042 MCQ

Select the correct option: Expand $\sum_{k=1}^4 (3k - 5)$ into a sum of terms:

Choose the correct option.

A $-2 + 1 + 4 + 7$

B $-2 + 0 + 2 + 4$

C $3 + 6 + 9 + 12$

D $-5 - 2 + 1 + 4$

Classified Practice

Summation

Q043 MCQ

Select the correct option: Expand $\sum_{k=1}^3 (4^k - 2)$ into a sum of terms:

Choose the correct option.

A $2 + 14 + 62$

B $4 + 16 + 64$

C $2 + 12 + 60$

D $-2 + 2 + 14$

Q044 MCQ

Select the correct option: Expand $\sum_{k=1}^3 (2k^2 - k)$ into a sum of terms:

Choose the correct option.

A $1 + 6 + 15$

B $2 + 7 + 18$

C $1 + 3 + 5$

D $0 + 6 + 12$

Classified Practice

Summation

Q045 Written

Express $3 + 7 + 11 + \cdots + (4n - 1)$ using the summation symbol Σ .

Q046 Written**Express $5 + 8 + 11 + \cdots + (3n + 2)$ using Σ .**

Classified Practice

Summation

Q047 Written

Write the sum of the first n terms of the sequence $a_n = 2n + 1$ using Σ .

Q048 Written**Express $1 + 3 + 5 + \cdots + (2n - 1)$ using Σ .**

Classified Practice

Summation

Q049 Written

Write the sum of the first n terms of $a_n = 7n - 1$ using Σ .

Q050 Written**Write the sum of the first n terms of $a_n = 4n + 5$ using Σ .**

Classified Practice

Summation

Q051 Written

Write the sum of the first n terms of $a_n = 3n - 2$ using Σ .

Q052 MCQ

Select the correct option: Express $2 + 5 + 8 + \dots + (3n - 1)$ using the summation symbol Σ :

Choose the correct option.

- A $\sum_{k=1}^n (3k - 1)$
 - B $\sum_{k=0}^n (3k - 1)$
 - C $\sum_{k=1}^n (3k + 1)$
 - D $\sum_{k=1}^n (2k - 1)$
-
-
-

Classified Practice

Summation

Q053 MCQ

Select the correct option: Express $7 + 11 + 15 + \dots + 43$ using the summation symbol Σ :**Choose the correct option.**

A $\sum_{k=1}^{10} (4k + 3)$

B $\sum_{k=1}^9 (4k + 3)$

C $\sum_{k=0}^{10} (4k + 3)$

D $\sum_{k=1}^{10} (4k - 3)$

Q054 MCQ

Select the correct option: Express $6 + 10 + 14 + \dots + (4n + 2)$ using the summation symbol Σ :

Choose the correct option.

- A $\sum_{k=1}^n (4k + 2)$
- B $\sum_{k=0}^n (4k + 2)$
- C $\sum_{k=1}^n (4k - 2)$
- D $\sum_{k=1}^n (3k + 2)$

Classified Practice

Summation

Q055 MCQ

Select the correct option: Express $8 + 13 + 18 + \dots + 103$ using the summation symbol Σ :**Choose the correct option.**

A $\sum_{k=1}^{20} (5k + 3)$

B $\sum_{k=1}^{19} (5k + 3)$

C $\sum_{k=0}^{20} (5k + 3)$

D $\sum_{k=1}^{20} (5k - 3)$

Q056 MCQ

Select the correct option: Express $11^2 + 12^2 + \dots + 19^2$ using the summation symbol Σ :

Choose the correct option.

- A $\sum_{k=11}^{19} k^2$
- B $\sum_{k=1}^9 k^2$
- C $\sum_{k=11}^{19} k$
- D $\sum_{k=12}^{19} k^2$

Classified Practice

Summation

Q057 MCQ

Select the correct option: Express $2^2 + 3^2 + \dots + 10^2$ using the summation symbol Σ :

Choose the correct option.

A $\sum_{k=2}^{10} k^2$

B $\sum_{k=1}^9 k^2$

C $\sum_{k=2}^{10} k$

D $\sum_{k=3}^{10} k^2$

Q058 MCQ

Select the correct option: Express $1 + 3 + 9 + \dots + 3^{n-1}$ using the summation symbol Σ :

Choose the correct option.

A $\sum_{k=0}^{n-1} 3^k$

B $\sum_{k=1}^n 3^k$

C $\sum_{k=1}^{n-1} 3^k$

D $\sum_{k=0}^n 3^k$

Classified Practice

Summation

Q059 MCQ

Select the correct option: Express $2 + 7 + 12 + \dots + (5n - 3)$ using the summation symbol Σ :

Choose the correct option.

A $\sum_{k=1}^n (5k - 3)$

B $\sum_{k=0}^n (5k - 3)$

C $\sum_{k=1}^n (5k + 3)$

D $\sum_{k=1}^n (4k - 3)$

Q060 Written

Express $6^2 + 7^2 + 8^2 + \dots + 15^2$ using Σ .

Classified Practice

Summation

Q061 Written

Express $4^2 + 5^2 + 6^2 + \dots + 13^2$ using Σ .

Q062 Written**Express $1 + 7 + 25 + 79 + \dots + (3^n - 2)$ using Σ .**

Classified Practice

Summation

Q063 Written

Express $1 + 2 + 4 + 8 + \dots + 2^{n-1}$ using Σ .

Q064 Written**Express $7^2 + 8^2 + \dots + 21^2$ using Σ .**

Classified Practice

Summation

Q065 Written

Express $12^2 + 13^2 + \dots + 18^2$ using Σ .

Q066 Written**Express $10^2 + 11^2 + \dots + 20^2$ using Σ .**

Classified Practice

Summation

Q067 MCQ

Select the correct option: Express $4^2 + 5^2 + \dots + 12^2$ using the summation symbol Σ :

Choose the correct option.

A $\sum_{k=4}^{12} k^2$

B $\sum_{k=1}^9 k^2$

C $\sum_{k=4}^{12} k$

D $\sum_{k=5}^{12} k^2$

Q068 MCQ

Select the correct option: Express $3^1 + 3^2 + \dots + 3^n$ using the summation symbol Σ :

Choose the correct option.

A $\sum_{k=1}^n 3^k$

B $\sum_{k=0}^n 3^k$

C $\sum_{k=1}^n 3^{k-1}$

D $\sum_{k=1}^n k^3$

Classified Practice

Summation

Q069 MCQ

Select the correct option: Express $2 + 4 + 8 + \dots + 2^n$ using the summation symbol Σ :

Choose the correct option.

- A $\sum_{k=1}^n 2^k$
- B $\sum_{k=0}^n 2^k$
- C $\sum_{k=1}^n 2^{k-1}$
- D $\sum_{k=1}^n k^2$

Q070 MCQ

Select the correct option: Express $1 + 2^1 + 2^2 + \dots + 2^{n-1}$ using the summation symbol Σ :

Choose the correct option.

- A $\sum_{k=0}^{n-1} 2^k$
- B $\sum_{k=1}^n 2^k$
- C $\sum_{k=1}^{n-1} 2^k$
- D $\sum_{k=0}^n 2^k$

Classified Practice

Summation

Q071 MCQ

Select the correct option: Express $7 + 14 + 21 + \dots + 7n$ using the summation symbol Σ :

Choose the correct option.

A $\sum_{k=1}^n 7k$

B $\sum_{k=0}^n 7k$

C $\sum_{k=1}^n (7k + 1)$

D $\sum_{k=1}^n k^7$

Q072 MCQ

Select the correct option: Express $10 + 15 + 20 + \dots + 5n$ using the summation symbol Σ :**Choose the correct option.**

A $\sum_{k=2}^n 5k$

B $\sum_{k=1}^n 5k$

C $\sum_{k=2}^n (5k + 1)$

D $\sum_{k=0}^n 5k$

Classified Practice

Summation

Q073 MCQ

Select the correct option: Express $6^2 + 7^2 + \dots + 16^2$ using the summation symbol Σ :

Choose the correct option.

A $\sum_{k=6}^{16} k^2$

B $\sum_{k=1}^{11} k^2$

C $\sum_{k=6}^{16} k$

D $\sum_{k=7}^{16} k^2$

Q074 Written**Express $2 + 5 + 8 + \cdots + 29$ using Σ .**

Classified Practice

Summation

Q075 Written

Express $8 + 16 + 24 + \cdots + 56$ using Σ .

Q076 Written**Express $7 + 13 + 19 + \cdots + 73$ using Σ .**

Classified Practice

Summation

Q077 Written

Express $4 + 9 + 14 + \cdots + 119$ using Σ .

Classified Practice

Summation

Q078 Challenge

A green-tech startup installs 1 solar panel in Week 1, 4 in Week 2, 7 in Week 3, and $3n - 2$ in Week n . Write one expression using Σ for the total panels installed from Week 1 through Week n .

Classified Practice

Summation

Q079 Written

Find $\sum_{k=1}^n (-3)$.

Q080 Written

Find $\sum_{k=1}^n 3$.

Classified Practice

Summation

Q081 Written

Find $\sum_{k=1}^n 6$.

Q082 Written

Find $\sum_{k=1}^n (-4)$.

Classified Practice

Summation

Q083 Written

Find $\sum_{k=1}^n 1$.

Q084 MCQ

Select the correct option: Find $\sum_{k=1}^n 9$:

Choose the correct option.

A $9n$

B $n + 9$

C $9n^2$

D $\frac{9n(n+1)}{2}$

Classified Practice

Summation

Q085 MCQ

Select the correct option: Find $\sum_{k=1}^n (-2)$:

Choose the correct option.

A $-2n$

B $2n$

C $-n - 2$

D $-n^2$

Q086 MCQ

Select the correct option: Find $\sum_{k=1}^7 4$:

Choose the correct option.

A 28

B 11

C 16

D 49

Classified Practice

Summation

Q087 MCQ

Select the correct option: Find $\sum_{k=1}^{12} (-5)$:

Choose the correct option.

A -60

B 60

C -17

D -144

Q088 MCQ

Select the correct option: Find $\sum_{k=1}^n k$:

Choose the correct option.

A $\frac{n(n+1)}{2}$

B $n(n+1)$

C $\frac{n(n-1)}{2}$

D n^2

Classified Practice

Summation

Q089 MCQ

Select the correct option: Find $\sum_{k=1}^n 4k$:

Choose the correct option.

A $2n(n+1)$

B $4n(n+1)$

C $2n(n-1)$

D $4n$

Q090 MCQ

Select the correct option: Find $\sum_{k=1}^{10} k$:

Choose the correct option.

A 55

B 45

C 100

D 110

Classified Practice

Summation

Q091 MCQ

Select the correct option: Find $\sum_{k=1}^8 3k$:

Choose the correct option.

A 108

B 72

C 96

D 120

Q092 MCQ

Select the correct option: Find $\sum_{k=1}^6 5k$:

Choose the correct option.

A 105

B 75

C 90

D 120

Classified Practice

Summation

Q093 MCQ

Select the correct option: Find $\sum_{k=1}^{20} k$:

Choose the correct option.

- A 210
- B 190
- C 200
- D 420

Q094 MCQ

Select the correct option: Find $\sum_{k=1}^n k^2$:

Choose the correct option.

A $\frac{n(n+1)(2n+1)}{6}$

B $\frac{n(n+1)^6}{2}$

C $\frac{n(n+1)(2n-1)}{6}$

D $\frac{n^2(n+1)^2}{4}$

Classified Practice

Summation

Q095 MCQ

Select the correct option: Find $\sum_{k=1}^n 6k^2$:

Choose the correct option.

A $n(n+1)(2n+1)$

B $3n(n+1)$

C $n(n+1)(2n-1)$

D $6n^2$

Q096 MCQ

Select the correct option: Find $\sum_{k=1}^5 k^2$:

Choose the correct option.

A 55

B 25

C 30

D 50

Classified Practice

Summation

Q097 MCQ

Select the correct option: Find $\sum_{k=1}^4 2k^2$:

Choose the correct option.

A 60

B 30

C 40

D 80

Q098 MCQ

Select the correct option: Find $\sum_{k=1}^3 4k^2$:

Choose the correct option.

A 56

B 36

C 48

D 72

Classified Practice

Summation

Q099 MCQ

Select the correct option: Find $\sum_{k=1}^7 k^2$:

Choose the correct option.

A 140

B 49

C 91

D 196

Q100 Written

Find $\sum_{k=1}^n (5k + 4)$.

Classified Practice

Summation

Q101 Written

Find $\sum_{k=1}^n (2k - 3)$.

Q102 Written

Find $\sum_{k=1}^n (4k - 1)$.

Classified Practice

Summation

Q103 Written

Find $\sum_{k=1}^n (3k + 2)$.

Q104 Written

Find $\sum_{k=1}^n (6k - 3)$.

Classified Practice

Summation

Q105 MCQ

Select the correct option: Find $\sum_{k=1}^n (3k - 1)$:

Choose the correct option.

A $\frac{n(3n+1)}{2}$

B $\frac{n(3n-1)}{2}$

C $n(3n + 1)$

D $\frac{3n(n+1)}{2}$

Q106 MCQ

Select the correct option: Find $\sum_{k=1}^n (2k + 5)$:

Choose the correct option.

A $n(n + 6)$

B $n(n + 5)$

C $n(n + 1) + 5$

D $2n(n + 6)$

Classified Practice

Summation

Q107 MCQ

Select the correct option: Find $\sum_{k=1}^n (8k - 4)$:

Choose the correct option.

A $4n^2$

B $4n(n + 1)$

C $8n^2$

D $4n(n - 1)$

Q108 MCQ

Select the correct option: Find $\sum_{k=1}^n (5k + 3)$:

Choose the correct option.

A $\frac{n(5n+11)}{2}$

B $\frac{n(5n+3)}{2}$

C $\frac{n(5n+5)}{2}$

D $n(5n + 11)$

Classified Practice

Summation

Q109 Written

Find $\sum_{k=1}^n (k-1)^2$.

Q110 MCQ

Select the correct option: Find $\sum_{k=1}^n (k^2 + k)$:

Choose the correct option.

A $\frac{n(n+1)(n+2)}{6}$

B $\frac{n(n+1)(2n+1)}{6}$

C $\frac{n(n+1)(n+2)}{6}$

D $n(n+1)(n+2)$

Classified Practice

Summation

Q111 MCQ

Select the correct option: Find $\sum_{k=1}^n (2k^2 - k)$:

Choose the correct option.

A $\frac{n(n+1)(4n-1)}{6}$

B $\frac{n(n+1)(4n+1)}{6}$

C $\frac{n(n+1)(2n-1)}{6}$

D $\frac{n(n-1)(4n-1)}{6}$

Q112 MCQ

Select the correct option: Find $\sum_{k=1}^n (k^2 - 1)$:

Choose the correct option.

A $\frac{n(2n^2+3n-5)}{6}$

B $\frac{n(2n^2+3n-1)}{6}$

C $\frac{n(2n^2-3n+1)}{6}$

D $\frac{n(n+1)(2n+1)}{6}$

Classified Practice

Summation

Q113 MCQ

Select the correct option: Find $\sum_{k=1}^n (3k^2 + 2k)$:

Choose the correct option.

A $\frac{n(n+1)(2n+3)}{2}$

B $\frac{n(n+1)(2n+1)}{2}$

C $n(n+1)(2n+3)$

D $\frac{n(n-1)(2n+3)}{2}$

Q114 MCQ

Select the correct option: Find $\sum_{k=1}^n (4k^2 - 3k + 2)$:

Choose the correct option.

A $\frac{n(8n^2+3n+7)}{6}$

B $\frac{n(8n^2+9n+7)}{6}$

C $\frac{n(4n^2+3n+2)}{6}$

D $\frac{n(8n^2-3n+7)}{6}$

Classified Practice

Summation

Q115 Written

Find $1 \cdot 3 + 2 \cdot 8 + 3 \cdot 13 + \cdots + n(5n - 2)$.

Q116 Written

Find $\sum_{k=1}^n (k-1)(3k-1)$.

Classified Practice

Summation

Q117 Written

Find $1 \cdot 3 + 2 \cdot 5 + 3 \cdot 7 + \cdots + n(2n + 1)$.

Q118 Written

Find $\sum_{k=1}^n k(k-2)$.

Classified Practice

Summation

Q119 Written

Find $\sum_{k=1}^n k(2k - 2)$.

Q120 Written

Find $1 \cdot 4 + 2 \cdot 5 + 3 \cdot 6 + \cdots + n(n + 3)$.

Classified Practice

Summation

Q121 Written

Find $1 \cdot 5 + 2 \cdot 8 + 3 \cdot 11 + \cdots + n(3n + 2)$.

Q122 Written**Find** $2 \cdot 5 + 3 \cdot 6 + 4 \cdot 7 + \cdots + (n + 1)(n + 4)$.

Classified Practice

Summation

Q123 Written

Find $\sum_{k=1}^4 (2k - 1)$.

Q124 Written

Find $\sum_{k=1}^5 (3k - 2)$.

Classified Practice

Summation

Q125 Written

Find $\sum_{k=1}^{20} (2k - 7)$.

Q126 Written

Find $\sum_{k=1}^8 (4k + 5)$.

Classified Practice

Summation

Q127 Written

Find $\sum_{k=1}^{10} (2k - 1)$.

Classified Practice

Summation

Q128 **Challenge**

Find $\sum_{r=4}^8 (r^2 + r + 5)$.

Classified Practice

Summation

Q129 Written

Find the sum $\sum_{k=1}^n k^2(k+1)$.

Q130 Written

Find the sum $\sum_{k=1}^n k^2(k-2)$.

Classified Practice

Summation

Q131 Written

Find the sum $\sum_{k=1}^n k^2(k+2)$.

Q132 Written

Find the sum $\sum_{k=1}^n k^2(k-3)$.

Classified Practice

Summation

Q133 MCQ

Select the correct option: Find the sum $\sum_{k=1}^n k^2(k+1)$:

Choose the correct option.

A $\frac{n(n+1)(n+2)(3n+1)}{12}$

B $\frac{n(n+1)(n+2)(3n-1)}{12}$

C $\frac{n(n+1)(3n+1)}{12}$

D $\frac{n(n+1)(n+2)(3n+2)}{6}$

Q134 MCQ

Select the correct option: Find the sum $\sum_{k=1}^n k^2(k-2)$:

Choose the correct option.

- A $\frac{n(n+1)(3n^2-5n-4)}{12}$
- B $\frac{n(n+1)(3n^2+5n-4)}{12}$
- C $\frac{n(n+1)(3n^2-5n+4)}{12}$
- D $\frac{n(n+1)(3n^2-4n-5)}{12}$

Classified Practice

Summation

Q135 MCQ

Select the correct option: Find the sum $\sum_{k=1}^n k^2(k+2)$:

Choose the correct option.

A $\frac{n(n+1)(3n^2+11n+4)}{12}$

B $\frac{n(n+1)(3n^2+11n-4)}{12}$

C $\frac{n(n+1)(3n^2+9n+4)}{12}$

D $\frac{n(n+1)(3n^2+4n+11)}{12}$

Q136 MCQ

Select the correct option: Find the sum $\sum_{k=1}^n k^2(k-3)$:

Choose the correct option.

A $\frac{n(n+1)(n^2-3n-2)}{4}$

B $\frac{n(n+1)(n^2+3n-2)}{4}$

C $\frac{n(n+1)(n^2-3n+2)}{4}$

D $\frac{n(n+1)(n^2-2n-3)}{4}$

Classified Practice

Summation

Q137 MCQ

Select the correct option: Find the sum $\sum_{k=1}^n k^2(k+3)$:

Choose the correct option.

A $\frac{n(n+1)(n^2+5n+2)}{4}$

B $\frac{n(n+1)(n^2+5n-2)}{4}$

C $\frac{n(n+1)(n^2+3n+2)}{4}$

D $\frac{n(n+1)(n^2+2n+5)}{4}$

Q138 MCQ

Select the correct option: Find the sum $\sum_{k=1}^n k^2(k-4)$:

Choose the correct option.

A $\frac{n(n+1)(3n^2-13n-8)}{12}$

B $\frac{n(n+1)(3n^2-13n+8)}{12}$

C $\frac{n(n+1)(3n^2-11n-8)}{12}$

D $\frac{n(n+1)(3n^2-8n-13)}{12}$

Classified Practice

Summation

Q139 MCQ

Select the correct option: Find the sum $\sum_{k=1}^n k^2(k+4)$:

Choose the correct option.

A $\frac{n(n+1)(3n^2+19n+8)}{12}$

B $\frac{n(n+1)(3n^2+19n-8)}{12}$

C $\frac{n(n+1)(3n^2+17n+8)}{12}$

D $\frac{n(n+1)(3n^2+8n+19)}{12}$

Q140 MCQ

Select the correct option: Find the sum $\sum_{k=1}^n k^2(k-1)$:

Choose the correct option.

- A $\frac{n(n+1)(3n^2-5n-4)}{12}$
- B $\frac{n(n+1)(3n^2-5n+4)}{12}$
- C $\frac{n(n+1)(3n^2-4n-5)}{12}$
- D $\frac{n(n+1)(3n^2-n-4)}{12}$

Classified Practice

Summation

Q141 Written

Find the sum $\sum_{k=1}^n k(k^2 - 1)$.

Q142 Written

Find the sum $\sum_{k=1}^n k(k^2 + 2)$.

Classified Practice

Summation

Q143 MCQ

Select the correct option: Find the sum $\sum_{k=1}^n k(k^2 - 1)$:

Choose the correct option.

A $\frac{n(n+1)(n-1)(n+2)}{4}$

B $\frac{n(n+1)(n+1)(n+2)}{4}$

C $\frac{n(n-1)(n+2)}{4}$

D $\frac{n(n+1)(n-2)(n+3)}{4}$

Q144 MCQ

Select the correct option: Find the sum $\sum_{k=1}^n k(k^2 + 2)$:

Choose the correct option.

A $\frac{n(n+1)(n^2+n+4)}{4}$

B $\frac{n(n+1)(n^2+n-4)}{4}$

C $\frac{n(n+1)(n^2+2n+4)}{4}$

D $\frac{n(n+1)(n^2+n+2)}{4}$

Classified Practice

Summation

Q145 MCQ

Select the correct option: Find the sum $\sum_{k=1}^n k(k^2 + 3)$:

Choose the correct option.

- A $\frac{n(n+1)(n^2+n+6)}{4}$
- B $\frac{n(n+1)(n^2+n-6)}{4}$
- C $\frac{n(n+1)(n^2+3n+6)}{4}$
- D $\frac{n(n+1)(n^2+n+3)}{4}$

Q146 MCQ

Select the correct option: Find the sum $\sum_{k=1}^n k(k^2 - 2)$:

Choose the correct option.

A $\frac{n(n+1)(n^2+n-4)}{4}$

B $\frac{n(n+1)(n^2+n+4)}{4}$

C $\frac{n(n+1)(n^2-2n-4)}{4}$

D $\frac{n(n+1)(n^2+n-2)}{4}$

Classified Practice

Summation

Q147 MCQ

Select the correct option: Find the sum $\sum_{k=1}^n k(k^2 + 4)$:

Choose the correct option.

- A $\frac{n(n+1)(n^2+n+8)}{4}$
- B $\frac{n(n+1)(n^2+n-8)}{4}$
- C $\frac{n(n+1)(n^2+4n+8)}{4}$
- D $\frac{n(n+1)(n^2+n+4)}{4}$

Q148 MCQ

Select the correct option: Find the sum $\sum_{k=1}^n k(k^2 - 3)$:

Choose the correct option.

A $\frac{n(n+1)(n^2+n-6)}{4}$

B $\frac{n(n+1)(n^2+n+6)}{4}$

C $\frac{n(n+1)(n^2-3n-6)}{4}$

D $\frac{n(n+1)(n^2+n-3)}{4}$

Classified Practice

Summation

Q149 MCQ

Select the correct option: Find the sum $\sum_{k=1}^n k^2(k+5)$:

Choose the correct option.

A $\frac{n(n+1)(3n^2+23n+10)}{12}$

B $\frac{n(n+1)(3n^2+23n-10)}{12}$

C $\frac{n(n+1)(3n^2+21n+10)}{12}$

D $\frac{n(n+1)(3n^2+10n+23)}{12}$

Q150 MCQ

Select the correct option: Find the sum $\sum_{k=1}^n k^2(k-5)$:

Choose the correct option.

A $\frac{n(n+1)(3n^2-17n-10)}{12}$

B $\frac{n(n+1)(3n^2-17n+10)}{12}$

C $\frac{n(n+1)(3n^2-15n-10)}{12}$

D $\frac{n(n+1)(3n^2-10n-17)}{12}$

Classified Practice

Summation

Q151 MCQ

Select the correct option: Find the sum $\sum_{k=1}^n k(k^2 + 5)$:

Choose the correct option.

A $\frac{n(n+1)(n^2+n+10)}{4}$

B $\frac{n(n+1)(n^2+n-10)}{4}$

C $\frac{n(n+1)(n^2+5n+10)}{4}$

D $\frac{n(n+1)(n^2+n+5)}{4}$

Q152 MCQ

Select the correct option: Find the sum $\sum_{k=1}^n k(k^2 - 5)$:

Choose the correct option.

A $\frac{n(n+1)(n^2+n-10)}{4}$

B $\frac{n(n+1)(n^2+n+10)}{4}$

C $\frac{n(n+1)(n^2-5n-10)}{4}$

D $\frac{n(n+1)(n^2+n-5)}{4}$

Classified Practice

Summation

Q153 MCQ

Select the correct option: Find the sum $\sum_{k=1}^n k^2(k+6)$:

Choose the correct option.

A $\frac{n(n+1)(3n^2+27n+12)}{12}$

B $\frac{n(n+1)(3n^2+27n-12)}{12}$

C $\frac{n(n+1)(3n^2+25n+12)}{12}$

D $\frac{n(n+1)(3n^2+12n+27)}{12}$

Q154 MCQ

Select the correct option: Find the sum $\sum_{k=1}^n k(k^2 + 6)$:

Choose the correct option.

- A $\frac{n(n+1)(n^2+n+12)}{4}$
- B $\frac{n(n+1)(n^2+n-12)}{4}$
- C $\frac{n(n+1)(n^2+6n+12)}{4}$
- D $\frac{n(n+1)(n^2+n+6)}{4}$

Classified Practice

Summation

Q155 MCQ

Select the correct option: Find the sum $\sum_{k=1}^n k^2(k-6)$:

Choose the correct option.

A $\frac{n(n+1)(3n^2-25n-12)}{12}$

B $\frac{n(n+1)(3n^2-25n+12)}{12}$

C $\frac{n(n+1)(3n^2-23n-12)}{12}$

D $\frac{n(n+1)(3n^2-12n-25)}{12}$

Q156 MCQ

Select the correct option: Find the sum $\sum_{k=1}^n k(k^2 - 6)$:

Choose the correct option.

A $\frac{n(n+1)(n^2+n-12)}{4}$

B $\frac{n(n+1)(n^2+n+12)}{4}$

C $\frac{n(n+1)(n^2-6n-12)}{4}$

D $\frac{n(n+1)(n^2+n-6)}{4}$

Classified Practice

Summation

Q157 MCQ

Select the correct option: Find the sum $\sum_{k=1}^n k^2(k+7)$:

Choose the correct option.

A $\frac{n(n+1)(3n^2+31n+14)}{12}$

B $\frac{n(n+1)(3n^2+31n-14)}{12}$

C $\frac{n(n+1)(3n^2+29n+14)}{12}$

D $\frac{n(n+1)(3n^2+14n+31)}{12}$

Q158 MCQ

Select the correct option: Find the sum $\sum_{k=1}^n k^2(k+8)$:

Choose the correct option.

A $\frac{n(n+1)(3n^2+35n+16)}{12}$

B $\frac{n(n+1)(3n^2+35n-16)}{12}$

C $\frac{n(n+1)(3n^2+33n+16)}{12}$

D $\frac{n(n+1)(3n^2+16n+35)}{12}$

Classified Practice

Summation

Q159 Written

Find the sum $\sum_{k=1}^n 6 \times 3^{k-1}$.

Q160 Written

Find the sum $\sum_{k=1}^n 4 \times 2^{k-1}$.

Classified Practice

Summation

Q161 Written

Find the sum $\sum_{k=1}^n 2^k$.

Q162 Written

Find the sum $\sum_{k=1}^n 2 \times (-3)^{k-1}$.

Classified Practice

Summation

Q163 Written

Find the sum $\sum_{k=1}^n 3 \times (-2)^{k-1}$.

Q164 Written

Find the sum $\sum_{k=1}^n (-5)^k$.

Classified Practice

Summation

Q165 Written

Find the sum $\sum_{k=1}^n 2 \times (-1)^{k-1}$.

Q166 Written

Find the sum $\sum_{k=1}^n 8 \times (-3)^{k-1}$.

Classified Practice

Summation

Q167 Written

Find the sum $\sum_{k=1}^n (-2)^{k-1}$.

Q168 Written

Find the sum $\sum_{k=1}^n (-3)^k$.

Classified Practice

Summation

Q169 Written

Find the sum $\sum_{k=1}^n 4^k$.

Q170 Written

Find the sum $\sum_{k=1}^n 4^{k-1}$.

Classified Practice

Summation

Q171 Written

Find the sum $\sum_{k=1}^n 3^{k-1}$.

Q172 Written

Find the sum $\sum_{k=1}^4 (-2)^{k-1}$.

Classified Practice

Summation

Q173 Written

Find the sum $\sum_{k=1}^3 3^k$.

Q174 Written

Find the sum $\sum_{k=1}^5 2^k$.

Classified Practice

Summation

Q175 Written

Find the sum $\sum_{k=1}^4 (-3)^k$.

Classified Practice

Summation

Q176 **Challenge**

Find the finite sum $\sum_{r=5}^9 (2r + 7)$.

Classified Practice

Summation

Q177 Written

Find $\frac{1}{2 \times 3} + \frac{1}{3 \times 4} + \frac{1}{4 \times 5} + \cdots + \frac{1}{(n+1)(n+2)}$.

Q178 Written

Find $\frac{1}{1 \times 2} + \frac{1}{2 \times 3} + \frac{1}{3 \times 4} + \cdots + \frac{1}{n(n+1)}$.

Classified Practice

Summation

Q179 Written

Find $\frac{1}{3 \times 4} + \frac{1}{4 \times 5} + \frac{1}{5 \times 6} + \cdots + \frac{1}{(n+2)(n+3)}$.

Q180 MCQ

Select the correct option: Find the sum $\frac{1}{1 \times 2} + \cdots + \frac{1}{n(n+1)}$:

Choose the correct option.

- A $\frac{n}{n+1}$
B $\frac{1}{n+1}$
C $\frac{2(n+1)}{n}$
D $\frac{n+1}{n}$

Classified Practice

Summation

Q181 MCQ

Select the correct option: Find the sum $\frac{1}{2 \times 3} + \cdots + \frac{1}{(n+1)(n+2)}$:

Choose the correct option.

A $\frac{n}{2(n+2)}$

B $\frac{1}{2(n+2)}$

C $\frac{n}{n+2}$

D $\frac{n+1}{2(n+2)}$

Q182 MCQ

Select the correct option: Find the sum $\frac{1}{3 \times 4} + \cdots + \frac{1}{(n+2)(n+3)}$:

Choose the correct option.

A $\frac{n}{3(n+3)}$

B $\frac{n}{n+3}$

C $\frac{1}{3(n+3)}$

D $\frac{n+1}{3(n+3)}$

Classified Practice

Summation

Q183 MCQ

Select the correct option: Find the sum $\frac{1}{4 \times 5} + \cdots + \frac{1}{(n+3)(n+4)}$:

Choose the correct option.

A $\frac{n}{4(n+4)}$

B $\frac{n}{3(n+4)}$

C $\frac{n+1}{4(n+4)}$

D $\frac{1}{4(n+4)}$

Q184 MCQ

Select the correct option: Find the sum $\frac{1}{6 \times 7} + \cdots + \frac{1}{(n+5)(n+6)}$:

Choose the correct option.

A $\frac{n}{6(n+6)}$

B $\frac{n}{5(n+6)}$

C $\frac{n+1}{6(n+6)}$

D $\frac{1}{6(n+6)}$

Classified Practice

Summation

Q185 MCQ

Select the correct option: Find the sum $\frac{1}{7 \times 8} + \cdots + \frac{1}{(n+6)(n+7)}$:

Choose the correct option.

A $\frac{n}{7(n+7)}$

B $\frac{n}{6(n+7)}$

C $\frac{n+1}{7(n+7)}$

D $\frac{1}{7(n+7)}$

Q186 MCQ

Select the correct option: Find the sum $\frac{1}{9 \times 10} + \cdots + \frac{1}{(n+8)(n+9)}$:

Choose the correct option.

A $\frac{n}{9(n+9)}$

B $\frac{n}{8(n+9)}$

C $\frac{n+1}{9(n+9)}$

D $\frac{1}{9(n+9)}$

Classified Practice

Summation

Q187 MCQ

Select the correct option: Find the sum $\frac{1}{10 \times 11} + \cdots + \frac{1}{(n+9)(n+10)}$:

Choose the correct option.

A $\frac{n}{10(n+10)}$

B $\frac{n}{9(n+10)}$

C $\frac{n+1}{10(n+10)}$

D $\frac{1}{10(n+10)}$

Q188 MCQ

Select the correct option: Find the sum $\frac{1}{11 \times 12} + \dots + \frac{1}{(n+10)(n+11)}$:

Choose the correct option.

A $\frac{n}{11(n+11)}$

B $\frac{n}{10(n+11)}$

C $\frac{n+1}{11(n+11)}$

D $\frac{1}{11(n+11)}$

Classified Practice

Summation

Q189 Written

Find the following sum: $\frac{1}{2 \times 5} + \frac{1}{5 \times 8} + \frac{1}{8 \times 11} + \cdots + \frac{1}{(3n-1)(3n+2)}$.

Q190 Written

Find $\frac{1}{1 \times 3} + \frac{1}{3 \times 5} + \frac{1}{5 \times 7} + \cdots + \frac{1}{(2n-1)(2n+1)}$.

Classified Practice

Summation

Q191 Written

Find $\frac{1}{1 \times 3} + \frac{1}{2 \times 4} + \frac{1}{3 \times 5} + \cdots + \frac{1}{n(n+2)}$.

Q192 Written

Find $\frac{1}{1 \times 4} + \frac{1}{4 \times 7} + \frac{1}{7 \times 10} + \cdots + \frac{1}{(3n-2)(3n+1)}$.

Classified Practice

Summation

Q193 Written

Find $\frac{1}{1 \times 7} + \frac{1}{4 \times 10} + \frac{1}{7 \times 13} + \cdots + \frac{1}{(3n-2)(3n+4)}$.

Q194 MCQ

Select the correct option: Find the sum $\frac{1}{1 \times 3} + \cdots + \frac{1}{(2n-1)(2n+1)}$:

Choose the correct option.

- A $\frac{n}{2n+1}$
B $\frac{n}{2(2n+1)}$
C $\frac{1}{2n+1}$
D $\frac{n+1}{2n+1}$

Classified Practice

Summation

Q195 MCQ

Select the correct option: Find the sum $\frac{1}{1 \times 4} + \cdots + \frac{1}{(3n-2)(3n+1)}$:

Choose the correct option.

- A $\frac{n}{3n+1}$
- B $\frac{n}{3(3n+1)}$
- C $\frac{1}{3n+1}$
- D $\frac{n+1}{3n+1}$

Q196 MCQ

Select the correct option: Find the sum $\frac{1}{1 \times 7} + \cdots + \frac{1}{(3n-2)(3n+4)}$:

Choose the correct option.

A $\frac{1}{6} \left(1 + \frac{1}{4} - \frac{1}{3n+1} - \frac{1}{3n+4} \right)$

B $\frac{1}{6} \left(1 - \frac{1}{3n+4} \right)$

C $\frac{1}{2(3n+4)}$

D $\frac{n}{2(3n+4)}$

Classified Practice

Summation

Q197 MCQ

Select the correct option: Find the sum $\frac{1}{2 \times 6} + \cdots + \frac{1}{(4n-2)(4n+2)}$:

Choose the correct option.

A $\frac{n}{2(4n+2)}$

B $\frac{n}{4(4n+2)}$

C $\frac{1}{4n+2}$

D $\frac{n+1}{2(4n+2)}$

Q198 MCQ

Select the correct option: Find the sum $\frac{1}{3 \times 8} + \cdots + \frac{1}{(5n-2)(5n+3)}$:

Choose the correct option.

A $\frac{n}{3(5n+3)}$

B $\frac{n}{5(5n+3)}$

C $\frac{n+1}{3(5n+3)}$

D $\frac{1}{5n+3}$

Classified Practice

Summation

Q199 MCQ

Select the correct option: Find the sum $\frac{1}{1 \times 5} + \cdots + \frac{1}{(4n-3)(4n+1)}$:

Choose the correct option.

- A $\frac{n}{4n+1}$
- B $\frac{n}{4(4n+1)}$
- C $\frac{1}{4n+1}$
- D $\frac{n+1}{4n+1}$

Q200 MCQ

Select the correct option: Find the sum $\frac{1}{2 \times 7} + \cdots + \frac{1}{(5n-3)(5n+2)}$:

Choose the correct option.

A $\frac{n}{2(5n+2)}$

B $\frac{n}{5(5n+2)}$

C $\frac{n+1}{2(5n+2)}$

D $\frac{1}{5n+2}$

Classified Practice

Summation

Q201 MCQ

Select the correct option: Find the sum $\frac{1}{4 \times 10} + \cdots + \frac{1}{(6n-2)(6n+4)}$:

Choose the correct option.

- A $\frac{n}{4(6n+4)}$
B $\frac{n}{6(6n+4)}$
C $\frac{n+1}{4(6n+4)}$
D $\frac{1}{6n+4}$

Q202 Written

Find $\frac{1}{4 \times 6} + \frac{1}{5 \times 7} + \frac{1}{6 \times 8} + \cdots + \frac{1}{(n+3)(n+5)}$.

Classified Practice

Summation

Q203 Written

Find $\frac{1}{2 \times 4} + \frac{1}{4 \times 6} + \frac{1}{6 \times 8} + \cdots + \frac{1}{2n(2n+2)}$.

Q204 Written

Find the following sum: $\sum_{k=1}^n \frac{1}{\sqrt{k+2} + \sqrt{k+1}}.$

Classified Practice

Summation

Q205 Written

Find $\sum_{k=1}^n \frac{1}{\sqrt{k+1} + \sqrt{k}}.$

Q206 Written

Find $\sum_{k=1}^n \frac{1}{\sqrt{k} + \sqrt{k-1}}.$

Classified Practice

Summation

Q207 Written

Find $\frac{1}{\sqrt{1} + \sqrt{2}} + \frac{1}{\sqrt{2} + \sqrt{3}} + \cdots + \frac{1}{\sqrt{n} + \sqrt{n+1}}$.

Q208 Written

Find $\sum_{k=1}^n \frac{1}{\sqrt{2k-1} + \sqrt{2k+1}}.$

Classified Practice

Summation

Q209 Written

Find $\frac{1}{\sqrt{2} + \sqrt{3}} + \frac{1}{\sqrt{3} + \sqrt{4}} + \cdots + \frac{1}{\sqrt{n+1} + \sqrt{n+2}}.$

Classified Practice

Summation

Q210 Challenge

Find the sum, showing your steps: $\sum_{r=1}^{30} \frac{1}{r(r+2)}.$

Classified Practice

Summation

Q211 Written

Find the following sum $S = 3 \times 1 + 5 \times 3 + 7 \times 3^2 + 9 \times 3^3 + \cdots + (2n + 1) \times 3^{n-1}$.

Q212 Written

Find the sum $\sum_{k=1}^n (2k + 1) \times (3)^{k-1}$.

Classified Practice

Summation

Q213 Written

Find the sum $\sum_{k=1}^4 (2k + 1) \times (3)^{k-1}$.

Q214 Written

Find the sum $\sum_{k=1}^n (2k - 1) \times (4)^{k-1}$.

Classified Practice

Summation

Q215 Written

Find the sum $\sum_{k=1}^6 (2k - 1) \times (4)^{k-1}$.

Q216 MCQ

Select the correct option: Find the sum $\sum_{k=1}^n (2k + 1) \times (3)^{k-1}$:

Choose the correct option.

A $3^n n$

B $-(3^n n)$

C $(3^n n) + 1$

D $(3^n n) - 1$

Classified Practice

Summation

Q217 MCQ

Select the correct option: Find the sum $\sum_{k=1}^n (2k - 1) \times (4)^{k-1}$:

Choose the correct option.

A $\frac{6 \cdot 2^{2n} n - 5 \cdot 2^{2n+5}}{9}$

B $-\left(\frac{6 \cdot 2^{2n} n - 5 \cdot 2^{2n+5}}{9}\right)$

C $\left(\frac{6 \cdot 2^{2n} n - 5 \cdot 2^{2n+5}}{9}\right) + 1$

D $\left(\frac{6 \cdot 2^{2n} n - 5 \cdot 2^{2n+5}}{9}\right) - 1$

Q218 MCQ

Select the correct option: Find the sum $\sum_{k=1}^n (k + 4) \times (2)^{k-1}$:

Choose the correct option.

A $2^n n + 3 \cdot 2^n - 3$

B $-(2^n n + 3 \cdot 2^n - 3)$

C $(2^n n + 3 \cdot 2^n - 3) + 1$

D $(2^n n + 3 \cdot 2^n - 3) - 1$

Classified Practice

Summation

Q219 Written

Find the following sum $S = 2 \times 1 + 3 \times 3 + 4 \times 3^2 + 5 \times 3^3 + \cdots + (n + 1) \times 3^{n-1}$.

Q220 MCQ

Select the correct option: Find the sum $\sum_{k=1}^n (k+1) \times (3)^{k-1}$:

Choose the correct option.

A $\frac{2 \cdot 3^n n + 3^n - 1}{4}$

B $-\left(\frac{2 \cdot 3^n n + 3^n - 1}{4}\right)$

C $\left(\frac{2 \cdot 3^n n + 3^n - 1}{4}\right) + 1$

D $\left(\frac{2 \cdot 3^n n + 3^n - 1}{4}\right) - 1$

Classified Practice

Summation

Q221 MCQ

Select the correct option: Find the sum $\sum_{k=1}^n (k-2) \times (3)^{k-1}$:

Choose the correct option.

A $\frac{2 \cdot 3^n n - 5 \cdot 3^n + 5}{4}$

B $-\left(\frac{2 \cdot 3^n n - 5 \cdot 3^n + 5}{4}\right)$

C $\left(\frac{2 \cdot 3^n n - 5 \cdot 3^n + 5}{4}\right) + 1$

D $\left(\frac{2 \cdot 3^n n - 5 \cdot 3^n + 5}{4}\right) - 1$

Q222 MCQ

Select the correct option: Find the sum $\sum_{k=1}^n (2k - 3) \times (3)^{k-1}$:

Choose the correct option.

A $3^n n - 2 \cdot 3^n + 2$

B $-(3^n n - 2 \cdot 3^n + 2)$

C $(3^n n - 2 \cdot 3^n + 2) + 1$

D $(3^n n - 2 \cdot 3^n + 2) - 1$

Classified Practice

Summation

Q223 MCQ

Select the correct option: Find the sum $\sum_{k=1}^n (2k + 1) \times (-2)^{k-1}$:

Choose the correct option.

A $-\frac{6(-2)^n n + 5(-2)^n - 5}{9}$

B $-\left(-\frac{6(-2)^n n + 5(-2)^n - 5}{9}\right)$

C $\left(-\frac{6(-2)^n n + 5(-2)^n - 5}{9}\right) + 1$

D $\left(-\frac{6(-2)^n n + 5(-2)^n - 5}{9}\right) - 1$

Classified Practice

Summation

Q224 MCQ

Select the correct option: Find the sum $\sum_{k=1}^n (k+2) \times (-3)^{k-1}$:

Choose the correct option.

- A $-\frac{4(-3)^n n + 9(-3)^n - 9}{16}$
- B $-\left(-\frac{4(-3)^n n + 9(-3)^n - 9}{16}\right)$
- C $\left(-\frac{4(-3)^n n + 9(-3)^n - 9}{16}\right) + 1$
- D $\left(-\frac{4(-3)^n n + 9(-3)^n - 9}{16}\right) - 1$

Classified Practice

Summation

Q225 Written

Find the following sum $S = 1 \times 1 + 2 \times 2 + 3 \times 2^2 + 4 \times 2^3 + \cdots + n \times 2^{n-1}$.

Q226 Written**Find the following sum** $S = 1 \times 3 + 3 \times 3^2 + 5 \times 3^3 + 7 \times 3^4 + \cdots + (2n - 1) \times 3^n$.

Classified Practice

Summation

Q227 MCQ

Select the correct option: Find the sum $\sum_{k=1}^n (3k) \times (3)^{k-1}$:

Choose the correct option.

A $\frac{3(2 \cdot 3^n n - 3^n + 1)}{4}$

B $-\left(\frac{3(2 \cdot 3^n n - 3^n + 1)}{4}\right)$

C $\left(\frac{3(2 \cdot 3^n n - 3^n + 1)}{4}\right) + 1$

D $\left(\frac{3(2 \cdot 3^n n - 3^n + 1)}{4}\right) - 1$

Classified Practice

Summation

Q228 **Challenge**

Find the sum of the partial sums $5 + (5 + 9) + (5 + 9 + 13) + \cdots + [5 + 9 + \cdots + (4k + 1)]$.

Classified Practice

Summation

Q229 Written

For the sequence $3, 3 + 5, 3 + 5 + 7, 3 + 5 + 7 + 9, \dots$, find the k -th term.

Q230 Written

For the sequence $1, 1 + 3, 1 + 3 + 5, 1 + 3 + 5 + 7, \dots$, find the k -th term.

Classified Practice

Summation

Q231 **Written**

For the sequence $2, 2 + 4, 2 + 4 + 6, 2 + 4 + 6 + 8, \dots$, find the k -th term.

Q232 Written

For the sequence $1, 1 + 5, 1 + 5 + 9, 1 + 5 + 9 + 13, \dots$, find the k -th term.

Classified Practice

Summation

Q233 MCQ

Select the correct option: For the cumulative sequence $2, 2 + 4, 2 + 4 + 6, 2 + 4 + 6 + 8, \dots$, what is a_k :

Choose the correct option.

A $k(k + 1)$

B $k(k + 3)$

C $2k^2$

D $k(k + 2)$

Q234 MCQ

Select the correct option: For the cumulative sequence $3, 3 + 6, 3 + 6 + 9, 3 + 6 + 9 + 12, \dots$, what is a_k :

Choose the correct option.

A $\frac{3k(k+1)}{2}$

B $\frac{3k(k+3)}{2}$

C $3k^2$

D $\frac{3k(k+2)}{2}$

Classified Practice

Summation

Q235 MCQ

Select the correct option: For the cumulative sequence $4, 4 + 6, 4 + 6 + 8, 4 + 6 + 8 + 10, \dots$, what is a_k :

Choose the correct option.

A $k(k + 3)$

B $k(k + 5)$

C $2k(k + 1)$

D $k(k + 4)$

Q236 MCQ

Select the correct option: For the cumulative sequence $2, 2 + 6, 2 + 6 + 10, 2 + 6 + 10 + 14, \dots$, what is a_k :

Choose the correct option.

A $2k^2$

B $2k(k + 2)$

C $2k(2k - 1)$

D $2k(k + 1)$

Classified Practice

Summation

Q237 MCQ

Select the correct option: For the cumulative sequence $5, 5 + 8, 5 + 8 + 11, 5 + 8 + 11 + 14, \dots$, what is a_k :

Choose the correct option.

A $\frac{k(3k+7)}{2}$

B $\frac{k(3k+13)}{2}$

C $k(3k+2)$

D $\frac{k(3k+10)}{2}$

Q238 MCQ

Select the correct option: For the cumulative sequence $3, 3 + 8, 3 + 8 + 13, 3 + 8 + 13 + 18, \dots$, what is a_k :

Choose the correct option.

A $\frac{k(5k+1)}{2}$

B $\frac{k(5k+11)}{2}$

C $k(5k - 2)$

D $\frac{k(5k+6)}{2}$

Classified Practice

Summation

Q239 MCQ

Select the correct option: For the cumulative sequence $6, 6 + 8, 6 + 8 + 10, 6 + 8 + 10 + 12, \dots$, what is a_k :

Choose the correct option.

A $k(k + 5)$

B $k(k + 7)$

C $2k(k + 2)$

D $k(k + 6)$

Q240 MCQ

Select the correct option: For the cumulative sequence $4, 4 + 8, 4 + 8 + 12, 4 + 8 + 12 + 16, \dots$, what is a_k :

Choose the correct option.

A $2k(k + 1)$

B $2k(k + 3)$

C $4k^2$

D $2k(k + 2)$

Classified Practice

Summation

Q241 Written

For the sequence $3, 3 + 5, 3 + 5 + 7, 3 + 5 + 7 + 9, \dots$, find the sum from the first term to the n -th term.

Q242 Written

For the sequence $1, 1 + 3, 1 + 3 + 5, 1 + 3 + 5 + 7, \dots$, find the sum from the first term to the n -th term.

Classified Practice

Summation

Q243 Written

For the sequence $1, 1 + 5, 1 + 5 + 9, 1 + 5 + 9 + 13, \dots$, find the sum from the first term to the n -th term.

Q244 Written

For the sequence $2, 2 + 4, 2 + 4 + 6, 2 + 4 + 6 + 8, \dots$, find the sum from the first term to the n -th term.

Classified Practice

Summation

Q245 MCQ

Select the correct option: For the cumulative sequence $2, 2 + 5, 2 + 5 + 8, 2 + 5 + 8 + 11, \dots$, find a_5 :

Choose the correct option.

A 40

B 46

C 34

D 52

Classified Practice

Summation

Q246 **Challenge**

The sum of the first n terms of $\{a_n\}$ is $S_n = 2n^2 + 5n$. Find the general term.

Classified Practice

Summation

Q247 **Written**

Find the general term of the sequence $\{a_n\}$ where the sum S_n from the first term to the n -th term is given by $S_n = n(n - 3)$.

Classified Practice

Summation

Q248 **Written**

Find the general term of the sequence $\{a_n\}$ where the sum S_n from the first term to the n -th term is given by $S_n = n(2n + 5)$.

Classified Practice

Summation

Q249 **Written**

Find the general term of the sequence $\{a_n\}$ where the sum S_n from the first term to the n -th term is given by $S_n = n^2$.

Classified Practice

Summation

Q250 Written

Find the general term of the sequence $\{a_n\}$ where the sum S_n from the first term to the n -th term is given by $S_n = n(n - 4)$.

Classified Practice

Summation

Q251 MCQ

Select the correct option: If $S_n = n(n + 2)$, the general term a_n is:

Choose the correct option.

A $2n + 3$

B $2n - 1$

C $2(n + 2)$

D $a_n = 2n + 1$

Classified Practice

Summation

Q252 MCQ

Select the correct option: If $S_n = 2n(n + 2)$, the general term a_n is:

Choose the correct option.

A $4n + 1$

B $2(2n + 3)$

C $a_n = 2(2n + 1)$

D $4n + 5$

Classified Practice

Summation

Q253 MCQ

Select the correct option: If $S_n = 3n(n - 1)$, the general term a_n is:

Choose the correct option.

A $6n$

B $a_n = 6(n - 1)$

C $6n - 5$

D $2(3n - 4)$

Classified Practice

Summation

Q254 MCQ

Select the correct option: If $S_n = 2n(2n - 1)$, the general term a_n is:

Choose the correct option.

A $a_n = 2(4n - 3)$

B $4(2n - 1)$

C $8n - 7$

D $2(4n + 1)$

Classified Practice

Summation

Q255 MCQ

Select the correct option: If $S_n = 5n(n + 1)$, the general term a_n is:

Choose the correct option.

A $10n + 3$

B $2(5n - 1)$

C $10(n + 1)$

D $a_n = 10n$

Classified Practice

Summation

Q256 MCQ

Select the correct option: If $S_n = n(2n + 3)$, the general term a_n is:

Choose the correct option.

A $4n - 1$

B $4n + 5$

C $a_n = 4n + 1$

D $4n + 3$

Classified Practice

Summation

Q257 MCQ

Select the correct option: If $S_n = n(3n + 1)$, the general term a_n is:

Choose the correct option.

A $3(2n - 1)$

B $2(3n + 2)$

C $a_n = 2(3n - 1)$

D $6n + 1$

Classified Practice

Summation

Q258 MCQ

Select the correct option: If $S_n = n(n^2 + 2n + 3)$, the general term a_n is:

Choose the correct option.

A $3n^2 + n + 1$

B $3n^2 + 7n + 6$

C $a_n = 3n^2 + n + 2$

D $3n^2 + n + 3$

Classified Practice

Summation

Q259 MCQ

Select the correct option: If $S_n = n(n+1)(2n-1)$, the general term a_n is:

Choose the correct option.

A $2(n+1)(3n+1)$

B $a_n = 2n(3n-2)$

C $2(3n^2 - 2n + 1)$

D $2(n-1)(3n+1)$

Classified Practice

Summation

Q260 MCQ

Select the correct option: If $S_n = n(3n^2 - 2n + 4)$, the general term a_n is:

Choose the correct option.

A $a_n = 9n^2 - 13n + 9$

B $9n^2 - 13n + 12$

C $9n^2 - 13n + 8$

D $9n^2 + 5n + 5$

Classified Practice

Summation

Q261 MCQ

Select the correct option: If $S_n = n(n^2 - n + 2)$, the general term a_n is:

Choose the correct option.

A $3n^2 - 5n + 5$

B $(n - 1)(3n - 2)$

C $3n^2 + n + 2$

D $a_n = 3n^2 - 5n + 4$

Classified Practice

Summation

Q262 MCQ

Select the correct option: If $S_n = n(n+1)(2n+1)$, the general term a_n is:

Choose the correct option.

A $2(3n^2 + 1)$

B $6n^2 - 1$

C $6(n+1)^2$

D $a_n = 6n^2$

Classified Practice

Summation

Q263 MCQ

Select the correct option: If $S_n = n(n+1)(n+2)$, the general term a_n is:

Choose the correct option.

A $3(n+1)(n+2)$

B $a_n = 3n(n+1)$

C $3n^2 + 3n + 1$

D $3n^2 + 3n - 2$

Classified Practice

Summation

Q264 **Written**

Find the general term of the sequence $\{a_n\}$ where the sum S_n from the first term to the n -th term is given by $S_n = n^3 + 2n + 6$.

Classified Practice

Summation

Q265 Written

Find the general term of the sequence $\{a_n\}$ where the sum S_n from the first term to the n -th term is given by $S_n = n^2 + 2n - 1$.

Classified Practice

Summation

Q266 Written

Find the general term of the sequence $\{a_n\}$ where the sum S_n from the first term to the n -th term is given by $S_n = (n-1)(n^2 + n + 1)$.

Classified Practice

Summation

Q267 **Written**

Find the general term of the sequence $\{a_n\}$ where the sum S_n from the first term to the n -th term is given by $S_n = 5^n - 1$.

Classified Practice

Summation

Q268 **Written**

Find the general term of the sequence $\{a_n\}$ where the sum S_n from the first term to the n -th term is given by $S_n = 3^n - 1$.

Classified Practice

Summation

Q269 Written

Find the general term of the sequence $\{a_n\}$ where the sum S_n from the first term to the n -th term is given by $S_n = 2^n - 1$.

Classified Practice

Summation

Q270 **Written**

Find the general term of the sequence $\{a_n\}$ where the sum S_n from the first term to the n -th term is given by $S_n = 7^n - 1$.

Classified Practice

Summation

Q271 Written

Find the general term of the sequence $\{a_n\}$ where the sum S_n from the first term to the n -th term is given by $S_n = 5^n - 2$.

Classified Practice

Summation

Q272 Written

Find the general term of the sequence $\{a_n\}$ where the sum S_n from the first term to the n -th term is given by $S_n = 4 \cdot 2^n - 3$.

Classified Practice

Summation

Q273 MCQ

Select the correct option: If $S_n = n(n + 4)$, find a_5 :**Choose the correct option.**

A -13

B 13

C 14

D 12

Classified Practice

Summation

Q274 **Challenge**

Find the general term of the sequence $\{a_n\}$ where S_n is given by $S_n = 3 \cdot 3^n - 2n + 1$.

Classified Practice

Summation

Quiz MCQ 1 [Concept Quiz](#) · [MCQ](#)

Select the correct option: Expand $\sum_{k=1}^4 (3k - 1)$ into a sum of terms:

Choose the correct option.

- A $2 + 5 + 8 + 11$
- B $3 + 6 + 9 + 12$
- C $1 + 4 + 7 + 10$
- D $2 + 6 + 10 + 14$

Quiz MCQ 2 Concept Quiz · MCQ

Select the correct option: Find the sum $\sum_{k=1}^n k(k^2 - 1)$:

Choose the correct option.

- A $\frac{n(n+1)(n-1)(n+2)}{4}$
- B $\frac{n(n+1)(n+1)(n+2)}{4}$
- C $\frac{n(n-1)(n+2)}{4}$
- D $\frac{n(n+1)(n-2)(n+3)}{4}$

Classified Practice

Summation

Quiz MCQ 3 Concept Quiz · MCQ

Select the correct option: Find the sum $\sum_{k=1}^n k \times 2^{k-1}$:

Choose the correct option.

A $(n-1) \times 2^n + 1$

B $-((n-1) \times 2^n + 1)$

C $((n-1) \times 2^n + 1) + 1$

D $((n-1) \times 2^n + 1) - 1$

Classified Practice

Summation

Quiz MCQ 4 **Concept Quiz · MCQ**

Select the correct option: If $S_n = 2^n - 3$, find a_6 :

Choose the correct option.

A 30

B -32

C 32

D 34

Classified Practice

Summation

Quiz WRITTEN 5 Concept Quiz · Written

Expand $\sum_{k=1}^5 (k^2 - 2)$ into a sum of terms.

Quiz WRITTEN 6 Concept Quiz · Written

Find the sum $\sum_{k=1}^n k^2(k+5)$.

Classified Practice

Summation

Quiz WRITTEN 7 Concept Quiz · Written**Find the sum** $\sum_{k=1}^n (4k - 1) \times 3^k$.

Classified Practice

Summation

Quiz WRITTEN 8 Concept Quiz · Written

Find the general term of the sequence $\{a_n\}$ when S_n is $2 \cdot 2^n - 2n + 1$.

Answer key

Use the question number shown on each page.

Q001 $2 + 4 + 6 + 8 + 10$

Q002 $3 + 5 + 7 + 9 + 11$

Q003 $3 + 6 + 9 + 12 + 15$

Q004 $2 + 6 + 10 + 14$

Q005 $-1 + 1 + 3 + 5 + 7$

Q006 A

Q007 A

Q008 A

Q009 A

Q010 A

Q011 A

Q012 A

Q013 A

Q014 $-2 + 1 + 6$

Q015 $5 + 14 + 29$

Q016 $1 + 4 + 9 + 16$

Q017 $5 + 11 + 21$

Q018 $-1 + 0 + 3 + 8$

Q019 A

Q020 A

Q021 A

Q022 A

Q023 A

Q024 A

Q025 A

Q026 A

Q027 $3 + 9 + 27 + 81$

Q028 $2 + 2 + 2$

Q029 $2 + 4 + 8 + 16$

Q030 $3 + 3 + 3 + 3 + 3 + 3$

Q031 $3 + 9 + 27$

Q032 $8 + 28 + 128$

Q033 $-1 + 1 + 5 + 13$

Q034 $5 + 5 + 5 + 5$

Q035 $3 + 3 + 3$

Q036 $4 + 4 + 4 + 4 + 4$

Q037 A

Q038 A

Q039 A

Q040 A

Q041 A

Q042 A

Q043 A

Q044 A

Q045 $\sum_{k=1}^n (4k - 1)$

Q046 $\sum_{k=1}^n (3k + 2)$

Q047 $\sum_{k=1}^n (2k + 1)$

Q048 $\sum_{k=1}^n (2k - 1)$

Q049 $\sum_{k=1}^n (7k - 1)$

Q050 $\sum_{k=1}^n (4k + 5)$

Q051 $\sum_{k=1}^n (3k - 2)$

Q052 A

Q053 A

Q054 A

Q055 A

Q056 A

Q057 A

Q058 A

Q059 A

Q060 $\sum_{k=6}^{15} k^2$

Q061 $\sum_{k=4}^{13} k^2$

Q062 $\sum_{k=1}^n (3^k - 2)$

Q063 $\sum_{k=1}^n 2^{k-1}$

Q064 $\sum_{k=7}^{21} k^2$

Q065 $\sum_{k=12}^{18} k^2$

Q066 $\sum_{k=10}^{20} k^2$

Q067 A

Q068 A

Q069 A

Q070 A

Q071 A

Q072 A

Q073 A

Q074 $\sum_{k=1}^{10} (3k - 1)$

Q075 $\sum_{k=1}^7 8k$

Q076 $\sum_{k=1}^{12} (6k + 1)$

Q077 $\sum_{k=1}^{24} (5k - 1)$

Q078 $\sum_{k=1}^n (3k - 2)$

Q079 $-3n$

Q080 $3n$

Q081 $6n$

Q082 $-4n$

Q083 n

Q084 A

Q085 A

Q086 A

Q087 A

Q088 A

Q089 A	Q118 $\frac{n(n+1)(2n-5)}{6}$	Q145 A	Q174 62
Q090 A	Q119 $\frac{2n(n+1)(n-1)}{3}$	Q146 A	Q175 60
Q091 A	Q120 $\frac{n(n+1)(n+5)}{3}$	Q147 A	Q176 105
Q092 A	Q121 $\frac{n(n+1)(2n+3)}{2}$	Q148 A	Q177 $\frac{n}{2(n+2)}$
Q093 A	Q122 $\frac{n(n+4)(n+5)}{3}$	Q149 A	Q178 $\frac{n}{n+1}$
Q094 A	Q123 16	Q150 A	Q179 $\frac{n}{3(n+3)}$
Q095 A	Q124 35	Q151 A	Q180 A
Q096 A	Q125 280	Q152 A	Q181 A
Q097 A	Q126 184	Q153 A	Q182 A
Q098 A	Q127 100	Q154 A	Q183 A
Q099 A	Q128 245	Q155 A	Q184 A
Q100 $\frac{n(5n+13)}{2}$	Q129 $\frac{n(n+1)(n+2)(3n+1)}{12}$	Q156 A	Q185 A
Q101 $n(n-2)$	Q130 $\frac{n(n+1)(3n^2-5n-4)}{12}$	Q157 A	Q186 A
Q102 $n(2n+1)$	Q131 $\frac{n(n+1)(3n^2+11n+4)}{12}$	Q158 A	Q187 A
Q103 $\frac{n(3n+7)}{2}$	Q132 $\frac{n(n+1)(n^2-3n-2)}{4}$	Q159 $3(3^n-1)$	Q188 A
Q104 $3n^2$	Q133 A	Q160 $4(2^n-1)$	Q189 $\frac{n}{2(3n+2)}$
Q105 A	Q134 A	Q161 $2(2^n-1)$	Q190 $\frac{n}{2n+1}$
Q106 A	Q135 A	Q162 $\frac{1-(-3)^n}{2}$	Q191 $\frac{1}{2}(1+\frac{1}{2}-\frac{1}{n+1}-\frac{1}{n+2})$
Q107 A	Q136 A	Q163 $1-(-2)^n$	Q192 $\frac{n}{3n+1}$
Q108 A	Q137 A	Q164 $\frac{5((-5)^n-1)}{6}$	Q193 $\frac{1}{6}(1+\frac{1}{4}-\frac{1}{3n+1}-\frac{1}{3n+4})$
Q109 $\frac{n(n-1)(2n-1)}{6}$	Q138 A	Q165 $1-(-1)^n$	Q194 A
Q110 A	Q139 A	Q166 $2(1-(-3)^n)$	Q195 A
Q111 A	Q140 A	Q167 $\frac{1-(-2)^n}{3}$	Q196 A
Q112 A	Q141 $\frac{n(n+1)(n-1)(n+2)}{4}$	Q168 $\frac{3((-3)^n-1)}{4}$	Q197 A
Q113 A	Q142 $\frac{n(n+1)(n^2+n+4)}{4}$	Q169 $\frac{4(4^n-1)}{3}$	Q198 A
Q114 A	Q143 A	Q170 $\frac{4^n-1}{3}$	Q199 A
Q115 $\frac{n(n+1)(10n-1)}{6}$	Q144 A	Q171 $\frac{3^n-1}{2}$	Q200 A
Q116 $\frac{n(n-1)(2n+1)}{2}$		Q172 -5	Q201 A
Q117 $\frac{n(n+1)(4n+5)}{6}$		Q173 39	Q202 $\frac{1}{2}(\frac{1}{4}+\frac{1}{5}-\frac{1}{n+4}-\frac{1}{n+5})$

$$\text{Q203 } \frac{n}{4(n+1)}$$

$$\text{Q204 } \sqrt{n+2} - \sqrt{2}$$

$$\text{Q205 } \sqrt{n+1} - 1$$

$$\text{Q206 } \sqrt{n}$$

$$\text{Q207 } \sqrt{n+1} - 1$$

$$\text{Q208 } \frac{\sqrt{2n+1} - 1}{2}$$

$$\text{Q209 } \sqrt{n+2} - \sqrt{2}$$

$$\text{Q210 } \frac{1425}{1984}$$

$$\text{Q211 } n \times 3^n$$

$$\text{Q212 } 3^n n$$

$$\text{Q213 } 324$$

$$\text{Q214 } \frac{6 \cdot 2^{2n} n - 5 \cdot 2^{2n} + 5}{9}$$

$$\text{Q215 } 14109$$

$$\text{Q216 } A$$

$$\text{Q217 } A$$

$$\text{Q218 } A$$

$$\text{Q219 } \frac{(2n+1) \times 3^n - 1}{4}$$

$$\text{Q220 } A$$

$$\text{Q221 } A$$

$$\text{Q222 } A$$

$$\text{Q223 } A$$

$$\text{Q224 } A$$

$$\text{Q225 } (n-1) \times 2^n + 1$$

$$\text{Q226 } (n-1) \times 3^{n+1} + 3$$

$$\text{Q227 } A$$

$$\text{Q228 } \frac{k(k+1)(4k+11)}{6}$$

$$\text{Q229 } a_k = k(k+2)$$

$$\text{Q230 } a_k = k^2$$

$$\text{Q231 } a_k = k(k+1)$$

$$\text{Q232 } a_k = k(2k-1)$$

$$\text{Q233 } A$$

$$\text{Q234 } A$$

$$\text{Q235 } A$$

$$\text{Q236 } A$$

$$\text{Q237 } A$$

$$\text{Q238 } A$$

$$\text{Q239 } A$$

$$\text{Q240 } A$$

$$\text{Q241 } S_n = \frac{n(n+1)(2n+7)}{6}$$

$$\text{Q242 } S_n = \frac{n(n+1)(2n+1)}{6}$$

$$\text{Q243 } S_n = \frac{n(n+1)(4n-1)}{6}$$

$$\text{Q244 } S_n = \frac{n(n+1)(n+2)}{3}$$

$$\text{Q245 } A$$

$$\text{Q246 } a_n = 4n + 3$$

$$\text{Q247 } a_n = 2(n-2)$$

$$\text{Q248 } a_n = 4n + 3$$

$$\text{Q249 } a_n = 2n - 1$$

$$\text{Q250 } a_n = 2n - 5$$

$$\text{Q251 } D$$

$$\text{Q252 } C$$

$$\text{Q253 } B$$

$$\text{Q254 } A$$

$$\text{Q255 } D$$

$$\text{Q256 } C$$

$$\text{Q257 } C$$

$$\text{Q258 } C$$

$$\text{Q259 } B$$

$$\text{Q260 } A$$

$$\text{Q261 } D$$

$$\text{Q262 } D$$

$$\text{Q263 } B$$

$$\text{Q264 } a_1 = 9, \quad a_n = 3(n^2 - n + 1) \quad (n \geq 2)$$

$$\text{Q265 } a_1 = 2, \quad a_n = 2n + 1 \quad (n \geq 2)$$

$$\text{Q266 } a_1 = 0, \quad a_n = 3n^2 - 3n + 1 \quad (n \geq 2)$$

$$\text{Q267 } a_n = 4 \cdot 5^{n-1}$$

$$\text{Q268 } a_n = 2 \cdot 3^{n-1}$$

$$\text{Q269 } a_n = 2^{n-1}$$

$$\text{Q270 } a_n = 6 \cdot 7^{n-1}$$

$$\text{Q271 } a_1 = 3, \quad a_n = 4 \cdot 5^{n-1} \quad (n \geq 2)$$

$$\text{Q272 } a_1 = 5, \quad a_n = 2^{n+1} \quad (n \geq 2)$$

$$\text{Q273 } B$$

$$\text{Q274 } a_1 = 8, \quad a_n = 2(3^n - 1) \quad (n \geq 2)$$

$$\text{Quiz MCQ 1 } A$$

$$\text{Quiz MCQ 2 } A$$

$$\text{Quiz MCQ 3 } A$$

$$\text{Quiz MCQ 4 } C$$

$$\text{Quiz WRITTEN 5 } -1 + 2 + 7 + 14 + 23$$

$$\text{Quiz WRITTEN 6 } \frac{n(n+1)(3n^2 + 23n + 10)}{12}$$

$$\text{Quiz WRITTEN 7 } \frac{3(4 \cdot 3^n n - 3 \cdot 3^n + 3)}{2}$$

$$\text{Quiz WRITTEN 8 } a_1 = 3, \quad a_n = 2^n - 2 \quad (n \geq 2)$$

Concept 4 | Advanced Sequences

English Student Edition • Focused formulas and guided practice

Formula opener

Differences

$$d_n = a_{n+1} - a_n$$

Rebuild

$$a_n = a_1 + \sum_{k=1}^{n-1} d_k$$

Grouping

odd/even index $\rightarrow$ separate rules

Classified Practice

Advanced Sequences

Q001 Written

Find the general term of the sequence $\{2, 5, 12, 23, 38, 57, \dots\}$.

Classified Practice

Advanced Sequences

Q002 MCQ

Select the correct option: If $a_1 = 1$ and $b_k = 2k + 1$, then a_n is:

Choose the correct option.

A $n^2 - 3$

B $(n + 1)^2$

C $a_n = n^2$

D $n^2 + 3$

Q003 MCQ

Select the correct option: If $a_1 = 2$ and $b_k = 3k - 2$, then a_n is:

Choose the correct option.

A $\frac{3n^2 - n + 4}{2}$

B $a_n = \frac{3n^2 - 7n + 8}{2}$

C $\frac{3n^2 - 7n + 16}{2}$

D $\frac{3n^2 - 7n + 6}{2}$

Classified Practice

Advanced Sequences

Q004 MCQ

Select the correct option: If $a_1 = 7$ and $b_k = 6k - 5$, then a_n is:

Choose the correct option.

A $a_n = 3n^2 - 8n + 12$

B $3n^2 - 8n + 13$

C $3n^2 - 8n + 9$

D $3n^2 - 2n + 7$

Classified Practice

Advanced Sequences

Q005 MCQ

Select the correct option: If $a_1 = 1$ and $b_k = 2k + 1$, find a_5 :

Choose the correct option.

A -25

B 25

C 30

D 23

Classified Practice

Advanced Sequences

Q006 MCQ

Select the correct option: If $a_1 = 2$ and $b_k = 3k - 2$, find a_6 :**Choose the correct option.**

A 37

B 38

C 34

D -37

Classified Practice

Advanced Sequences

Q007 MCQ

Select the correct option: If $a_1 = 7$ and $b_k = 6k - 5$, find a_6 :

Choose the correct option.

A 73

B 70

C -72

D 72

Classified Practice

Advanced Sequences

Q008 **Written****Find the general term of the sequence $\{2, 10, 24, 44, 70, 102, \dots\}$.**

Classified Practice

Advanced Sequences

Q009 Written

Find the general term of the sequence $\{1, 6, 15, 28, 45, 66, \dots\}$.

Classified Practice

Advanced Sequences

Q010 MCQ

Select the correct option: If $a_1 = 3$ and $b_k = 5k + 4$, then a_n is:

Choose the correct option.

A $a_n = \frac{(n+1)(5n-2)}{2}$

B $\frac{n(5n+3)}{2}$

C $\frac{5n^2+3n-6}{2}$

D $\frac{(n+2)(5n+3)}{2}$

Q011 MCQ

Select the correct option: If $a_1 = 4$ and $b_k = 7k - 3$, then a_n is:

Choose the correct option.

A $\frac{7n^2 - 13n + 18}{2}$

B $\frac{7n^2 - 13n + 8}{2}$

C $\frac{7n^2 + n + 8}{2}$

D $a_n = \frac{7n^2 - 13n + 14}{2}$

Classified Practice

Advanced Sequences

Q012 MCQ

Select the correct option: If $a_1 = 6$ and $b_k = 2(2k + 3)$, then a_n is:

Choose the correct option.

A $2(n + 1)(n + 3)$

B $a_n = 2n(n + 2)$

C $2(n^2 + 2n + 2)$

D $2(n^2 + 2n - 1)$

Classified Practice

Advanced Sequences

Q013 Written

Find the general term of the sequence $\{5, 8, 9, 8, 5, 0, \dots\}$.

Classified Practice

Advanced Sequences

Q014 **Written****Find the general term of the sequence $\{2, 3, 6, 15, 42, 123, \dots\}$.**

Classified Practice

Advanced Sequences

Q015 Written

Find the general term of the sequence $\{1, 2, 5, 14, 41, 122, \dots\}$.

Classified Practice

Advanced Sequences

Q016 **Written****Find the general term of the sequence $\{5, 7, 11, 19, 35, \dots\}$.**

Classified Practice

Advanced Sequences

Q017 Written

Find the general term of the sequence $\{2, 3, 1, 5, -3, 13, \dots\}$.

Classified Practice

Advanced Sequences

Q018 **Written****Find the general term of the sequence $\{5, 4, 7, -2, 25, -56, \dots\}$.**

Classified Practice

Advanced Sequences

Q019 MCQ

Select the correct option: If $a_1 = 1$ and $b_k = k^2$, then a_n is:

Choose the correct option.

A $a_n = \frac{(n+1)(2n^2-5n+6)}{6}$

B $\frac{2n^3-3n^2+n+12}{6}$

C $\frac{2n^3-3n^2+n-12}{6}$

D $\frac{(n+2)(2n^2-n+3)}{6}$

Classified Practice

Advanced Sequences

Q020 Challenge

An oncology team records tissue volumes 1, 3, 7, 13, 21, 31, ... in mm^3 . Find the general formula a_n and calculate a_{12} to decide whether the 130 mm^3 threshold is exceeded.

Classified Practice

Advanced Sequences

Q021 Written

For groups with $a_j = 2 + (j - 1)2$ and $c_j = 2n - 1$ terms in group j , find the first term of group 10.

Q022 Written

For groups with $a_j = 1 + (j - 1)2$ and $c_j = 2n - 1$ terms in group j , find the first term of group 10.

Classified Practice

Advanced Sequences

Q023 Written

For groups with $a_j = 1 + (j - 1)1$ and $c_j = n$ terms in group j , find the first term of group 10.

Q024 MCQ**Select the correct option: find the first term of group 3:****Choose the correct option.**

A 6

B 4

C 5

D 3

Classified Practice

Advanced Sequences

Q025 MCQ

Select the correct option: find the first term of group 6:**Choose the correct option.**

- A 77
- B 108
- C 78
- D 79

Q026 MCQ**Select the correct option: find the first term of group 9:****Choose the correct option.**

A 38

B 36

C 45

D 37

Classified Practice

Advanced Sequences

Q027 MCQ

Select the correct option: find the first term of group 4:**Choose the correct option.**

A 48

B 30

C 31

D 29

Q028 MCQ**Select the correct option: find the first term of group 7:****Choose the correct option.**

A 21

B 28

C 22

D 23

Classified Practice

Advanced Sequences

Q029 MCQ

Select the correct option: find the first term of group 10:**Choose the correct option.**

A 247

B 245

C 300

D 246

Q030 MCQ**Select the correct option: find the first term of group 4:****Choose the correct option.**

A 20

B 21

C 19

D 32

Classified Practice

Advanced Sequences

Q031 MCQ

Select the correct option: find the first term of group 7:**Choose the correct option.**

A 138

B 110

C 111

D 109

Q032 MCQ**Select the correct option: find the first term of group 10:****Choose the correct option.**

A 163

B 200

C 164

D 165

Classified Practice

Advanced Sequences

Q033 MCQ

Select the correct option: find the first term of group 5:**Choose the correct option.**

A 58

B 59

C 57

D 78

Q034 MCQ**Select the correct option: find the first term of group 8:****Choose the correct option.**

A 128

B 100

C 101

D 99

Classified Practice

Advanced Sequences

Q035 Written

For groups with $a_j = 1 + (j - 1)3$ and $c_j = 3n - 1$ terms in group j , find the first term of group 10.

Q036 MCQ**Select the correct option: find the first term of group 5:****Choose the correct option.**

A 30

B 28

C 39

D 29

Classified Practice

Advanced Sequences

Q037 MCQ

Select the correct option: find the first term of group 8:**Choose the correct option.**

A 89

B 90

C 88

D 110

Q038 Written

For groups with $a_j = 2 + (j - 1)2$ and $c_j = 2n - 1$ terms in group j , derive the sum of the terms in group n .

Classified Practice

Advanced Sequences

Q039 **Written****For the same grouped sequence, find the sum of the terms in group 10.**

Q040 Written

For the same grouped sequence, find the sum of the terms in group 10.

Classified Practice

Advanced Sequences

Q041 **Written****For the same grouped sequence, find the sum of the terms in group 10.**

Q042 Written

In the grouped arithmetic sequence, locate the value 88: state its group and its position within that group.

Classified Practice

Advanced Sequences

Q043 Written

In the grouped arithmetic sequence, locate the value 101: state its group and its position within that group.

Q044 Written

In the grouped arithmetic sequence, locate the value 124: state its group and its position within that group.

Classified Practice

Advanced Sequences

Q045 **Written**

In the grouped arithmetic sequence, locate the value 100: state its group and its position within that group.

Classified Practice

Advanced Sequences

Q046 Challenge

For the grouped sequence $2, 6 \mid 10, 14, 18 \mid 22, 26, 30, 34 \mid \dots$, solve: (a) the first term of Group 12; (b) the sum in Group n ; (c) the group and position of 202; (d) the total through Group 12.

Classified Practice

Advanced Sequences

Quiz MCQ 1 **Concept Quiz · MCQ**

Select the correct option: If $a_1 = 2$ and $b_k = 3k + 1$, then a_n is:

Choose the correct option.

A $a_n = \frac{3n^2 - n + 2}{2}$

B $\frac{3n^2 - n + 4}{2}$

C $\frac{n(3n-1)}{2}$

D $\frac{3n^2 + 5n + 4}{2}$

Quiz MCQ 2 Concept Quiz · MCQ

Select the correct option: If $a_1 = 3$ and $b_k = (-2)^{k-1}$, then a_n is:

Choose the correct option.

A $\frac{(-2)^n + 2}{6}$

B $-\frac{(-2)^n - 10}{3}$

C $a_n = \frac{(-2)^n + 20}{6}$

D $\frac{(-2)^n + 38}{6}$

Classified Practice

Advanced Sequences

Quiz MCQ 3 [Concept Quiz](#) · [MCQ](#)

Select the correct option: find the first term of group 9:

Choose the correct option.

A 195

B 196

C 194

D 243

Quiz MCQ 4 **Concept Quiz · MCQ**

Select the correct option: the value 29 is in:

Choose the correct option.

- A Group 5, position 2
- B Group 5, position 1
- C Group 6, position 1
- D Group 4, position 2

Classified Practice

Advanced Sequences

Quiz WRITTEN 5 Concept Quiz · Written

A sequence has $a_1 = 1$ and difference sequence $b_k = 4k - 1$. Find the required general term a_n .

Quiz WRITTEN 6 Concept Quiz · Written

A sequence has $a_1 = 2$ and difference sequence $b_k = (-2)^{k-1}$. Find the required general term a_n .

Classified Practice

Advanced Sequences

Quiz WRITTEN 7 Concept Quiz · Written

For groups with $a_j = 5 + (j - 1)3$ and $c_j = n$ terms in group j , find the first term of group 12.

Quiz WRITTEN 8 Concept Quiz · Written

In the grouped arithmetic sequence, locate the value 66: state its group and its position within that group.

Answer key

Use the question number shown on each page.

Q001 $a_n = 2n^2 - 3n + 3$

Q002 C

Q003 B

Q004 A

Q005 B

Q006 A

Q007 D

Q008 $a_n = n(3n - 1)$

Q009 $a_n = n(2n - 1)$

Q010 A

Q011 D

Q012 B

Q013 $a_n = -n(n - 6)$

Q014 $a_n = \frac{3^n + 9}{6}$

Q015 $a_n = \frac{3^n + 3}{6}$

Q016 $a_n = 2^n + 3$

Q017 $a_n = \frac{(-2)^n + 14}{6}$

Q018 $a_n = -\frac{(-3)^n - 57}{12}$

Q019 A

Q020 $a_n = n^2 - n + 1$, $a_{12} = 133$; surgery required

Q021 164

Q022 163

Q023 46

Q024 B

Q025 C

Q026 D

Q027 B

Q028 C

Q029 D

Q030 A

Q031 B

Q032 C

Q033 A

Q034 B

Q035 379

Q036 D

Q037 A

Q038 $2(2n - 1)(n^2 - n + 1)$

Q039 3439

Q040 505

Q041 12209

Q042 Group 7, position 8

Q043 Group 8, position 2

Q044 Group 16, position 4

Q045 Group 5, position 8

Q046 (a) 310; (b) $2(n + 1)(n^2 + 2n - 1)$; (c) Group 9, position 7; (d) 16200

Quiz MCQ 1 A

Quiz MCQ 2 C

Quiz MCQ 3 A

Quiz MCQ 4 B

Quiz WRITTEN 5 $a_n = 2n^2 - 3n + 2$

Quiz WRITTEN 6 $a_n = \frac{(-2)^n + 14}{6}$

Quiz WRITTEN 7 203

Quiz WRITTEN 8 Group 5, position 3

Concept 5 | Recurrence and Induction

English Student Edition • Focused formulas and guided practice

Formula opener

Recurrence

$$a_{n+1} = f(a_n)$$

Induction

$$P(1), P(k) \Rightarrow P(k + 1)$$

Divisibility

$$P(k) = mq \Rightarrow P(k + 1) = mq'$$

Classified Practice

Recurrence and Induction

Q001 Written

For the sequence $\{a_n\}$ with $a_1 = 2$ and recurrence $a_{n+1} = 3a_n + (2)$. Find the 2nd through the 4th terms.

Q002 Written

For the sequence $\{a_n\}$ with $a_1 = -6$ and recurrence $a_{n+1} = 2a_n + (3)$. Find the 2nd through the 4th terms.

Classified Practice

Recurrence and Induction

Q003 Written

For the sequence $\{a_n\}$ with $a_1 = -6$ and recurrence $a_{n+1} = 1a_n + (n(n-2))$. Find the 2nd through the 4th terms.

Q004 Written

For the sequence $\{a_n\}$ with $a_1 = -2$ and recurrence $a_{n+1} = 1a_n + (-n)$. Find the 2nd through the 4th terms.

Classified Practice

Recurrence and Induction

Q005 Written

For the sequence $\{a_n\}$ with $a_1 = 2$ and recurrence $a_{n+1} = 1a_n + (2n)$. Find the 2nd through the 4th terms.

Q006 Written

For the sequence $\{a_n\}$ with $a_1 = -2$ and recurrence $a_{n+1} = 1a_n + (3^n)$. Find the 2nd through the 4th terms.

Classified Practice

Recurrence and Induction

Q007 MCQ

Select the correct option: For the sequence $\{a_n\}$ with $a_1 = 3$ and recurrence $a_{n+1} = 3a_n + (0)$. The fourth term is:

Choose the correct option.

A -81

B 81

C 82

D 80

Q008 MCQ

Select the correct option: For the sequence $\{a_n\}$ with $a_1 = 2$ and recurrence $a_{n+1} = 4a_n + (0)$. The fourth term is:

Choose the correct option.

A 133

B 127

C -128

D 128

Classified Practice

Recurrence and Induction

Q009 MCQ

Select the correct option: For the sequence $\{a_n\}$ with $a_1 = 2$ and recurrence $a_{n+1} = 2a_n + (n + 1)$. The fourth term is:

Choose the correct option.

A 34

B 37

C 33

D -34

Q010 MCQ

Select the correct option: For the sequence $\{a_n\}$ with $a_1 = 5$ and recurrence $a_{n+1} = 3a_n + (2n)$. The fourth term is:

Choose the correct option.

A -171

B 171

C 172

D 170

Classified Practice

Recurrence and Induction

Q011 MCQ

Select the correct option: For the sequence $\{a_n\}$ with $a_1 = 1$ and recurrence $a_{n+1} = 1a_n + (4)$. The fourth term is:

Choose the correct option.

A 17

B 11

C -13

D 13

Q012 MCQ

Select the correct option: For the sequence $\{a_n\}$ with $a_1 = 4$ and recurrence $a_{n+1} = -2a_n + (5)$. The fourth term is:

Choose the correct option.

A -17

B -15

C -19

D 17

Classified Practice

Recurrence and Induction

Q013 MCQ

Select the correct option: For the sequence $\{a_n\}$ with $a_1 = 1$ and recurrence $a_{n+1} = -3a_n + (2)$. The fourth term is:

Choose the correct option.

A -15

B 13

C -13

D -12

Q014 MCQ

Select the correct option: For the sequence $\{a_n\}$ with $a_1 = 4$ and recurrence $a_{n+1} = 1a_n + (n^2)$. The fourth term is:

Choose the correct option.

A 22

B 16

C -18

D 18

Classified Practice

Recurrence and Induction

Q015 MCQ

Select the correct option: For the sequence $\{a_n\}$ with $a_1 = 2$ and recurrence $a_{n+1} = 2a_n + (1)$. The fourth term is:

Choose the correct option.

A 20

B -23

C 23

D 28

Q016 MCQ

Select the correct option: For the sequence $\{a_n\}$ with $a_1 = 5$ and recurrence $a_{n+1} = 1a_n + (-3)$. The fourth term is:

Choose the correct option.

A -1

B -7

C 4

D -4

Classified Practice

Recurrence and Induction

Q017 MCQ

Select the correct option: For the sequence $\{a_n\}$ with $a_1 = 3$ and recurrence $a_{n+1} = 1a_n + (n)$. The fourth term is:

Choose the correct option.

A -9

B 9

C 11

D 6

Q018 MCQ

Select the correct option: For the sequence $\{a_n\}$ with $a_1 = 1$ and recurrence $a_{n+1} = 2a_n + (-n)$. The fourth term is:

Choose the correct option.

A -6

B 3

C -3

D 2

Classified Practice

Recurrence and Induction

Q019 Written

For the sequence $\{a_n\}$ with $a_1 = -1$ and recurrence $a_{n+1} = 1a_n + (5)$. Find the general term.

Q020 Written

For the sequence $\{a_n\}$ with $a_1 = 1$ and recurrence $a_{n+1} = 1a_n + (3)$. Find the general term.

Classified Practice

Recurrence and Induction

Q021 Written

For the sequence $\{a_n\}$ with $a_1 = 4$ and recurrence $a_{n+1} = 1a_n + (-2)$. Find the general term.

Q022 Written

For the sequence $\{a_n\}$ with $a_1 = 1$ and recurrence $a_{n+1} = 1a_n + (6)$. Find the general term.

Classified Practice

Recurrence and Induction

Q023 Written

For the sequence $\{a_n\}$ with $a_1 = 3$ and recurrence $a_{n+1} = 1a_n + (-5)$. Find the general term.

Q024 Written

For the sequence $\{a_n\}$ with $a_1 = -2$ and recurrence $a_{n+1} = 1a_n + (3)$. Find the general term.

Classified Practice

Recurrence and Induction

Q025 Written

For the sequence $\{a_n\}$ with $a_1 = 4$ and recurrence $a_{n+1} = 1a_n + (-4)$. Find the general term.

Q026 MCQ

Select the correct option: For the sequence $\{a_n\}$ with $a_1 = 3$ and recurrence $a_{n+1} = 3a_n + (0)$. The general term is:

Choose the correct option.

A $3^n + 2$

B $3^n - 1$

C $3^n + 3$

D $a_n = 3^n$

Classified Practice

Recurrence and Induction

Q027 MCQ

Select the correct option: For the sequence $\{a_n\}$ with $a_1 = 2$ and recurrence $a_{n+1} = 4a_n + (0)$. The general term is:

Choose the correct option.

A $a_n = 2^{2n-1}$

B $\frac{2^{2n+2}}{2}$

C $\frac{2^{2n-2}}{2}$

D $\frac{2^{2n+4}}{2}$

Q028 MCQ

Select the correct option: For the sequence $\{a_n\}$ with $a_1 = 4$ and recurrence $a_{n+1} = 5a_n + (0)$. The general term is:

Choose the correct option.

A $a_n = 4 \cdot 5^{n-1}$

B $\frac{4 \cdot 5^{n+5}}{5}$

C $\frac{4 \cdot 5^n - 5}{5}$

D $\frac{2(2 \cdot 5^n + 5)}{5}$

Classified Practice

Recurrence and Induction

Q029 MCQ

Select the correct option: For the sequence $\{a_n\}$ with $a_1 = 6$ and recurrence $a_{n+1} = 2a_n + (0)$. The general term is:

Choose the correct option.

A $a_n = 3 \cdot 2^n$

B $3 \cdot 2^n + 1$

C $3 \cdot 2^n - 1$

D $3 \cdot 2^n + 2$

Q030 Written

For the sequence $\{a_n\}$ with $a_1 = 2$ and recurrence $a_{n+1} = 3a_n + (0)$. Find the general term.

Classified Practice

Recurrence and Induction

Q031 Written

For the sequence $\{a_n\}$ with $a_1 = 3$ and recurrence $a_{n+1} = 5a_n + (0)$. Find the general term.

Q032 Written

For the sequence $\{a_n\}$ with $a_1 = 4$ and recurrence $a_{n+1} = 3a_n + (0)$. Find the general term.

Classified Practice

Recurrence and Induction

Q033 Written

For the sequence $\{a_n\}$ with $a_1 = 1$ and recurrence $a_{n+1} = 4a_n + (0)$. Find the general term.

Q034 MCQ

Select the correct option: For the sequence $\{a_n\}$ with $a_1 = 1$ and recurrence $a_{n+1} = 1a_n + (4)$. The general term is:

Choose the correct option.

A $4n - 1$

B $a_n = 4n - 3$

C $4n + 1$

D $4n - 5$

Classified Practice

Recurrence and Induction

Q035 MCQ

Select the correct option: For the sequence $\{a_n\}$ with $a_1 = 4$ and recurrence $a_{n+1} = 1a_n + (n^2)$. The general term is:

Choose the correct option.

A $\frac{2n^3 - 3n^2 + n + 36}{6}$

B $a_n = \frac{2n^3 - 3n^2 + n + 24}{6}$

C $\frac{2n^3 - 3n^2 + n + 48}{6}$

D $\frac{2n^3 - 3n^2 + n + 12}{6}$

Q036 MCQ

Select the correct option: For the sequence $\{a_n\}$ with $a_1 = 1$ and recurrence $a_{n+1} = -4a_n + (0)$. The general term is:

Choose the correct option.

A $-\frac{(-4)^n - 8}{4}$

B $-\frac{(-4)^n + 8}{4}$

C $-\frac{(-4)^n - 12}{4}$

D $a_n = (-4)^{n-1}$

Classified Practice

Recurrence and Induction

Q037 MCQ

Select the correct option: For the sequence $\{a_n\}$ with $a_1 = 5$ and recurrence $a_{n+1} = 1a_n + (3^n)$. The general term is:

Choose the correct option.

A $a_n = \frac{3^{n+7}}{2}$

B $\frac{3^{n+9}}{2}$

C $\frac{3^{n^2+3}}{2}$

D $\frac{3^{n^2+11}}{2}$

Q038 Written

For the sequence $\{a_n\}$ with $a_1 = -3$ and recurrence $a_{n+1} = -2a_n + (0)$. Find the general term.

Classified Practice

Recurrence and Induction

Q039 Written

For the sequence $\{a_n\}$ with $a_1 = 2$ and recurrence $a_{n+1} = -5a_n + (0)$. Find the general term.

Q040 Written

For the sequence $\{a_n\}$ with $a_1 = -2$ and recurrence $a_{n+1} = -4a_n + (0)$. Find the general term.

Classified Practice

Recurrence and Induction

Q041 MCQ

Select the correct option: For the sequence $\{a_n\}$ with $a_1 = 5$ and recurrence $a_{n+1} = 1a_n + (-3)$. The general term is:

Choose the correct option.

A $10 - 3n$

B $a_n = 8 - 3n$

C $-3(n - 4)$

D $5 - 3n$

Q042 MCQ

Select the correct option: For the sequence $\{a_n\}$ with $a_1 = 3$ and recurrence $a_{n+1} = 1a_n + (n)$. The general term is:

Choose the correct option.

A $\frac{n(n-1)}{2}$

B $\frac{n^2-n+10}{2}$

C $a_n = \frac{n^2-n+6}{2}$

D $\frac{n^2-n+12}{2}$

Classified Practice

Recurrence and Induction

Q043 MCQ

Select the correct option: For the sequence $\{a_n\}$ with $a_1 = 3$ and recurrence $a_{n+1} = 1a_n + (-5)$. The general term is:

Choose the correct option.

A $-5(n - 2)$

B $a_n = 8 - 5n$

C $12 - 5n$

D $-5(n - 1)$

Q044 MCQ

Select the correct option: For the sequence $\{a_n\}$ with $a_1 = 2$ and recurrence $a_{n+1} = 1a_n + (6)$. The general term is:

Choose the correct option.

A $6n - 7$

B $2(3n - 1)$

C $a_n = 2(3n - 2)$

D $6n - 1$

Classified Practice

Recurrence and Induction

Q045 Written

For the sequence $\{a_n\}$ with $a_1 = 2$ and recurrence $a_{n+1} = 3a_n + (-n)$. Find the 2nd through the 4th terms.

Q046 Written

For the sequence $\{a_n\}$ with $a_1 = 1$ and recurrence $a_{n+1} = -2a_n + (5)$. Find the 2nd through the 4th terms.

Classified Practice

Recurrence and Induction

Q047 Written

For the sequence $\{a_n\}$ with $a_1 = 3$ and recurrence $a_{n+1} = 3a_n + (-n)$. Find the 2nd through the 4th terms.

Classified Practice

Recurrence and Induction

Q048 Challenge

Solve the third-order recurrence $a_n = 6a_{n-1} - 11a_{n-2} + 6a_{n-3}$ for $n \geq 3$ with $a_0 = 0, a_1 = 1, a_2 = 6$.

Classified Practice

Recurrence and Induction

Q049 Written

Find the general term of the sequence $\{a_n\}$ determined by $a_1 = 1$ and $a_{n+1} = a_n + (2n - 3)$.

Q050 Written

Find the general term of the sequence $\{a_n\}$ determined by $a_1 = 1$ and $a_{n+1} = a_n + (3n)$.

Classified Practice

Recurrence and Induction

Q051 Written

Find the general term of the sequence $\{a_n\}$ determined by $a_1 = 2$ and $a_{n+1} = a_n + (2n)$.

Q052 Written

Find the general term of the sequence $\{a_n\}$ determined by $a_1 = 2$ and $a_{n+1} = a_n + (3n^2)$.

Classified Practice

Recurrence and Induction

Q053 Written

Find the general term of the sequence $\{a_n\}$ determined by $a_1 = 2$ and $a_{n+1} = a_n + (3n + 1)$.

Q054 Written

Find the general term of the sequence $\{a_n\}$ determined by $a_1 = 2$ and $a_{n+1} = a_n + (n(2n + 1))$.

Classified Practice

Recurrence and Induction

Q055 MCQ

Select the correct option: If $a_1 = 3$ and $a_{n+1} = a_n + (n^2)$, then a_n is:

Choose the correct option.

A $\frac{(n+1)(2n^2-5n+6)}{6}$

B $\frac{(n+2)(2n^2-7n+15)}{6}$

C $a_n = \frac{2n^3-3n^2+n+18}{6}$

D $\frac{2n^3-3n^2+n+36}{6}$

Q056 MCQ

Select the correct option: If $a_1 = 2$ and $a_{n+1} = a_n + (3n^2)$, then a_n is:

Choose the correct option.

A $a_n = \frac{2n^3 - 3n^2 + n + 4}{2}$

B $\frac{(n+1)(2n^2 - 5n + 6)}{2}$

C $\frac{n(n-1)(2n-1)}{2}$

D $\frac{2n^3 - 3n^2 + n + 8}{2}$

Classified Practice

Recurrence and Induction

Q057 MCQ

Select the correct option: If $a_1 = 2$ and $a_{n+1} = a_n + (n^2 + 1)$, then a_n is:

Choose the correct option.

A $\frac{(n-1)(2n^2-n+6)}{6}$

B $\frac{2n^3-3n^2+7n+18}{6}$

C $a_n = \frac{2n^3-3n^2+7n+6}{6}$

D $\frac{(2n+3)(n^2-3n+8)}{6}$

Q058 MCQ

Select the correct option: If $a_1 = 5$ and $a_{n+1} = a_n + (n(4n - 1))$, then a_n is:

Choose the correct option.

A $\frac{8n^3 - 15n^2 + 7n + 42}{6}$

B $\frac{8n^3 - 15n^2 + 7n + 18}{6}$

C $\frac{8n^3 - 15n^2 + 7n + 48}{6}$

D $a_n = \frac{(n+1)(8n^2 - 23n + 30)}{6}$

Classified Practice

Recurrence and Induction

Q059 MCQ

Select the correct option: If $a_1 = 2$ and $a_{n+1} = a_n + (8n)$, then a_n is:

Choose the correct option.

A $4n(n-1)$

B $4(n^2 - n + 1)$

C $a_n = 2(2n^2 - 2n + 1)$

D $4n^2 - 4n + 5$

Q060 MCQ**Select the correct option: find a_4 when $a_1 = 3$ and $a_{n+1} = a_n + (n^2)$:****Choose the correct option.**

A 17

B 19

C 15

D -17

Classified Practice

Recurrence and Induction

Q061 MCQ

Select the correct option: find a_5 when $a_1 = 2$ and $a_{n+1} = a_n + (3n^2)$:

Choose the correct option.

A 96

B 90

C -92

D 92

Q062 MCQ**Select the correct option: find a_3 when $a_1 = 2$ and $a_{n+1} = a_n + (n^2 + 1)$:****Choose the correct option.**

A 9

B 11

C 7

D -9

Classified Practice

Recurrence and Induction

Q063 MCQ

Select the correct option: find a_6 when $a_1 = 5$ and $a_{n+1} = a_n + (n(4n - 1))$:

Choose the correct option.

A -210

B 210

C 215

D 208

Q064 MCQ

Select the correct option: find a_2 when $a_1 = 2$ and $a_{n+1} = a_n + (8n)$:

Choose the correct option.

A 10

B 12

C 8

D -10

Classified Practice

Recurrence and Induction

Q065 Written

Find the general term of the sequence $\{a_n\}$ determined by $a_1 = 2$ and $a_{n+1} = a_n + (n^2)$.

Q066 MCQ

Select the correct option: If $a_1 = 1$ and $a_{n+1} = a_n + (2n + 1)$, then a_n is:

Choose the correct option.

A $a_n = n^2$

B $n^2 + 1$

C $n^2 - 3$

D $n^2 + 2$

Classified Practice

Recurrence and Induction

Q067 Written

Find the general term of the sequence $\{a_n\}$ determined by $a_1 = 1$ and $a_{n+1} = a_n + (2^n)$.

Q068 Written

Find the general term of the sequence $\{a_n\}$ determined by $a_1 = 2$ and $a_{n+1} = a_n + (-3^n)$.

Classified Practice

Recurrence and Induction

Q069 Written

Find the general term of the sequence $\{a_n\}$ determined by $a_1 = 4$ and $a_{n+1} = a_n + (-4^n)$.

Classified Practice

Recurrence and Induction

Q070 Challenge

A technology startup has a reserve of 400,000 LE, represented by $a_1 = 4$, and burns $0.04n$ hundred-thousand LE each quarter: $a_{n+1} = a_n - 0.04n$. Find an explicit formula for a_n .

Classified Practice

Recurrence and Induction

Q071 Written

Find the general term of $\{a_n\}$ given $a_1 = 1$ and $a_{n+1} = 3a_n + (4)$.

Classified Practice

Recurrence and Induction

Q072 **Written**

Find the general term of $\{a_n\}$ given $a_1 = 1$ and $a_{n+1} = 5a_n + (8)$.

Classified Practice

Recurrence and Induction

Q073 Written

Find the general term of $\{a_n\}$ given $a_1 = 2$ and $a_{n+1} = 2a_n + (-1)$.

Classified Practice

Recurrence and Induction

Q074 **Written**

Find the general term of $\{a_n\}$ given $a_1 = -3$ and $a_{n+1} = 4a_n + (3)$.

Classified Practice

Recurrence and Induction

Q075 Written

Find the general term of $\{a_n\}$ given $a_1 = 3$ and $a_{n+1} = 3a_n + (-2)$.

Classified Practice

Recurrence and Induction

Q076 Written

Find the general term of $\{a_n\}$ given $a_1 = -2$ and $a_{n+1} = -3a_n + (4)$.

Classified Practice

Recurrence and Induction

Q077 Written

Find the general term of $\{a_n\}$ given $a_1 = 1$ and $a_{n+1} = \frac{1}{3}a_n + (2)$.

Classified Practice

Recurrence and Induction

Q078 Written

Find the general term of $\{a_n\}$ given $a_1 = 2$ and $a_{n+1} = \frac{2}{3}a_n + (-\frac{1}{2})$.

Classified Practice

Recurrence and Induction

Q079 Written

Find the general term of $\{a_n\}$ given $a_1 = 4$ and $a_{n+1} = -2a_n + (3)$.

Classified Practice

Recurrence and Induction

Q080 MCQ

Select the correct option: If $a_1 = 3$ and $a_{n+1} = -2a_n + (5)$, the general term is:

Choose the correct option.

A $-\frac{2(-2)^n - 11}{3}$

B $a_n = -\frac{2(-2)^n - 5}{3}$

C $-\frac{2(-2)^n - 17}{3}$

D $-\frac{2((-2)^n + 2)}{3}$

Classified Practice

Recurrence and Induction

Q081 MCQ

Select the correct option: If $a_1 = 1$ and $a_{n+1} = \frac{2}{3}a_n + (-2)$, the general term is:

Choose the correct option.

A $\frac{3 \cdot 3^{-n}(7 \cdot 2^n - 6 \cdot 3^n)}{2}$

B $\frac{3^{-n}(21 \cdot 2^n - 8 \cdot 3^n)}{2}$

C $a_n = \frac{3 \cdot 3^{-n}(7 \cdot 2^n - 4 \cdot 3^n)}{2}$

D $\frac{3 \cdot 3^{-n}(7 \cdot 2^n - 2 \cdot 3^n)}{2}$

Classified Practice

Recurrence and Induction

Q082 MCQ

Select the correct option: If $a_1 = 2$ and $a_{n+1} = 7a_n + (-4)$, the general term is:

Choose the correct option.

A $\frac{4(7^n+14)}{21}$

B $\frac{4 \cdot 7^n - 49}{21}$

C $\frac{4 \cdot 7^n + 77}{21}$

D $a_n = \frac{2(2 \cdot 7^n + 7)}{21}$

Classified Practice

Recurrence and Induction

Q083 MCQ

Select the correct option: If $a_1 = 6$ and $a_{n+1} = 2a_n + (-5)$, the general term is:

Choose the correct option.

A $\frac{2^n+14}{2}$

B $a_n = \frac{2^n+10}{2}$

C $\frac{2^n+18}{2}$

D $\frac{2^n+4}{2}$

Classified Practice

Recurrence and Induction

Q084 MCQ

Select the correct option: If $a_1 = 5$ and $a_{n+1} = -2a_n + (7)$, the general term is:

Choose the correct option.

A $-\frac{2(2(-2)^n+1)}{3}$

B $-\frac{4(-2)^n-13}{3}$

C $a_n = -\frac{4(-2)^n-7}{3}$

D $-\frac{4((-2)^n-4)}{3}$

Classified Practice

Recurrence and Induction

Q085 MCQ

Select the correct option: If $a_1 = 4$ and $a_{n+1} = -5a_n + (-2)$, the general term is:

Choose the correct option.

A $-\frac{13(-5)^n - 25}{15}$

B $-\frac{13(-5)^n + 50}{15}$

C $-\frac{13(-5)^n - 40}{15}$

D $a_n = -\frac{13(-5)^n + 5}{15}$

Classified Practice

Recurrence and Induction

Q086 MCQ

Select the correct option: If $a_1 = 2$ and $a_{n+1} = -3a_n + (5)$, the general term is:

Choose the correct option.

A $-\frac{(-3)^n - 13}{4}$

B $a_n = -\frac{(-3)^n - 5}{4}$

C $-\frac{(-3)^n - 21}{4}$

D $-\frac{(-3)^n + 7}{4}$

Classified Practice

Recurrence and Induction

Q087 MCQ

Select the correct option: If $a_1 = 1$ and $a_{n+1} = \frac{1}{2}a_n + (7)$, the general term is:

Choose the correct option.

A $2^{-n} (11 \cdot 2^n - 26)$

B $2 \cdot 2^{-n} (8 \cdot 2^n - 13)$

C $a_n = 2 \cdot 2^{-n} (7 \cdot 2^n - 13)$

D $2^{-n} (17 \cdot 2^n - 26)$

Classified Practice

Recurrence and Induction

Q088 MCQ

Select the correct option: If $a_1 = 1$ and $a_{n+1} = 2a_n + (-1)$, the general term is:

Choose the correct option.

A 3

B 0

C 4

D $a_n = 1$

Classified Practice

Recurrence and Induction

Q089 MCQ

Select the correct option: If $a_1 = 3$ and $a_{n+1} = -2a_n + (5)$, the general term is:

Choose the correct option.

A $-\frac{2(-2)^n - 11}{3}$

B $-\frac{2((-2)^n + 2)}{3}$

C $-\frac{2((-2)^n - 7)}{3}$

D $a_n = -\frac{2(-2)^n - 5}{3}$

Classified Practice

Recurrence and Induction

Q090 Challenge

A sequence satisfies $a_{n+1} = 4a_n - 9$ with $a_1 = 2$. Find the equilibrium value, the explicit formula, and a_5 .

Classified Practice

Recurrence and Induction

Q091 Written

Use mathematical induction to prove $\sum_{i=1}^n i = \frac{n(n+1)}{2}$.

Classified Practice

Recurrence and Induction

Q092 Written

Use mathematical induction to prove $\sum_{i=1}^n (2i - 1)^2 = \frac{n(2n-1)(2n+1)}{3}$.

Classified Practice

Recurrence and Induction

Q093 Written

Use mathematical induction to prove $\sum_{i=1}^n i(i+1) = \frac{n(n+1)(n+2)}{3}$.

Classified Practice

Recurrence and Induction

Q094 MCQ

Select the correct option: the right-hand side equivalent to $\sum_{i=1}^n i(i+1)$ is:

Choose the correct option.

A $\frac{n(n+1)(n+2)}{3}$

B $\frac{n^3+3n^2+2n+3}{3}$

C $\frac{n^3+3n^2+2n-3}{3}$

D $\frac{(n+3)(n^2+2)}{3}$

Classified Practice

Recurrence and Induction

Q095 MCQ

Select the correct option: the right-hand side equivalent to $\sum_{i=1}^n 3 \cdot 4^{i-1}$ is:

Choose the correct option.

A $4^n + 1$

B $4^n - 1$

C $4^n + 3$

D $4^n - 2$

Classified Practice

Recurrence and Induction

Q096 MCQ

Select the correct option: the right-hand side equivalent to $\sum_{i=1}^n 2i + 3$ is:

Choose the correct option.

A $n^2 + 4n - 1$

B $n^2 + 4n + 2$

C $n(n + 4)$

D $(n + 1)(n + 3)$

Classified Practice

Recurrence and Induction

Q097 MCQ

Select the correct option: the right-hand side equivalent to $\sum_{i=1}^n 2i^2 + 3i$ is:

Choose the correct option.

A $\frac{4n^3+15n^2+11n+12}{6}$

B $\frac{n(n+1)(4n+11)}{6}$

C $\frac{4n^3+15n^2+11n+24}{6}$

D $\frac{(n+2)(4n^2+7n-3)}{6}$

Classified Practice

Recurrence and Induction

Q098 MCQ

Select the correct option: the right-hand side equivalent to $\sum_{i=1}^n 3i^2 + 2i$ is:

Choose the correct option.

A $\frac{2n^3 + 5n^2 + 3n - 2}{2}$

B $\frac{2n^3 + 5n^2 + 3n + 4}{2}$

C $\frac{n(n+1)(2n+3)}{2}$

D $\frac{2n^3 + 5n^2 + 3n + 6}{2}$

Classified Practice

Recurrence and Induction

Q099 MCQ

Select the correct option: the right-hand side equivalent to $\sum_{i=1}^n 7 \cdot 2^{i-1}$ is:

Choose the correct option.

A $7(2^n - 1)$

B $72^n - 6$

C $72^n - 8$

D $72^n - 5$

Classified Practice

Recurrence and Induction

Q100 MCQ

Select the correct option: the right-hand side equivalent to $\sum_{i=1}^n i(i+1)$ is:

Choose the correct option.

A $\frac{n^3+3n^2+2n-3}{3}$

B $\frac{(n+3)(n^2+2)}{3}$

C $\frac{n(n+1)(n+2)}{3}$

D $\frac{n^3+3n^2+2n+9}{3}$

Classified Practice

Recurrence and Induction

Q101 MCQ

Select the correct option: the right-hand side equivalent to $\sum_{i=1}^n 3 \cdot 4^{i-1}$ is:

Choose the correct option.

A $4^n - 1$

B 4^n

C $4^n - 2$

D $4^n + 1$

Classified Practice

Recurrence and Induction

Q102 MCQ

Select the correct option: the right-hand side equivalent to $\sum_{i=1}^n 2i + 3$ is:

Choose the correct option.

A $n^2 + 4n + 2$

B $n(n + 4)$

C $(n + 2)^2$

D $n^2 + 4n - 1$

Classified Practice

Recurrence and Induction

Q103 Written

Use mathematical induction to prove $\sum_{i=1}^n (2i - 1) = n^2$.

Classified Practice

Recurrence and Induction

Q104 **Written**

Use mathematical induction to prove $\sum_{i=1}^n i(i+1) = \frac{n(n+1)(n+2)}{3}$.

Classified Practice

Recurrence and Induction

Q105 Written

Use mathematical induction to prove $\sum_{i=1}^n i(i+2) = \frac{n(n+1)(2n+7)}{6}$.

Classified Practice

Recurrence and Induction

Q106 Written

Use mathematical induction to prove $\sum_{i=1}^n 3 \cdot 4^{i-1} = 4^n - 1$.

Classified Practice

Recurrence and Induction

Q107 Written

Use mathematical induction to prove $\sum_{i=0}^n 10^i = \frac{10^{n+1}-10}{9}$.

Classified Practice

Recurrence and Induction

Q108 Challenge

Use mathematical induction to prove $\sum_{i=1}^n (i^2 + i + 1) = \frac{n(n^2+3n+5)}{3}$.

Classified Practice

Recurrence and Induction

Q109 **Written****Prove by mathematical induction that $2^n > 3n$ for $n \geq 4$.**

Classified Practice

Recurrence and Induction

Q110 Written

Prove by mathematical induction that $3^n > 5n + 1$ for $n \geq 3$.

Classified Practice

Recurrence and Induction

Q111 **Written****Prove by mathematical induction that $3^n > 2n$ for $n \geq 1$.**

Classified Practice

Recurrence and Induction

Q112 Written

Prove by mathematical induction that $2^n > 1n + 2$ for $n \geq 3$.

Classified Practice

Recurrence and Induction

Q113 Written

Prove by mathematical induction that $3^n > 5n + 1$ for $n \geq 4$.

Classified Practice

Recurrence and Induction

Q114 **Written**

Prove by mathematical induction that $3^n > 5n + 2$ for $n \geq 4$.

Classified Practice

Recurrence and Induction

Q115 **Written****Prove by mathematical induction that $4^n > 2n + 3$ for $n \geq 4$.**

Classified Practice

Recurrence and Induction

Q116 MCQ

Select the correct option: which inequality is proved for $n \geq 3$?:**Choose the correct option.**

A $2^n \geq 1n + 2$

B $2^n > 2n + 2$

C $2^n + 1 > 1n + 2$

D $2^n > 1n + 2$

Classified Practice

Recurrence and Induction

Q117 MCQ

Select the correct option: which inequality is proved for $n \geq 2$:**Choose the correct option.**

A $3^n > 2n$

B $3^n \geq 2n + 0$

C $3^n > 3n + 0$

D $3^n + 1 > 2n + 0$

Classified Practice

Recurrence and Induction

Q118 MCQ

Select the correct option: which inequality is proved for $n \geq 4$?:**Choose the correct option.**

A $3^n + 1 > 5n + 1$

B $3^n > 5n + 1$

C $3^n \geq 5n + 1$

D $3^n > 6n + 1$

Classified Practice

Recurrence and Induction

Q119 MCQ

Select the correct option: which inequality is proved for $n \geq 3$?:

Choose the correct option.

A $3^n \geq 3n + 0$

B $3^n > 4n + 0$

C $3^n + 1 > 3n + 0$

D $3^n > 3n$

Classified Practice

Recurrence and Induction

Q120 MCQ

Select the correct option: which inequality is proved for $n \geq 4$?:**Choose the correct option.**

A $3^n > 5n + 2$

B $3^n \geq 5n + 2$

C $3^n > 6n + 2$

D $3^n + 1 > 5n + 2$

Classified Practice

Recurrence and Induction

Q121 MCQ

Select the correct option: which inequality is proved for $n \geq 3$?:**Choose the correct option.**

A $4^n + 1 > 3n + 1$

B $4^n > 3n + 1$

C $4^n \geq 3n + 1$

D $4^n > 4n + 1$

Classified Practice

Recurrence and Induction

Q122 MCQ

Select the correct option: which inequality is proved for $n \geq 4$?:**Choose the correct option.**

A $4^n \geq 2n + 3$

B $4^n > 3n + 3$

C $4^n + 1 > 2n + 3$

D $4^n > 2n + 3$

Classified Practice

Recurrence and Induction

Q123 MCQ

Select the correct option: which inequality is proved for $n \geq 4$?**Choose the correct option.**

A $4^n > 4n + 1$

B $4^n \geq 4n + 1$

C $4^n > 5n + 1$

D $4^n + 1 > 4n + 1$

Classified Practice

Recurrence and Induction

Q124 **Written**

Prove by mathematical induction that $2^n > 3n + 1$ for $n \geq 4$.

Classified Practice

Recurrence and Induction

Q125 MCQ

Select the correct option: which inequality is proved for $n \geq 3$?:**Choose the correct option.**

A $4^n + 1 > 3n + 0$

B $4^n > 3n$

C $4^n \geq 3n + 0$

D $4^n > 4n + 0$

Classified Practice

Recurrence and Induction

Q126 **Challenge**

When $h > 0$ and $n \geq 3$, prove $(1 + h)^n > 1 + nh^2$.

Classified Practice

Recurrence and Induction

Q127 Written

Prove that $n^3 + (n + 1)^3 + (n + 2)^3$ is a multiple of 9 for every natural number n , using mathematical induction.

Classified Practice

Recurrence and Induction

Q128 Written

Prove that $n^3 + 2n$ is a multiple of 3 for every natural number n , using mathematical induction.

Classified Practice

Recurrence and Induction

Q129 Written

Prove by mathematical induction that $n^2 + n$ is a multiple of 2.

Classified Practice

Recurrence and Induction

Q130 Written

Prove by mathematical induction that $n^2 + 5n + 8$ is a multiple of 2.

Classified Practice

Recurrence and Induction

Q131 **Written****Prove by mathematical induction that $n^2 + 7n + 12$ is a multiple of 2.**

Classified Practice

Recurrence and Induction

Q132 Written

Prove by mathematical induction that $n^2 + 11n + 32$ is a multiple of 2.

Classified Practice

Recurrence and Induction

Q133 MCQ

Select the correct option: which positive integer is guaranteed to divide $n^2 + n$ for every natural number n ?:

Choose the correct option.

A 2

B 3

C 4

D 5

Classified Practice

Recurrence and Induction

Q134 MCQ

Select the correct option: which positive integer is guaranteed to divide $n^2 + 5n + 8$ for every natural number n ?:

Choose the correct option.

A 3

B 4

C 5

D 2

Classified Practice

Recurrence and Induction

Q135 MCQ

Select the correct option: which positive integer is guaranteed to divide $n^2 + 7n + 12$ for every natural number n ?:

Choose the correct option.

A 4

B 5

C 2

D 3

Classified Practice

Recurrence and Induction

Q136 MCQ

Select the correct option: which positive integer is guaranteed to divide $n^2 + 11n + 32$ for every natural number n ?:

Choose the correct option.

A 4

B 5

C 2

D 3

Classified Practice

Recurrence and Induction

Q137 MCQ

Select the correct option: which positive integer is guaranteed to divide $n^3 + 3n^2 + 2n$ for every natural number n ?:

Choose the correct option.

A 9

B 6

C 7

D 8

Classified Practice

Recurrence and Induction

Q138 MCQ

Select the correct option: which positive integer is guaranteed to divide $n^3 + 9n^2 + 26n + 30$ for every natural number n ?:

Choose the correct option.

A 6

B 7

C 8

D 9

Classified Practice

Recurrence and Induction

Q139 MCQ

Select the correct option: which positive integer is guaranteed to divide $n^3 + 12n^2 + 47n + 60$ for every natural number n ?:

Choose the correct option.

A 7

B 8

C 9

D 6

Classified Practice

Recurrence and Induction

Q140 MCQ

Select the correct option: which positive integer is guaranteed to divide $n^3 + 18n^2 + 107n + 216$ for every natural number n ?:

Choose the correct option.

A 7

B 8

C 9

D 6

Classified Practice

Recurrence and Induction

Q141 MCQ

Select the correct option: which positive integer is guaranteed to divide $n^2 + n + 2$ for every natural number n ?:

Choose the correct option.

A 3

B 4

C 5

D 2

Classified Practice

Recurrence and Induction

Q142 **Written**

Prove by mathematical induction that $2n^3 - 3n^2 + n$ is a multiple of 6.

Classified Practice

Recurrence and Induction

Q143 **Written****Prove by mathematical induction that $4n^3 - n$ is a multiple of 3.**

Classified Practice

Recurrence and Induction

Q144 Challenge

For any integer $n \geq 1$, prove that $(n + 1)^n - 1$ is a multiple of n^2 .

Classified Practice

Recurrence and Induction

Q145 Written

For the sequence determined by $a_1 = 3$ and $a_{n+1} = \frac{a_n^2 - 1}{n+1}$, find a_2, a_3, a_4 and estimate the general term.

Q146 Written

For the sequence determined by $a_1 = -1$ and $a_{n+1} = a_n^2 + 2na_n - 2$, find a_2, a_3, a_4 and estimate the general term.

Classified Practice

Recurrence and Induction

Q147 Written

For the sequence determined by $a_1 = 1$ and $a_{n+1} = a_n^2 - n^2 a_n + (n+1)^2$, find a_2, a_3, a_4 and estimate the general term.

Q148 Written

For the sequence determined by $a_1 = 1$ and $a_{n+1} = \frac{a_n - 4}{a_n - 3}$, find a_2, a_3, a_4 and estimate the general term.

Classified Practice

Recurrence and Induction

Q149 Written

For a sequence with $a_1 = 3$ and $a_{n+1} = \frac{a_n^2 - 1}{n+1}$, prove by mathematical induction that $a_n = n + 2$.

Classified Practice

Recurrence and Induction

Q150 MCQ

Select the correct option: for $a_1 = 1$ and $a_{n+1} = a_n + 2$, the general term is:

Choose the correct option.

A $a_n = 2n$

B $a_n = 2(n - 1)$

C $a_n = 1 - 2n$

D $a_n = 2n - 1$

Classified Practice

Recurrence and Induction

Q151 MCQ

Select the correct option: for $a_1 = 2$ and $a_{n+1} = a_n + 2$, the general term is:

Choose the correct option.

A $a_n = 2n - 1$

B $a_n = -2n$

C $a_n = 2n$

D $a_n = 2n + 1$

Classified Practice

Recurrence and Induction

Q152 MCQ

Select the correct option: for $a_1 = 3$ and $a_{n+1} = a_n + 2$, the general term is:

Choose the correct option.

A $a_n = -2n - 1$

B $a_n = 2n + 1$

C $a_n = 2(n + 1)$

D $a_n = 2n$

Classified Practice

Recurrence and Induction

Q153 MCQ

Select the correct option: for $a_1 = 4$ and $a_{n+1} = a_n + 2$, the general term is:

Choose the correct option.

A $a_n = 2(n + 1)$

B $a_n = 2n + 3$

C $a_n = 2n + 1$

D $a_n = -2(n + 1)$

Classified Practice

Recurrence and Induction

Q154 MCQ

Select the correct option: for $a_1 = 1$ and $a_{n+1} = a_n + 3$, the general term is:

Choose the correct option.

A $a_n = 3n - 1$

B $a_n = 3(n - 1)$

C $a_n = 2 - 3n$

D $a_n = 3n - 2$

Classified Practice

Recurrence and Induction

Q155 MCQ

Select the correct option: for $a_1 = 2$ and $a_{n+1} = a_n + 3$, the general term is:

Choose the correct option.

A $a_n = 3n - 2$

B $a_n = 1 - 3n$

C $a_n = 3n - 1$

D $a_n = 3n$

Classified Practice

Recurrence and Induction

Q156 MCQ

Select the correct option: for $a_1 = 3$ and $a_{n+1} = a_n + 3$, the general term is:

Choose the correct option.

A $a_n = -3n$

B $a_n = 3n$

C $a_n = 3n + 1$

D $a_n = 3n - 1$

Classified Practice

Recurrence and Induction

Q157 MCQ

Select the correct option: for $a_1 = 4$ and $a_{n+1} = a_n + 3$, the general term is:

Choose the correct option.

A $a_n = 3n + 1$

B $a_n = 3n + 2$

C $a_n = 3n$

D $a_n = -3n - 1$

Classified Practice

Recurrence and Induction

Q158 **Written**

For a sequence with $a_1 = 1$ and $a_{n+1} = a_n^2 - n^2 a_n + (n+1)^2$, prove by mathematical induction that $a_n = n^2$.

Classified Practice

Recurrence and Induction

Q159 Written

For a sequence with $a_1 = -1$ and $a_{n+1} = a_n^2 + 2na_n - 2$, prove by mathematical induction that $a_n = 1 - 2n$.

Classified Practice

Recurrence and Induction

Q160 MCQ

Select the correct option: for $a_1 = 1$ and $a_{n+1} = 2a_n + 0$, the general term is:

Choose the correct option.

A $a_n = \frac{2^n - 2}{2}$

B $a_n = -2^{n-1}$

C $a_n = 2^{n-1}$

D $a_n = \frac{2^n + 2}{2}$

Classified Practice

Recurrence and Induction

Q161 Written

For a sequence with $a_1 = 1$ and $a_{n+1} = \frac{a_n - 4}{a_n - 3}$, prove by mathematical induction that $a_n = \frac{2n-1}{n}$.

Classified Practice

Recurrence and Induction

Q162 Challenge

Three numbers A, B, C in that order form a geometric sequence with common ratio r . If $A, 2B, C$ in that order form an arithmetic sequence, determine the possible values of r .

Classified Practice

Recurrence and Induction

Quiz MCQ 1 **Concept Quiz · MCQ**

Select the correct option: For the sequence $\{a_n\}$ with $a_1 = 2$ and recurrence $a_{n+1} = 1a_n + (4)$. The general term is:

Choose the correct option.

A $4n$

B $4n - 5$

C $4n + 1$

D $a_n = 2(2n - 1)$

Classified Practice

Recurrence and Induction

Quiz MCQ 2 **Concept Quiz · MCQ**

Select the correct option: If $a_1 = 2$ and $a_{n+1} = -2a_n + (3)$, the general term is:

Choose the correct option.

A $a_n = -\frac{(-2)^n - 2}{2}$

B $-\frac{(-2)^n - 4}{2}$

C $-\frac{(-2)^n + 4}{2}$

D $-\frac{(-2)^n - 6}{2}$

Classified Practice

Recurrence and Induction

Quiz MCQ 3 **Concept Quiz · MCQ**

Select the correct option: which inequality is proved for $n \geq 4$?:

Choose the correct option.

A $2^n + 1 > 3n + 1$

B $2^n > 3n + 1$

C $2^n \geq 3n + 1$

D $2^n > 4n + 1$

Classified Practice

Recurrence and Induction

Quiz MCQ 4 [Concept Quiz · MCQ](#)

Select the correct option: for $a_1 = 4$ and $a_{n+1} = a_n + 2$, the general term is:

Choose the correct option.

A $a_n = 2n + 1$

B $a_n = -2(n + 1)$

C $a_n = 2(n + 1)$

D $a_n = 2n + 3$

Classified Practice

Recurrence and Induction

Quiz WRITTEN 5 Concept Quiz · Written

For the sequence $\{a_n\}$ with $a_1 = 2$ and recurrence $a_{n+1} = 1a_n + (5)$. Find the general term.

Classified Practice

Recurrence and Induction

Quiz WRITTEN 6 Concept Quiz · Written

Find the general term of $\{a_n\}$ given $a_1 = 1$ and $a_{n+1} = 5a_n + (8)$.

Classified Practice

Recurrence and Induction

Quiz WRITTEN 7 Concept Quiz · Written

Prove by mathematical induction that $2^n > 4n + 1$ for $n \geq 4$.

Classified Practice

Recurrence and Induction

Quiz WRITTEN 8 Concept Quiz · Written

For a sequence with $a_1 = 5$ and $a_{n+1} = a_n + 1(2n + 1) + 2$, prove by mathematical induction that $a_n = n^2 + 2n + 2$.

Answer key

Use the question number shown on each page.

Q001 $a_2 = 8, a_3 = 26, a_4 = 80$

Q002 $a_2 = -9, a_3 = -15, a_4 = -27$

Q003 $a_2 = -7, a_3 = -7, a_4 = -4$

Q004 $a_2 = -3, a_3 = -5, a_4 = -8$

Q005 $a_2 = 4, a_3 = 8, a_4 = 14$

Q006 $a_2 = 1, a_3 = 10, a_4 = 37$

Q007 B

Q008 D

Q009 A

Q010 B

Q011 D

Q012 A

Q013 C

Q014 D

Q015 C

Q016 D

Q017 B

Q018 C

Q019 $a_n = 5n - 6$

Q020 $a_n = 3n - 2$

Q021 $a_n = -2(n - 3)$

Q022 $a_n = 6n - 5$

Q023 $a_n = 8 - 5n$

Q024 $a_n = 3n - 5$

Q025 $a_n = -4(n - 2)$

Q026 D

Q027 A

Q028 A

Q029 A

Q030 $a_n = 2 \cdot 3^{n-1}$

Q031 $a_n = 3 \cdot 5^{n-1}$

Q032 $a_n = 4 \cdot 3^{n-1}$

Q033 $a_n = 2^{2n-2}$

Q034 B

Q035 B

Q036 D

Q037 A

Q038 $a_n = \frac{3(-2)^n}{2}$

Q039 $a_n = 2(-5)^{n-1}$

Q040 $a_n = \frac{(-4)^n}{2}$

Q041 B

Q042 C

Q043 B

Q044 C

Q045 $a_2 = 5, a_3 = 13, a_4 = 36$

Q046 $a_2 = 3, a_3 = -1, a_4 = 7$

Q047 $a_2 = 8, a_3 = 22, a_4 = 63$

Q048 $a_n = -\frac{4 \cdot 2^n - 3 \cdot 3^n - 1}{2}$

Q049 $a_n = (n - 2)^2$

Q050 $a_n = \frac{3n^2 - 3n + 2}{2}$

Q051 $a_n = n^2 - n + 2$

Q052 $a_n = \frac{2n^3 - 3n^2 + n + 4}{2}$

Q053 $a_n = \frac{3n^2 - n + 2}{2}$

Q054 $a_n = \frac{4n^3 - 3n^2 - n + 12}{6}$

Q055 C

Q056 A

Q057 C

Q058 D

Q059 C

Q060 A

Q061 D

Q062 A

Q063 B

Q064 A

Q065 $a_n = \frac{2n^3 - 3n^2 + n + 12}{6}$

Q066 A

Q067 $a_n = 2^n - 1$

Q068 $a_n = -\frac{3^n - 7}{2}$

Q069 $a_n = -\frac{4^n - 16}{3}$

Q070 $a_n = -\frac{n^2 - n - 200}{50}$

Q071 $a_n = 3^n - 2$

Q072 $a_n = \frac{3 \cdot 5^n - 10}{5}$

Q073 $a_n = \frac{2^n + 2}{2}$

Q074 $a_n = -\frac{4^n + 2}{2}$

Q075 $a_n = \frac{2 \cdot 3^n + 3}{3}$

Q076 $a_n = (-3)^n + 1$

Q077 $a_n = 3 \cdot 3^{-n} (3^n - 2)$

Q078 $a_n = \frac{3 \cdot 3^{-n} (7 \cdot 2^n - 2 \cdot 3^n)}{4}$

Q079 $a_n = -\frac{3(-2)^n - 2}{2}$

Q080 B

Q081 C

Q082 D

Q083 B

Q084 C

Q085 D

Q086 B

Q087 C

Q088 D

Q089 D

Q090 $c = 3, a_n = -\frac{2^{2n} - 12}{4}, a_5 = -253$

Q091 $\frac{n(n+1)}{2}$

Q092 $\frac{n(2n-1)(2n+1)}{3}$

Q093 $\frac{n(n+1)(n+2)}{3}$

Q094 A

Q095 B	Q113 $3^n > 5n + 1$	Q133 A	Q152 B
Q096 C	Q114 $3^n > 5n + 2$	Q134 D	Q153 A
Q097 B	Q115 $4^n > 2n + 3$	Q135 C	Q154 D
Q098 C	Q116 D	Q136 C	Q155 C
Q099 A	Q117 A	Q137 B	Q156 B
Q100 C	Q118 B	Q138 A	Q157 A
Q101 A	Q119 D	Q139 D	Q158 $a_n = n^2$
Q102 B	Q120 A	Q140 D	Q159 $a_n = 1 - 2n$
Q103 n^2	Q121 B	Q141 D	Q160 C
Q104 $\frac{n(n+1)(n+2)}{3}$	Q122 D	Q142 $6 \mid (2n^3 - 3n^2 + n)$	Q161 $a_n = \frac{2n-1}{n}$
Q105 $\frac{n(n+1)(2n+7)}{6}$	Q123 A	Q143 $3 \mid (4n^3 - n)$	Q162 $r = 2 \pm \sqrt{3}$
Q106 $4^n - 1$	Q124 $2^n > 3n + 1$	Q144 $n^2 \mid [(n+1)^n - 1]$	Quiz MCQ 1 D
Q107 $\frac{10 \cdot 10^n - 1}{9}$	Q125 B	Q145 $a_2 = 4, a_3 = 5, a_4 = 6; a_n = n + 2$	Quiz MCQ 2 A
Q108 $\frac{n(n^2 + 3n + 5)}{3}$	Q126 $(1+h)^n > 1 + nh^2$	Q146 $a_2 = -3, a_3 = -5, a_4 = -7; a_n = 1 - 2n$	Quiz MCQ 3 B
Q109 $2^n > 3n$	Q127 $9 \mid [n^3 + (n+1)^3 + (n+2)^3]$	Q147 $a_2 = 4, a_3 = 9, a_4 = 16; a_n = n^2$	Quiz MCQ 4 C
Q110 $3^n > 5n + 1$	Q128 $3 \mid (n^3 + 2n)$	Q148 $a_2 = \frac{3}{2}, a_3 = \frac{5}{3}, a_4 = \frac{7}{4}; a_n = \frac{2n-1}{n}$	Quiz WRITTEN 5 $a_n = 5n - 3$
Q111 $3^n > 2n$	Q129 $2 \mid (n^2 + n)$	Q149 $a_n = n + 2$	Quiz WRITTEN 6 $a_n = \frac{3 \cdot 5^n - 10}{5}$
Q112 $2^n > 1n + 2$	Q130 $2 \mid (n^2 + 5n + 8)$	Q150 D	Quiz WRITTEN 7 $2^n > 4n + 1$
	Q131 $2 \mid (n^2 + 7n + 12)$	Q151 C	Quiz WRITTEN 8 $a_n = n^2 + 2n + 2$
	Q132 $2 \mid (n^2 + 11n + 32)$		